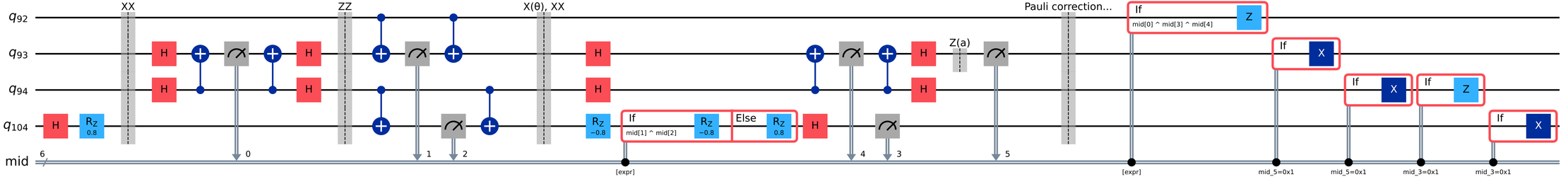
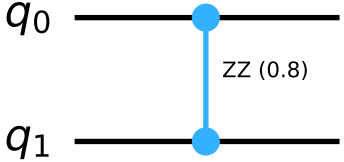
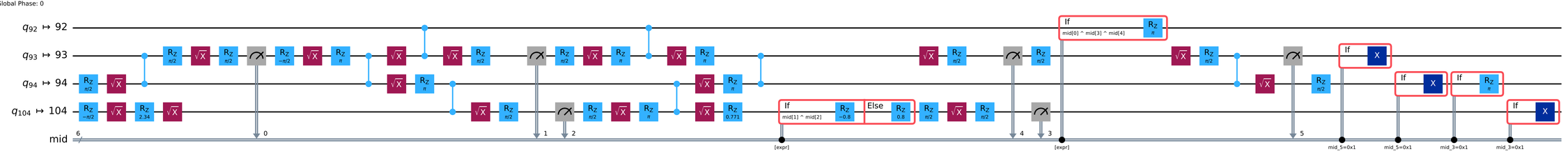
[Z2] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, 6 mid-circuit measurements, logical→physical [92, 94]



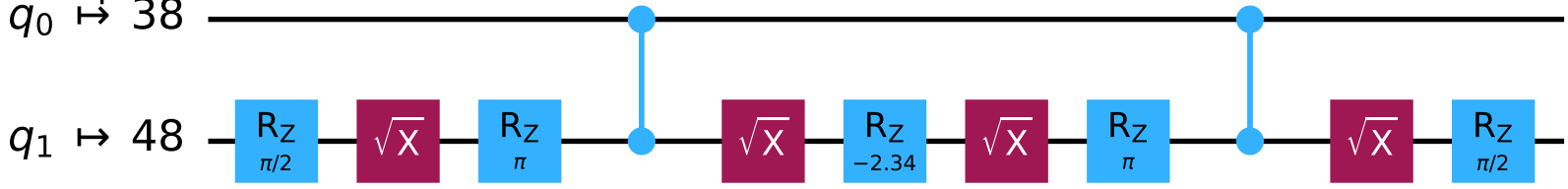
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1



[Z2] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:23 sx:18 cz:8 measure:6 if_else:6)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, total depth 36

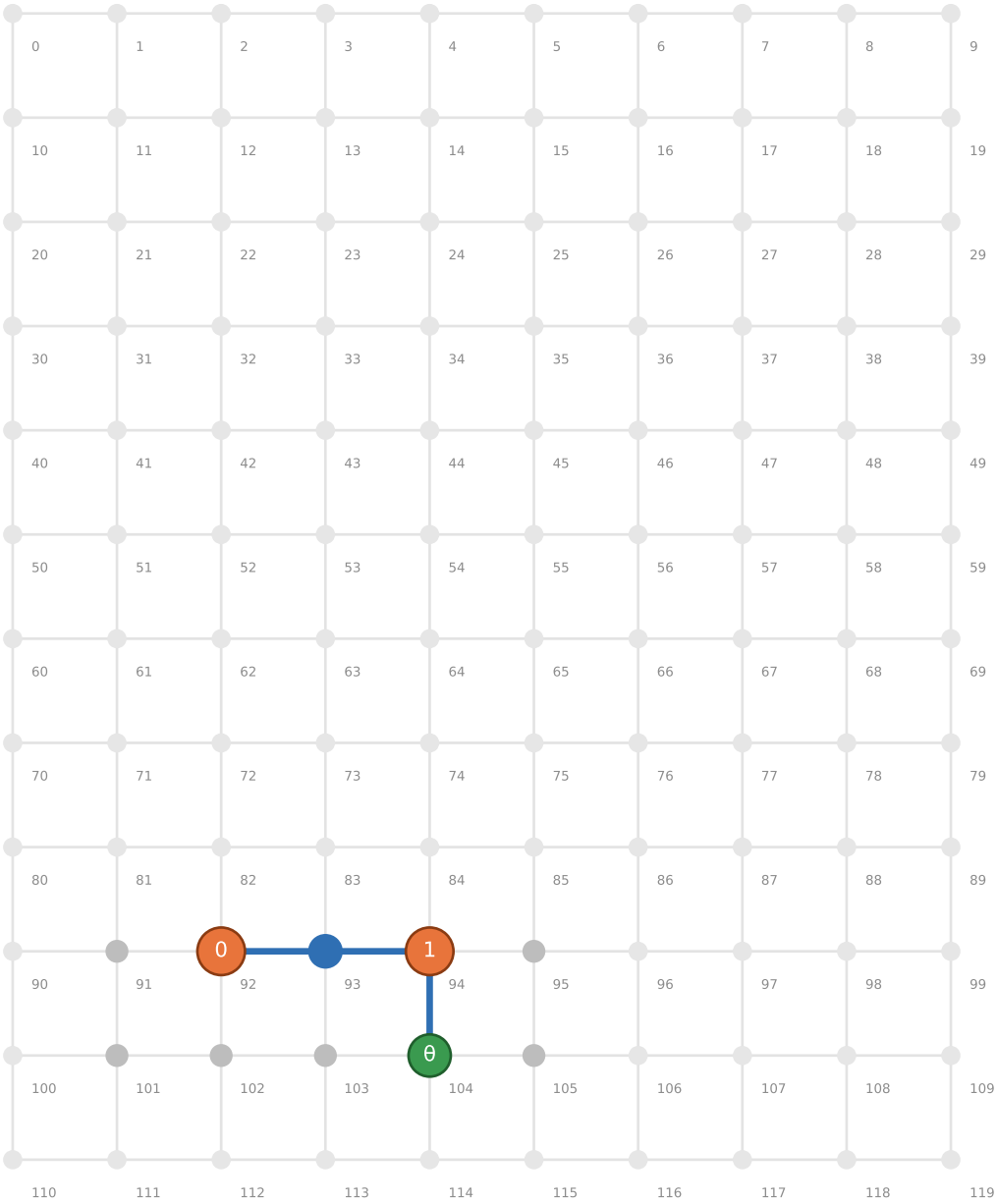


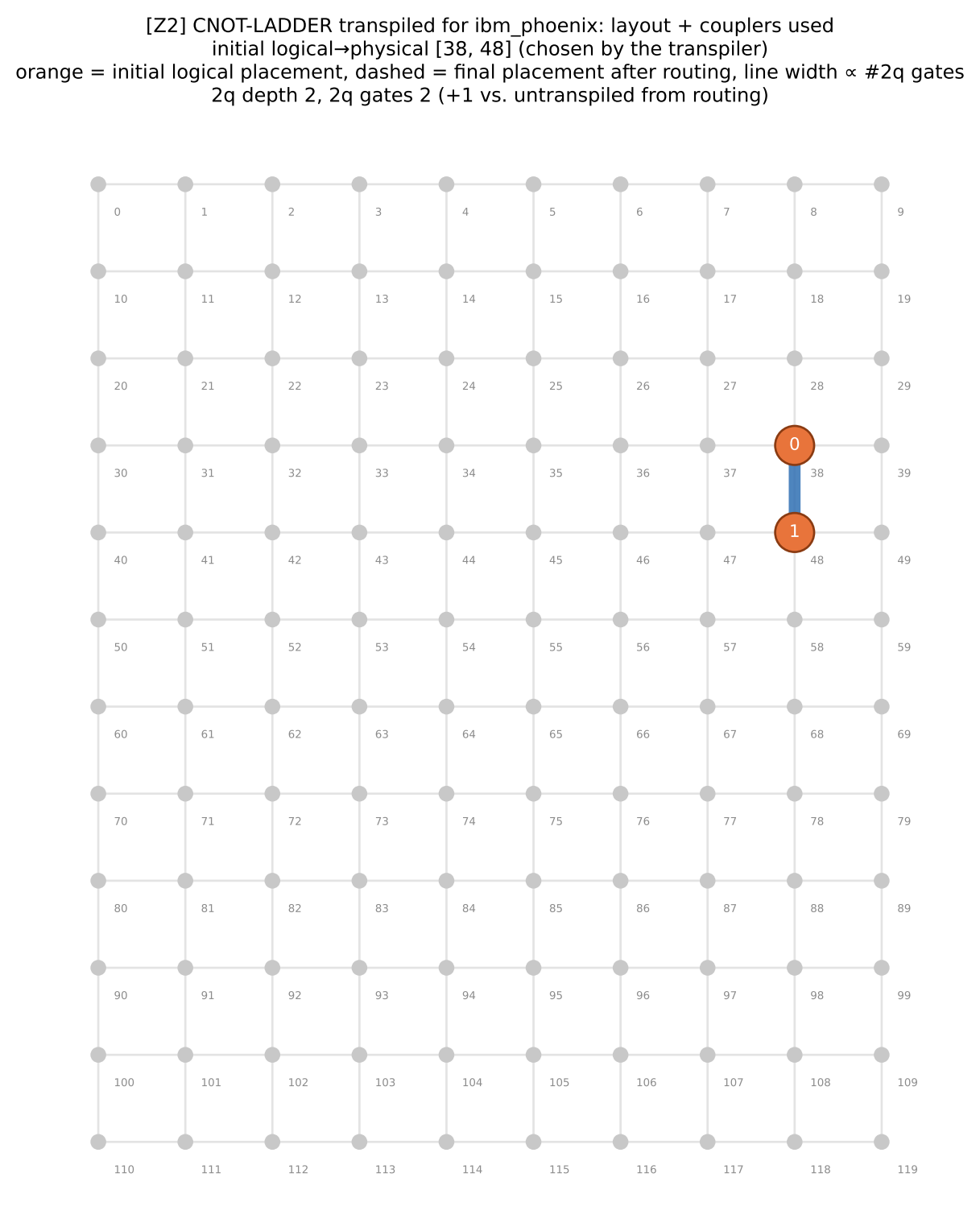
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11



[Z2] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [92, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers




[Z2] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
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98
99
100
101
102
103
104
105
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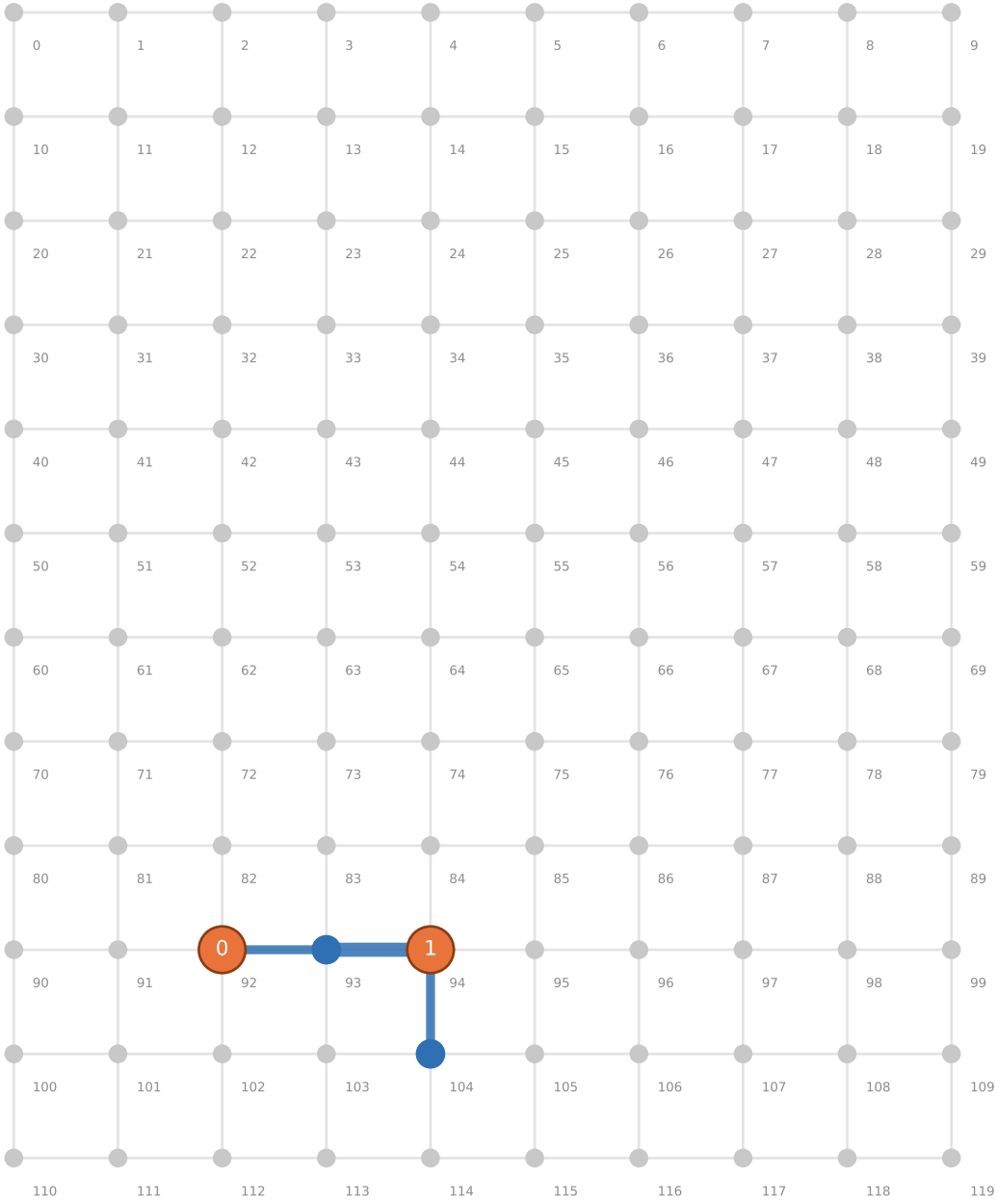
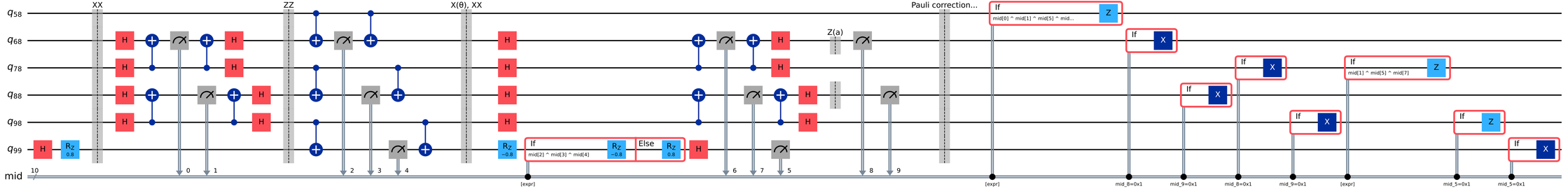
108
109
110
111
112
113
114
115
116
117
118
119
0
1

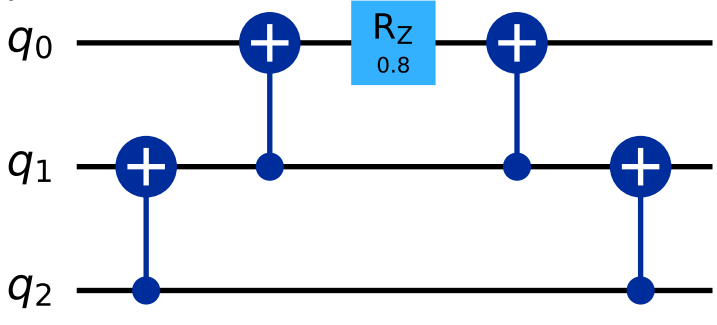
[Z2] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [92, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 8



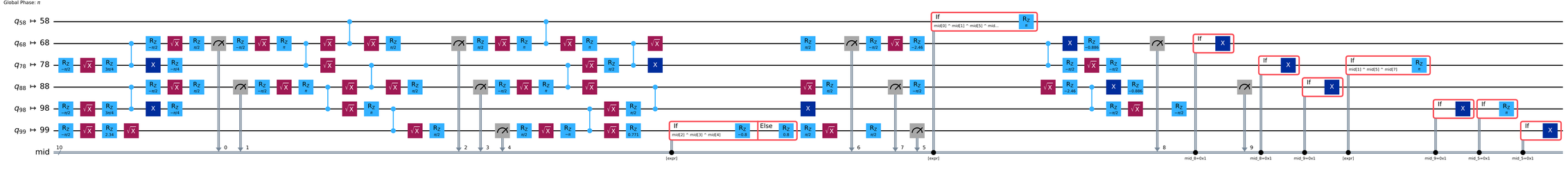
[Z3] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 (3 logical qubits)
2q depth 6, 10 mid-circuit measurements, logical→physical [58, 78, 98]

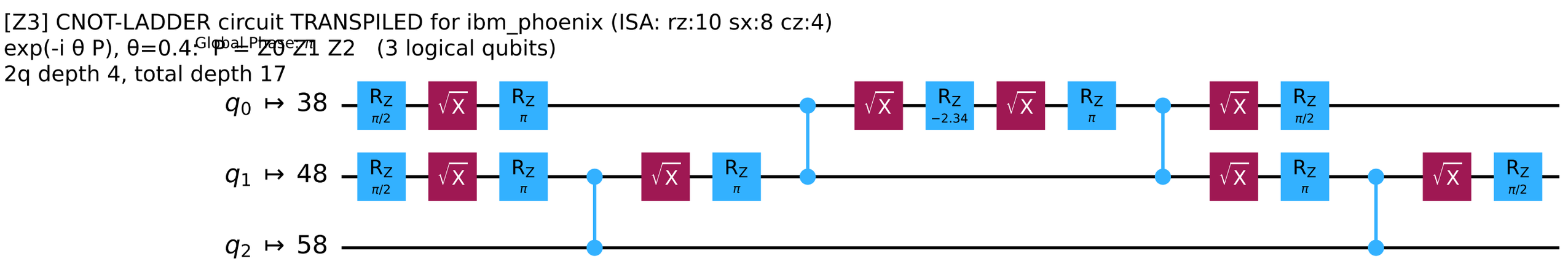


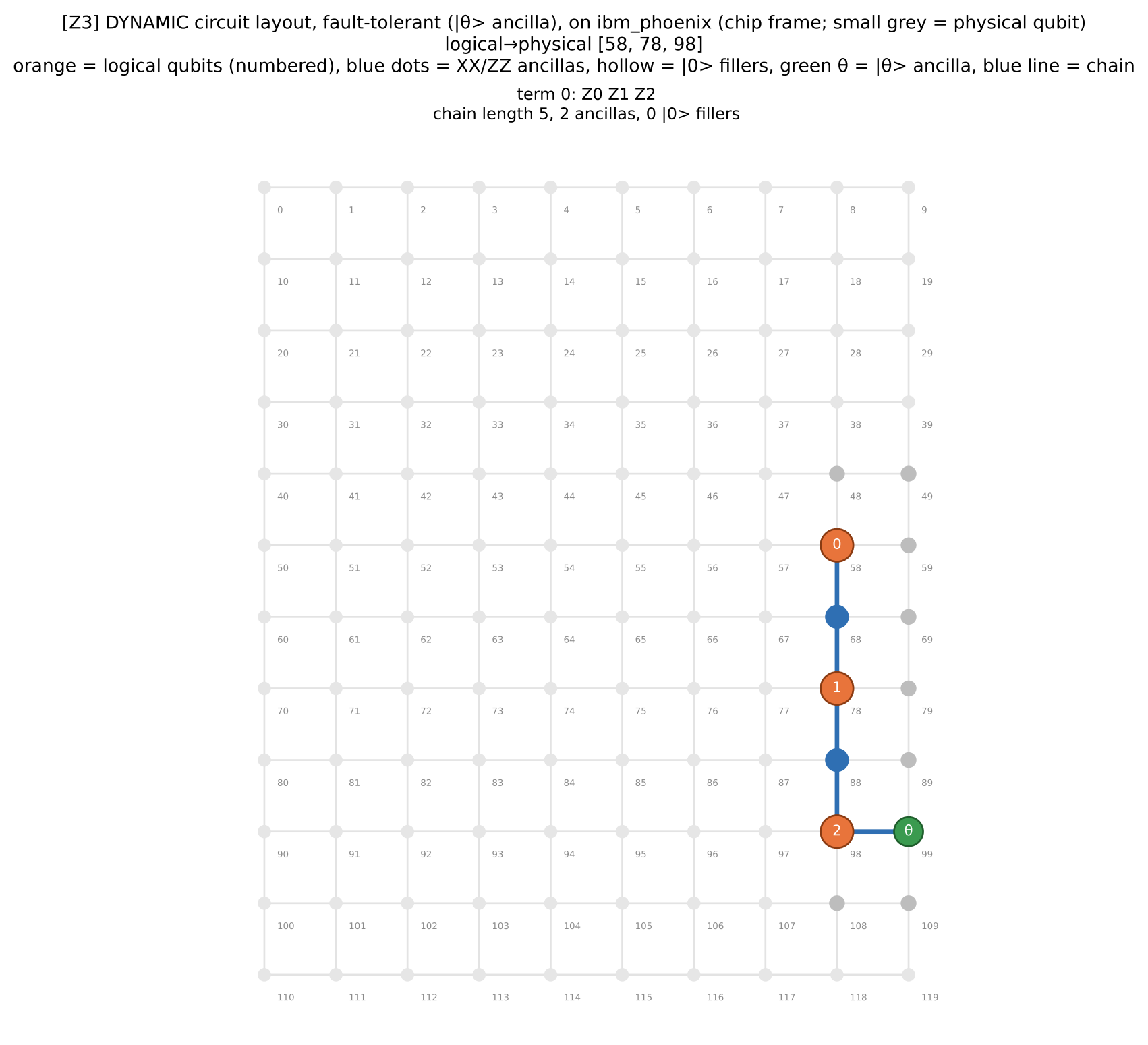
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



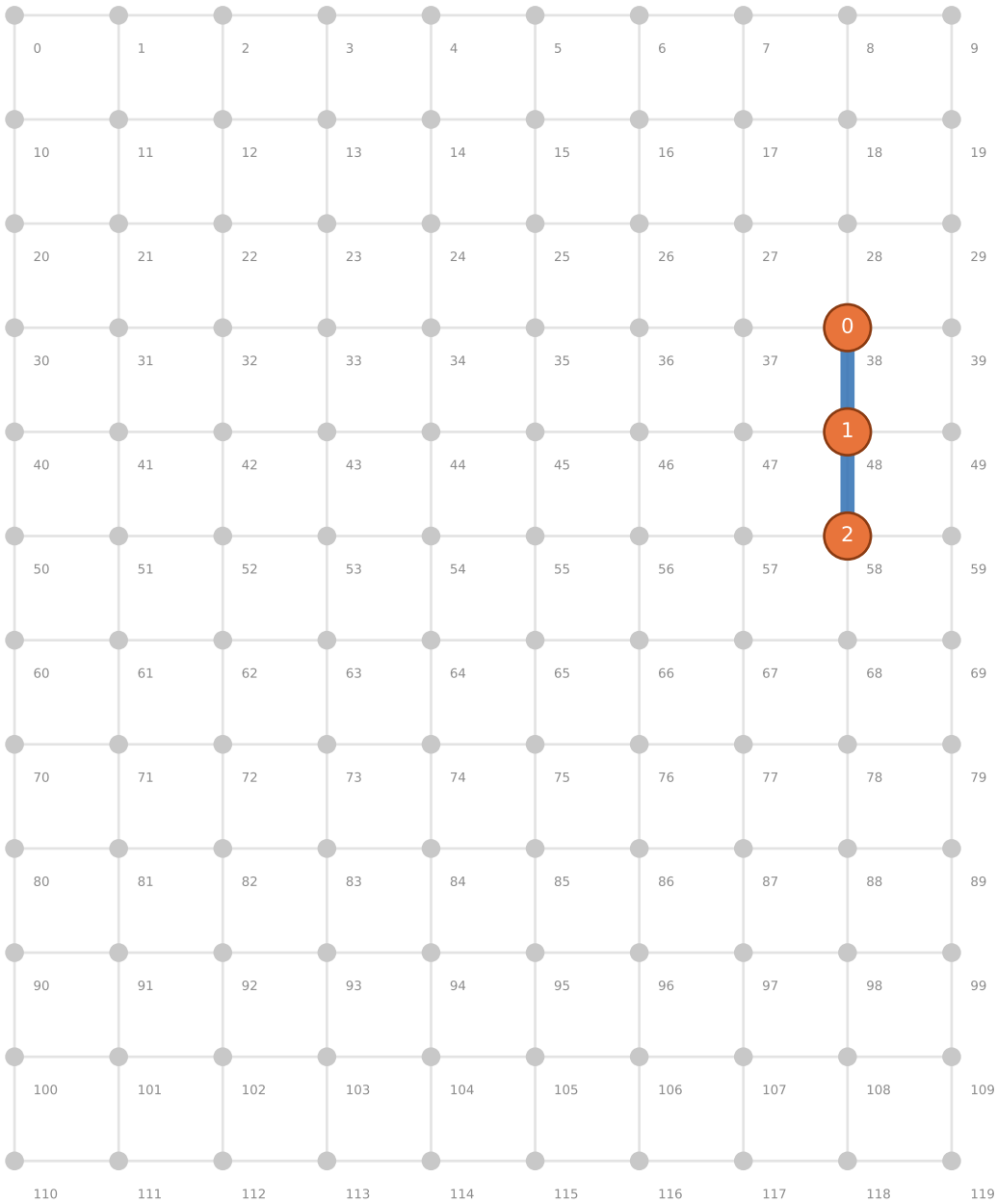
[Z3] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:44 sx:30 cz:14 measure:10 if_else:9 x:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, total depth 39



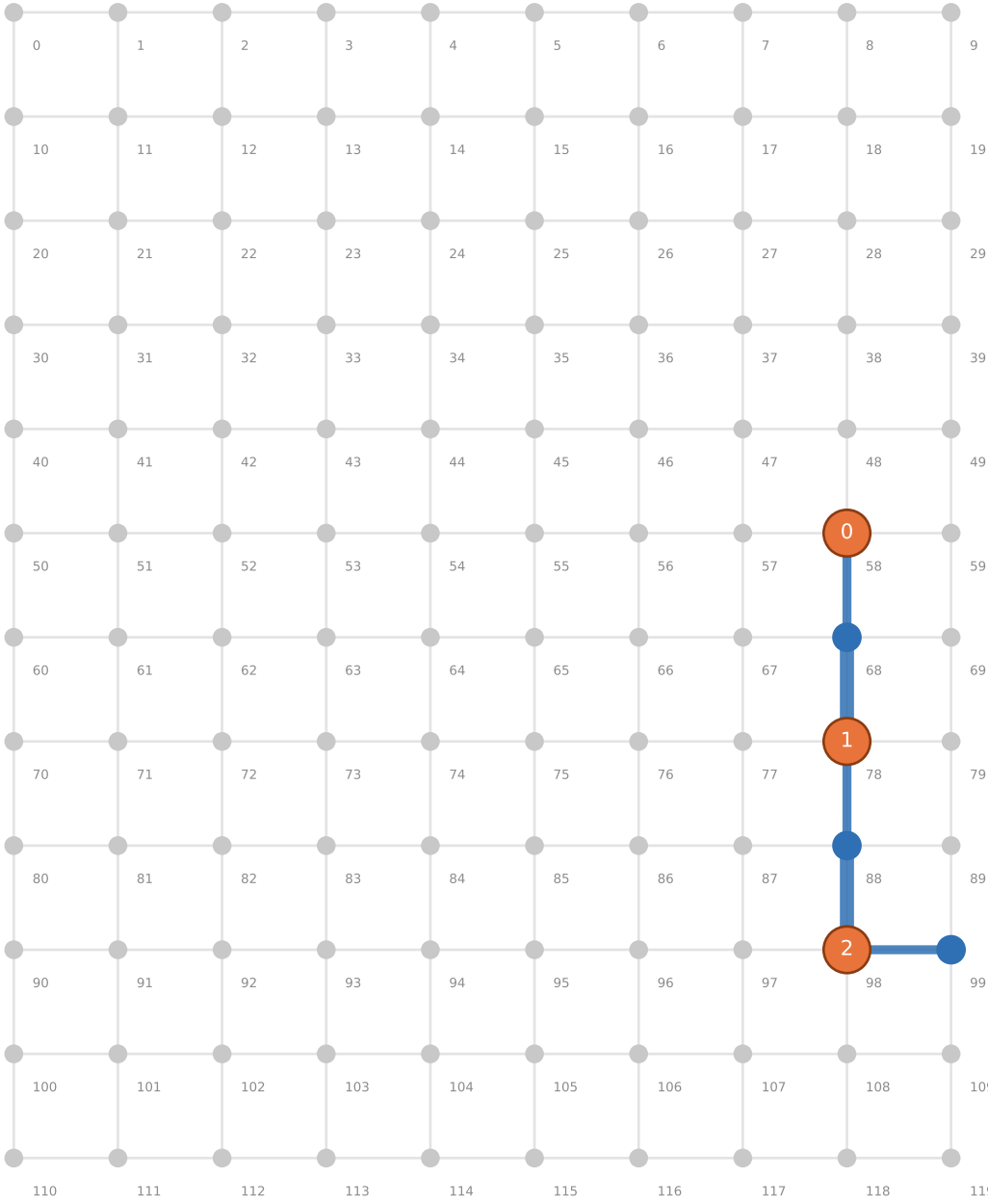

[Z3] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4, total depth 17
GlobalPhase: π
q0 ↦ 38
q1 ↦ 48
q2 ↦ 58
RZ π/2
√X
RZ π
RZ π/2
√X
RZ −2.34
√X
RZ π
√X
RZ π/2
RZ π/2
√X
RZ π
√X
RZ π
√X
RZ π/2


[Z3] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
θ

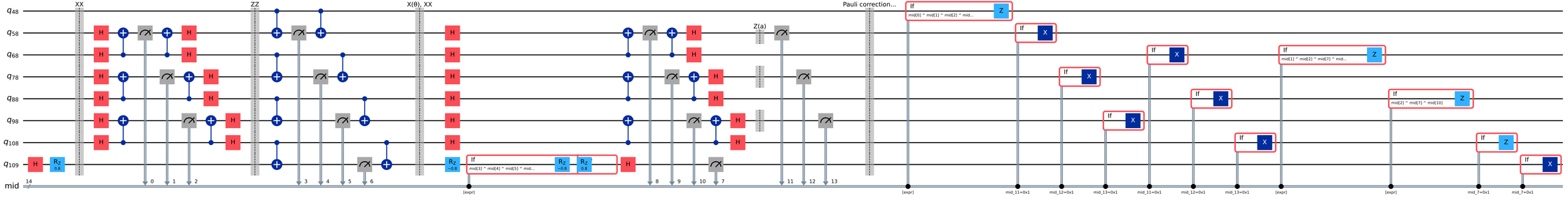
[Z3] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



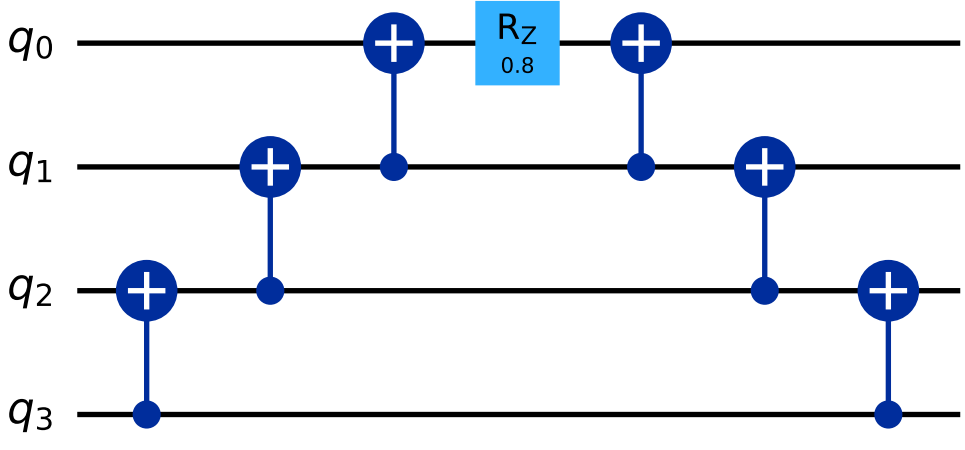
[Z3] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 14



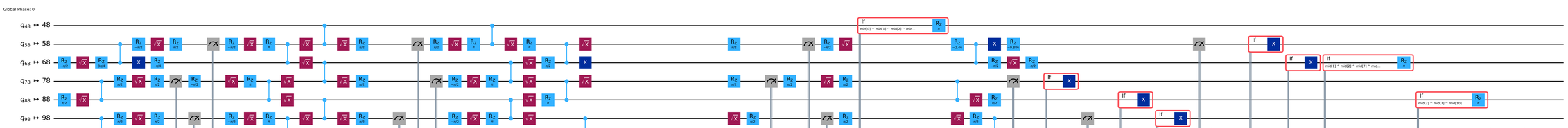
[Z4] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, 14 mid-circuit measurements, logical→physical [48, 68, 88, 108]



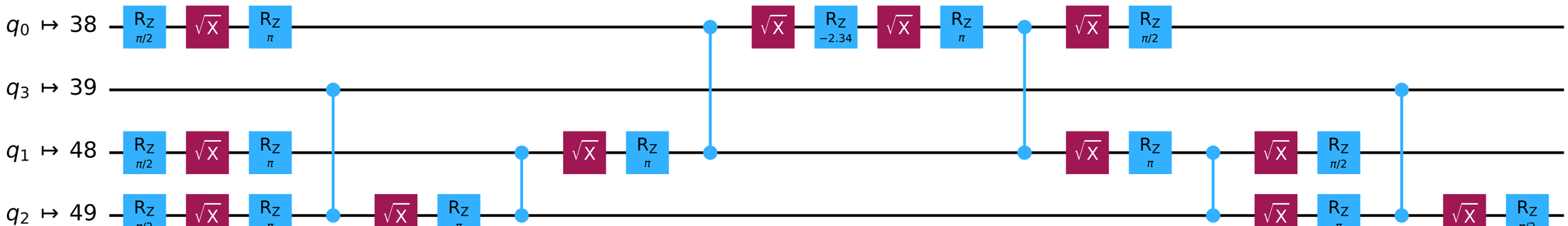
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6



[Z4] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:53 sx:42 cz:20 measure:14 if_else:12 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, total depth 40

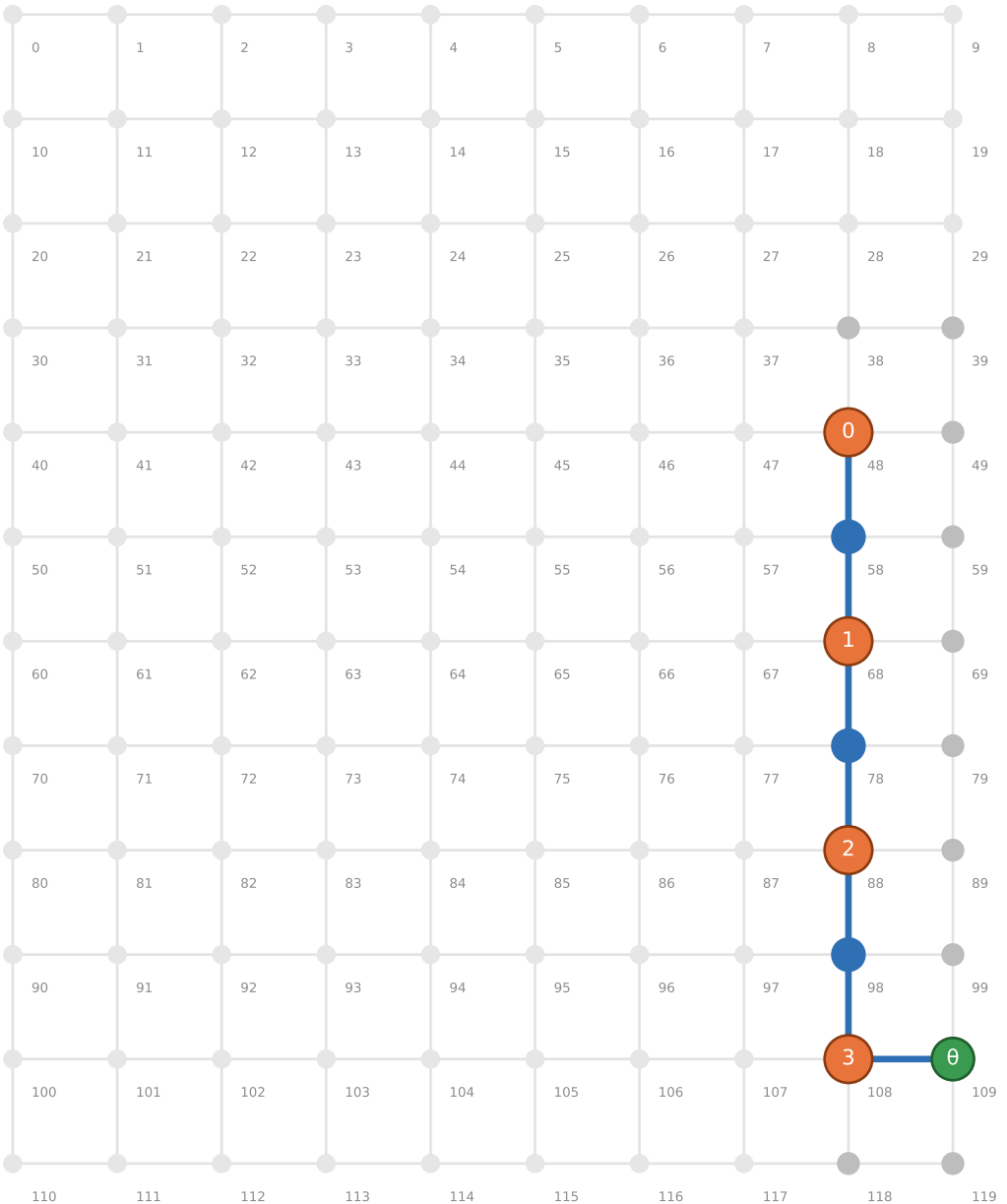


[Z4] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23

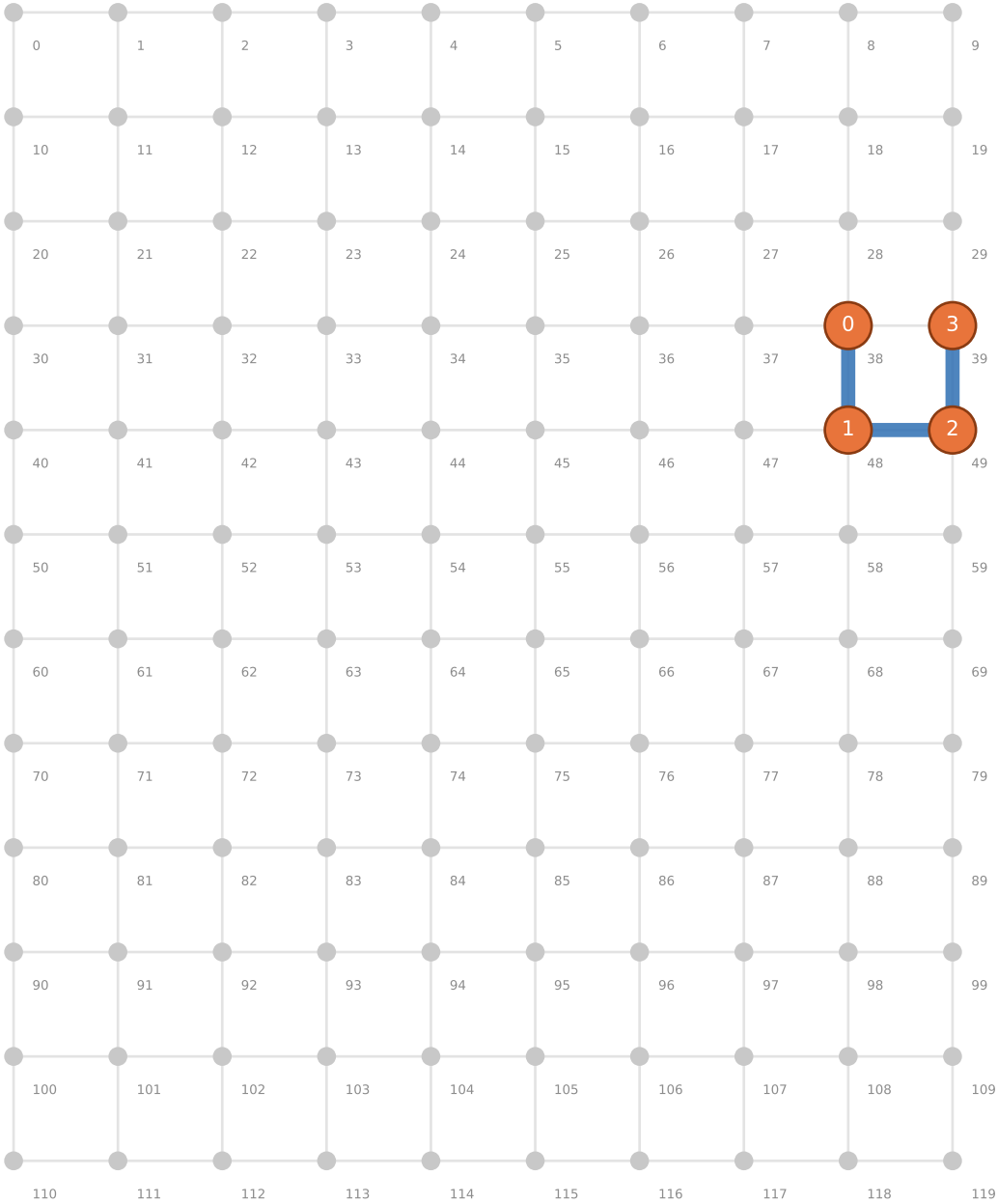


[Z4] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

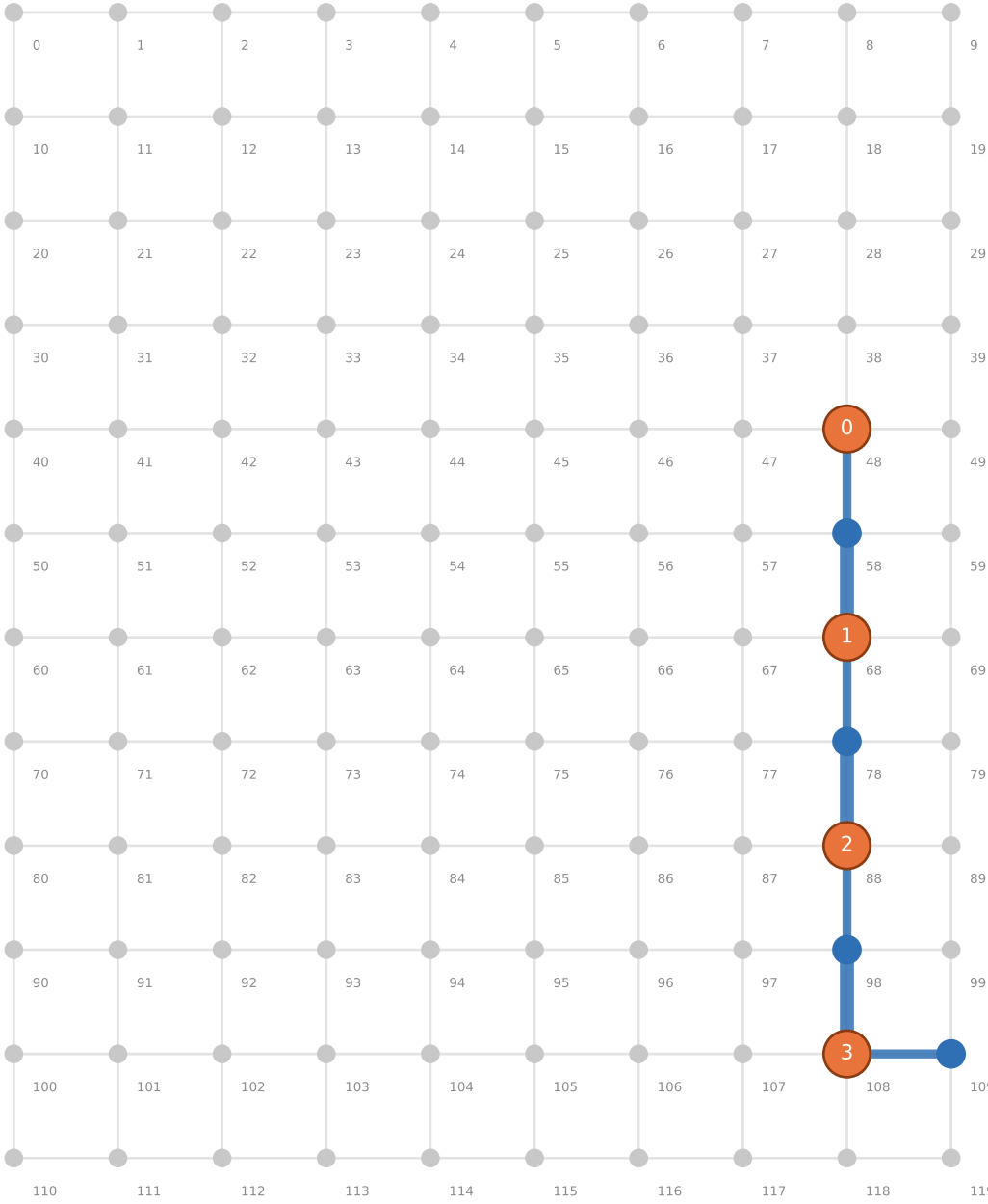
term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers



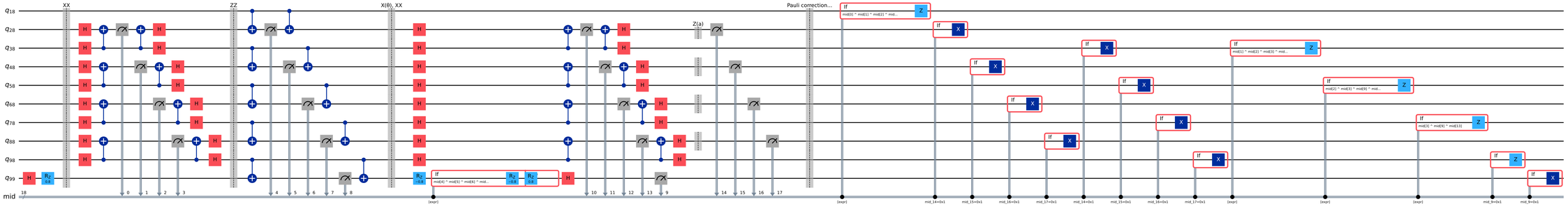
[Z4] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)



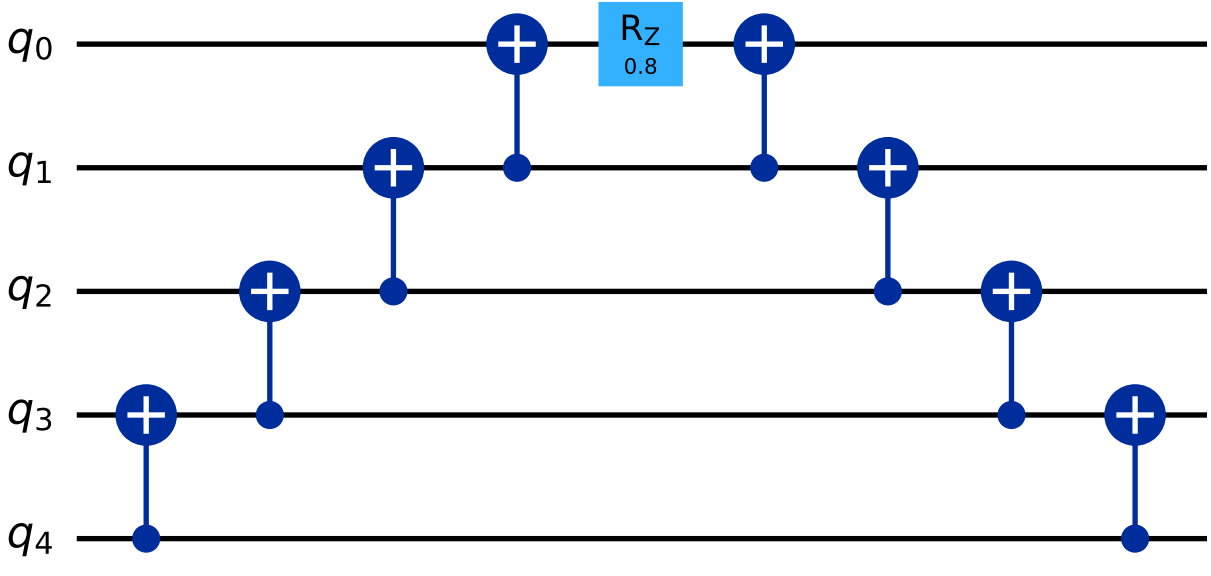
[Z4] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 20



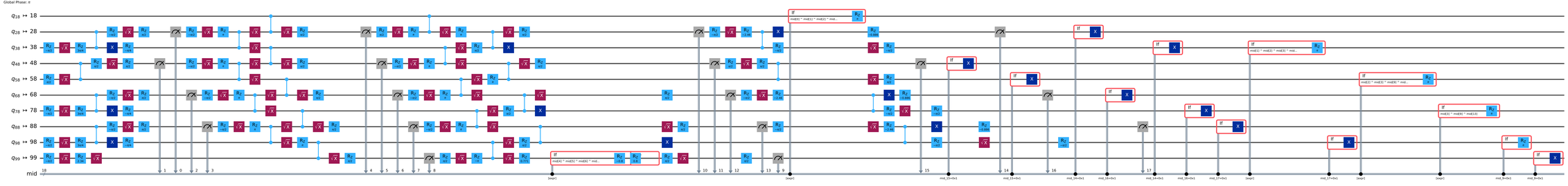
[Z5] DYNAMIC constant-depth circuit, fault-tolerant (|0> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, 18 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



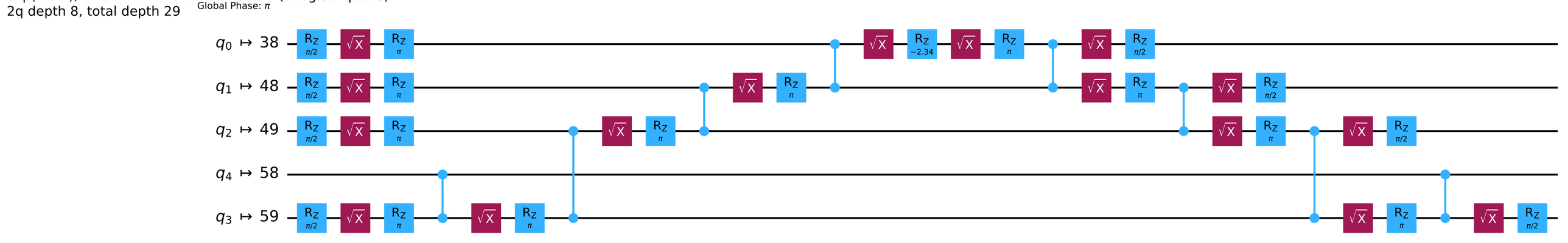
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8



[Z5] DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:74 sx:54 cz:26 measure:18 if_else:15 x:9)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, total depth 41

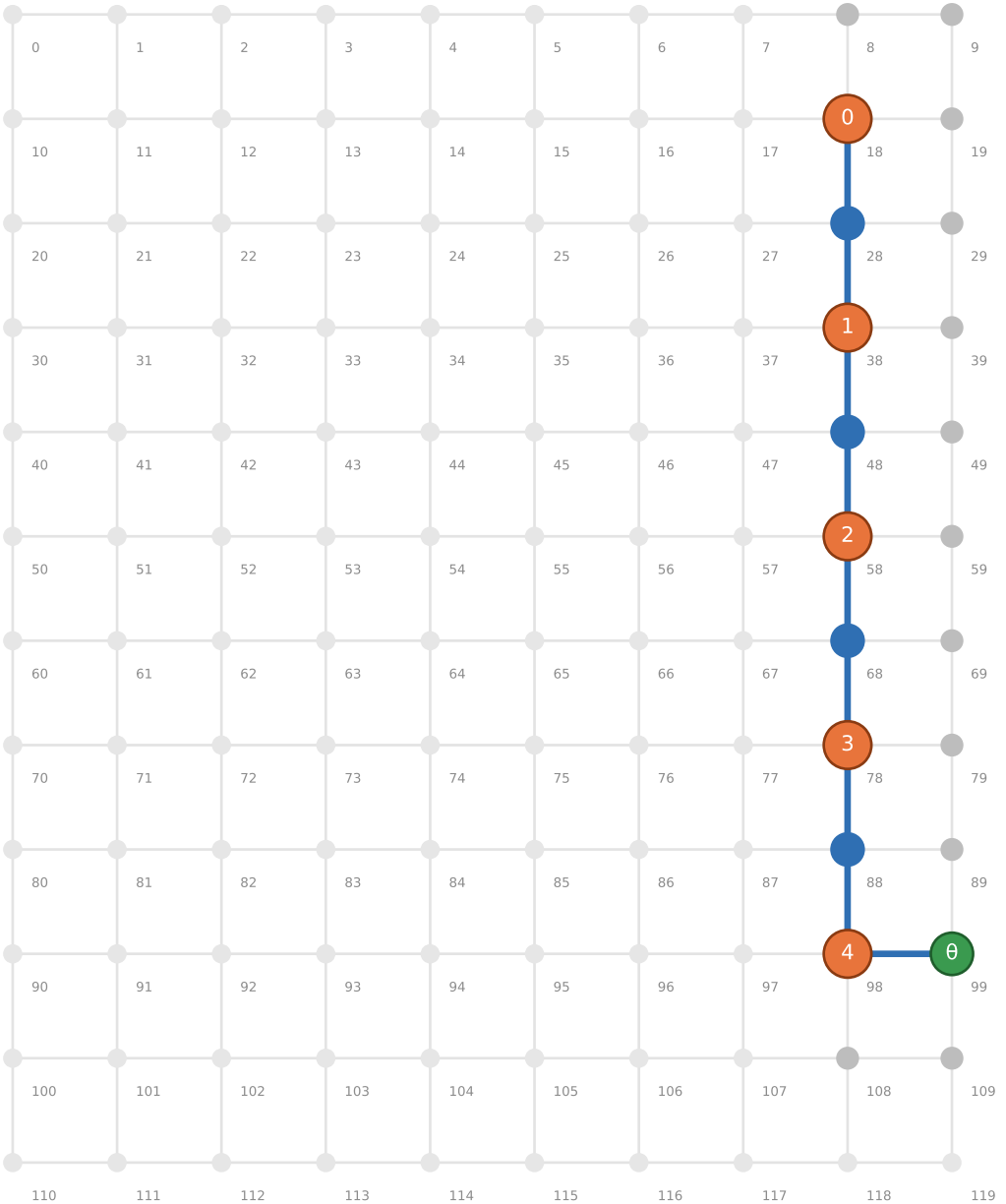


[Z5] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29

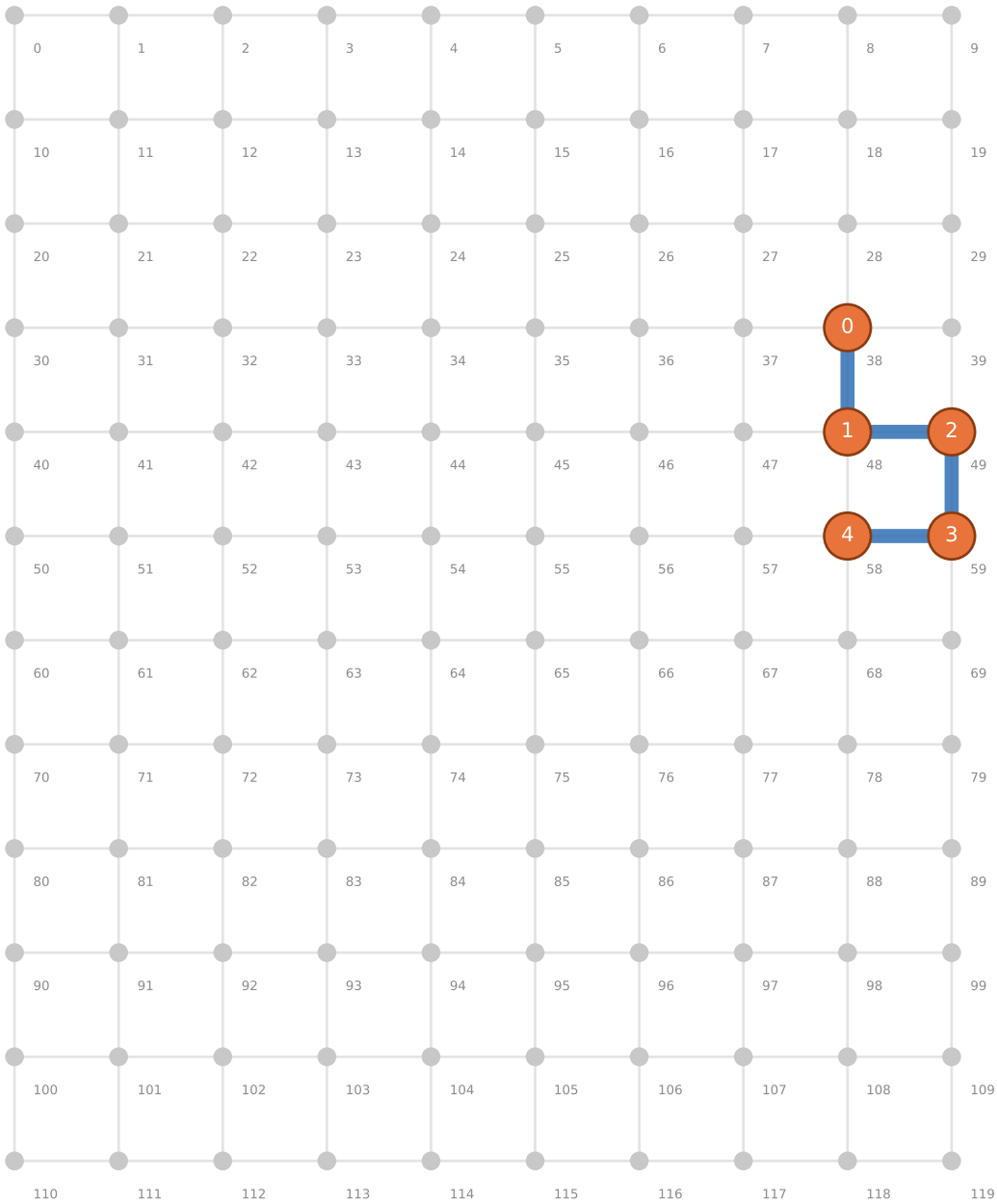


[Z5] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

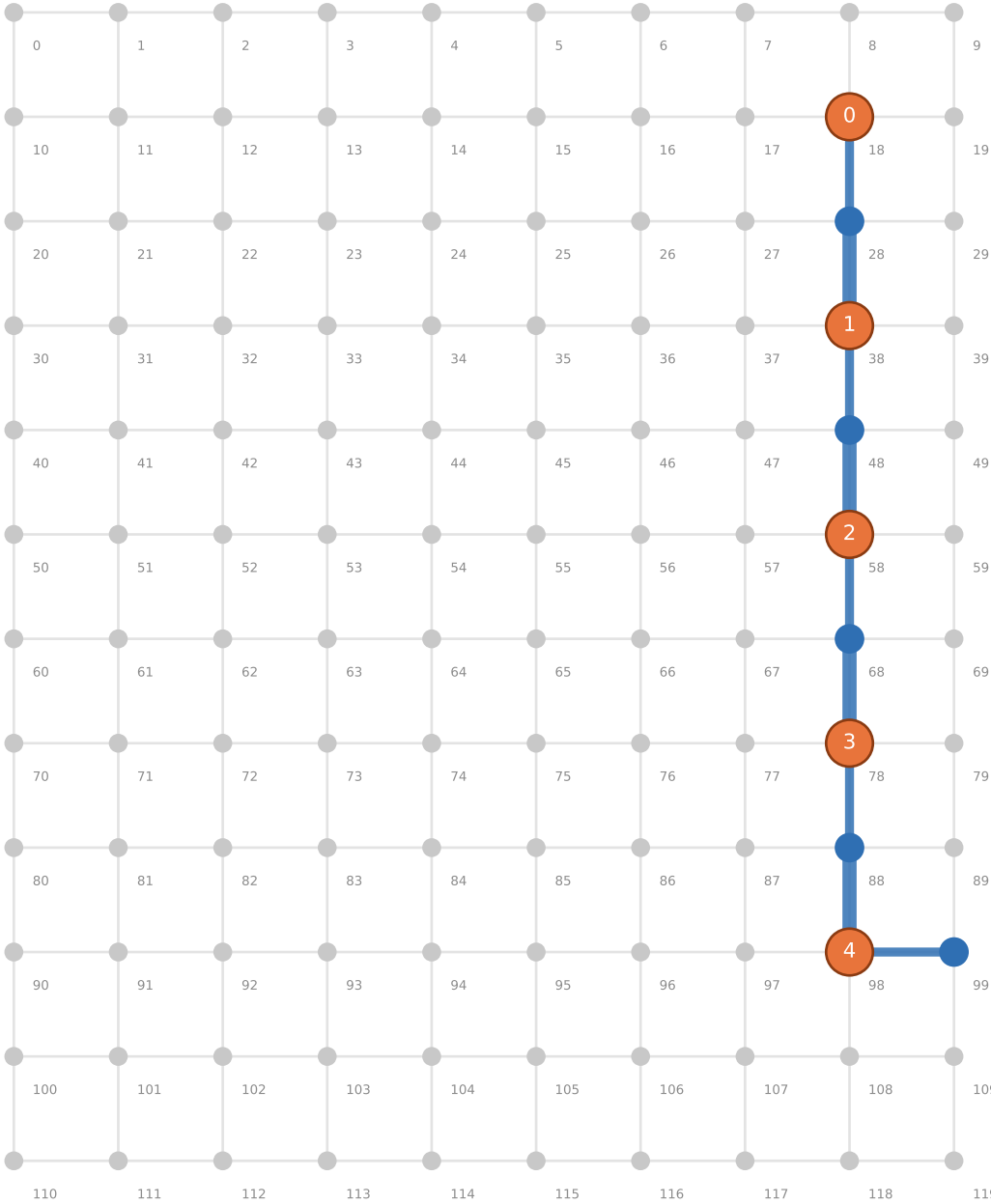
term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers



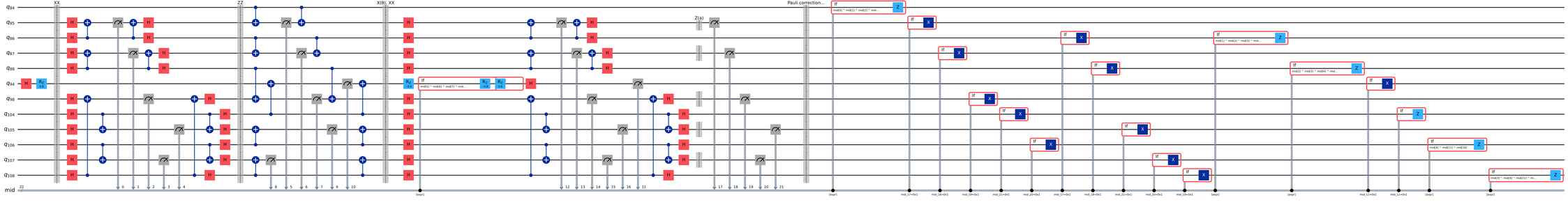
[Z5] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)



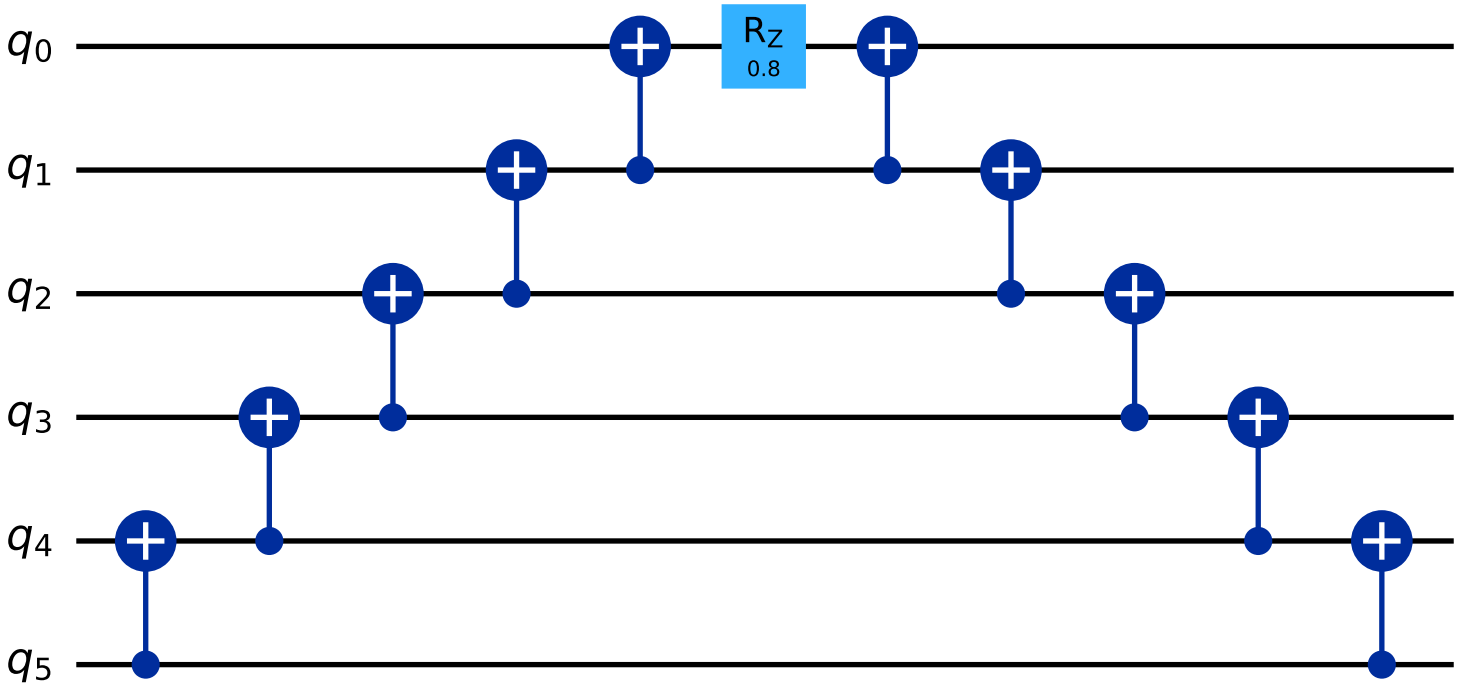
[Z5] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 26



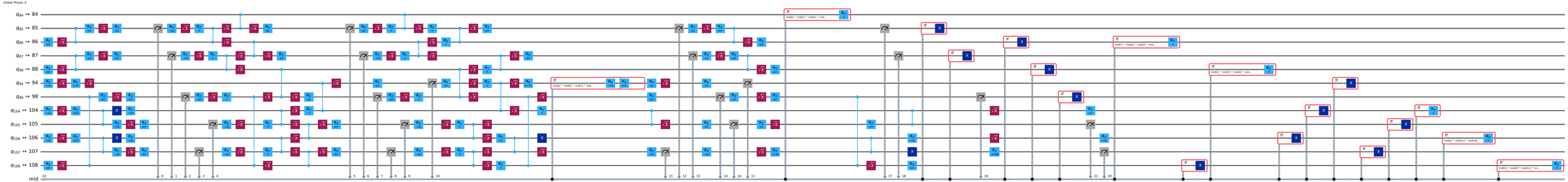
[Z6] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 6, 22 mid-circuit measurements, logical→physical [84, 86, 88, 108, 106, 104]



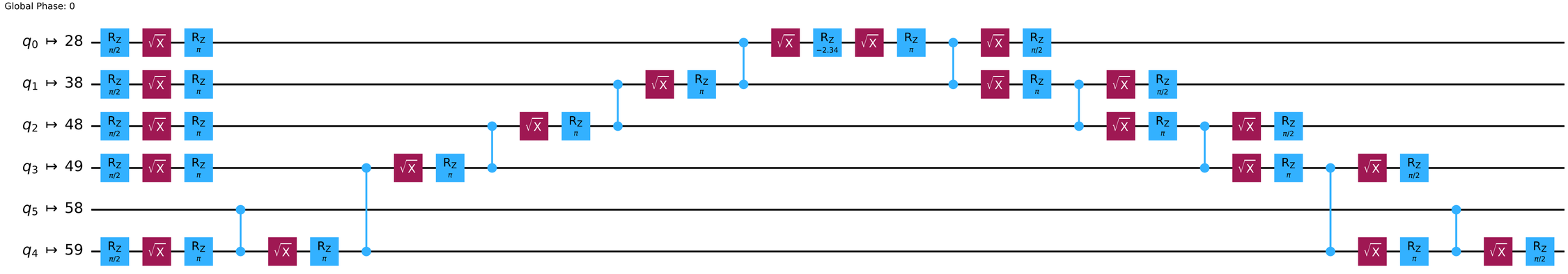
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10



{Z6} DYNAMIC circuit, fault-tolerant (|0> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:81 sx:66 cz:32 measure:22 if_else:18 x:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 6, total depth 39

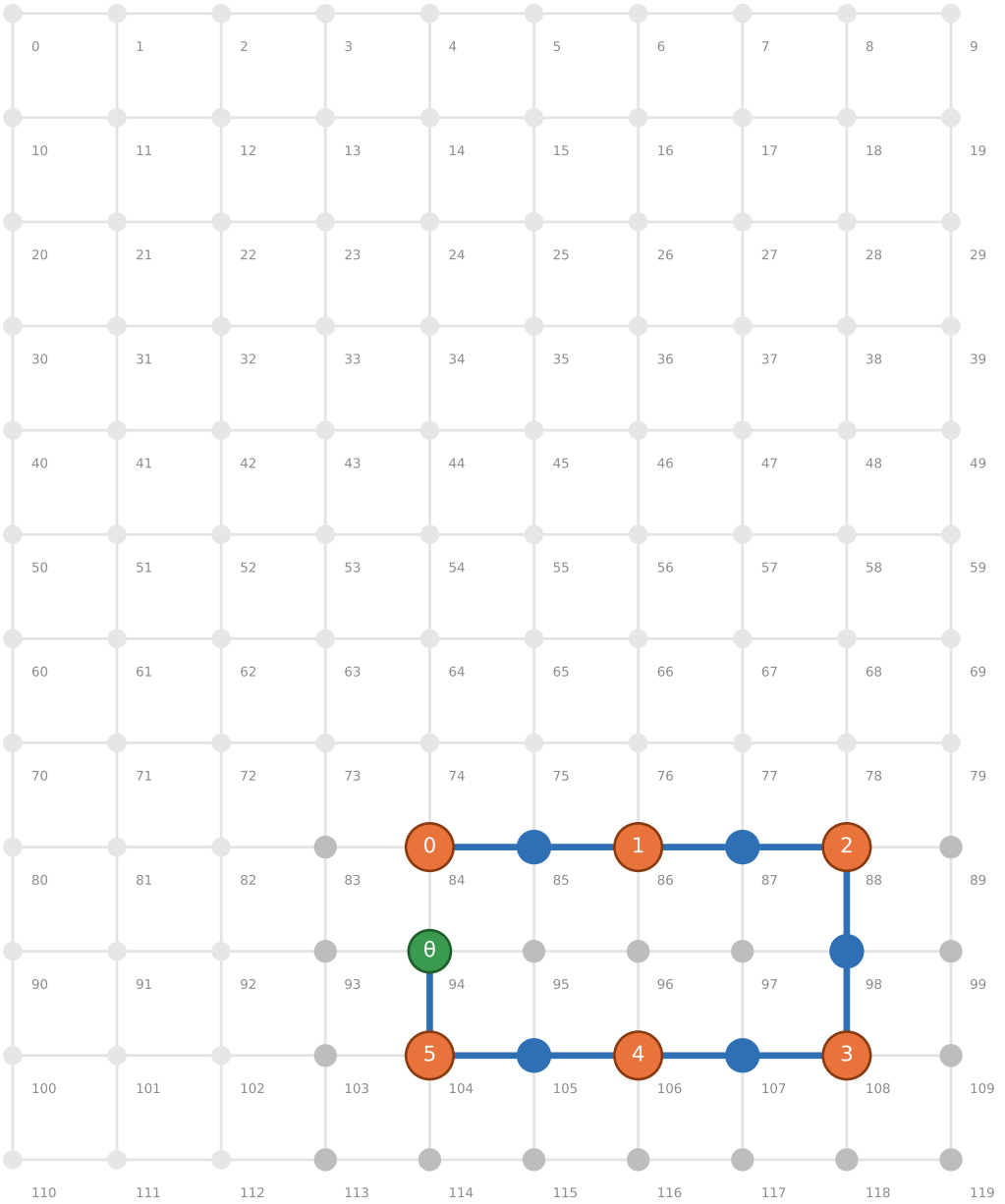


[Z6] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10, total depth 35

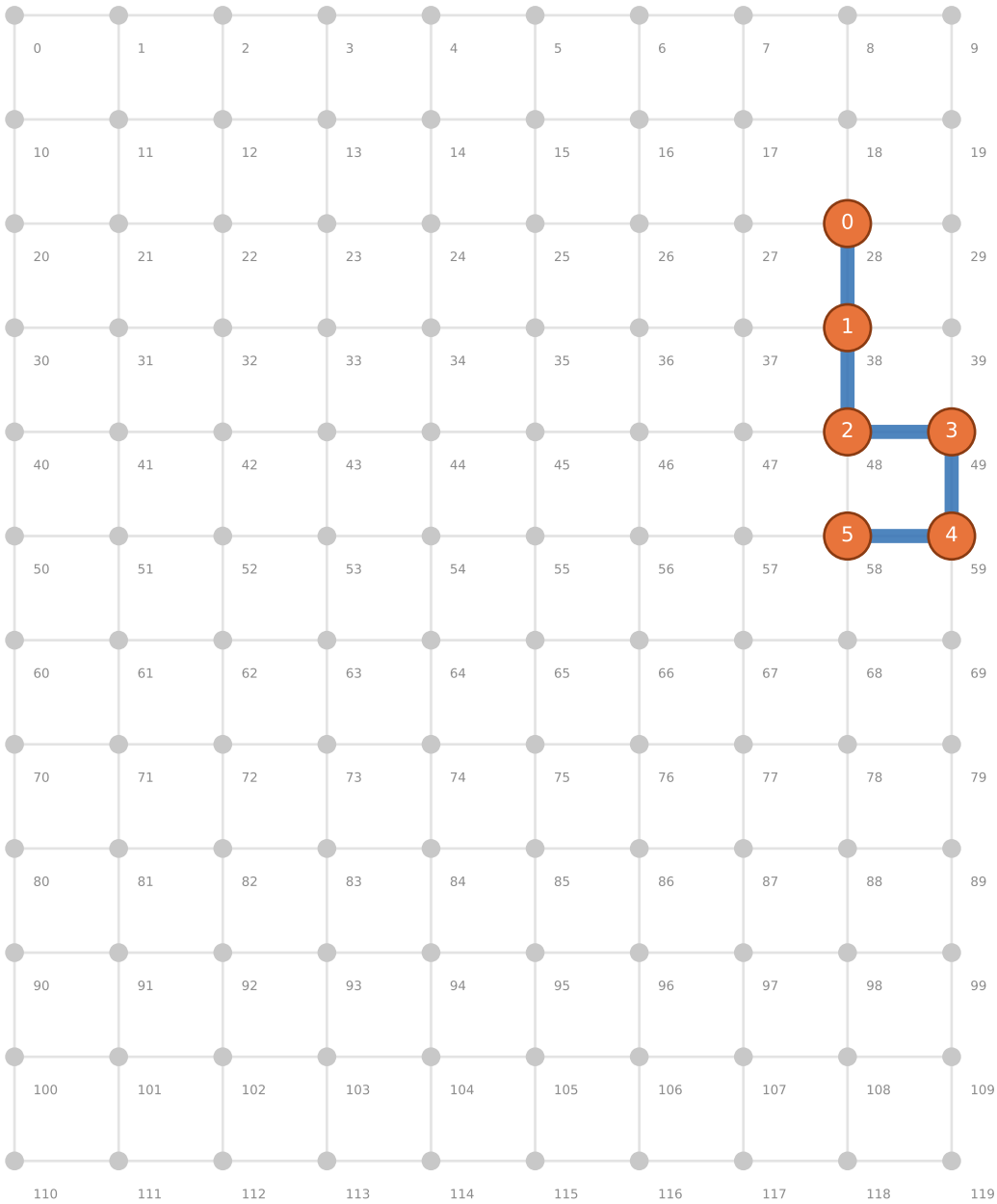


[Z6] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

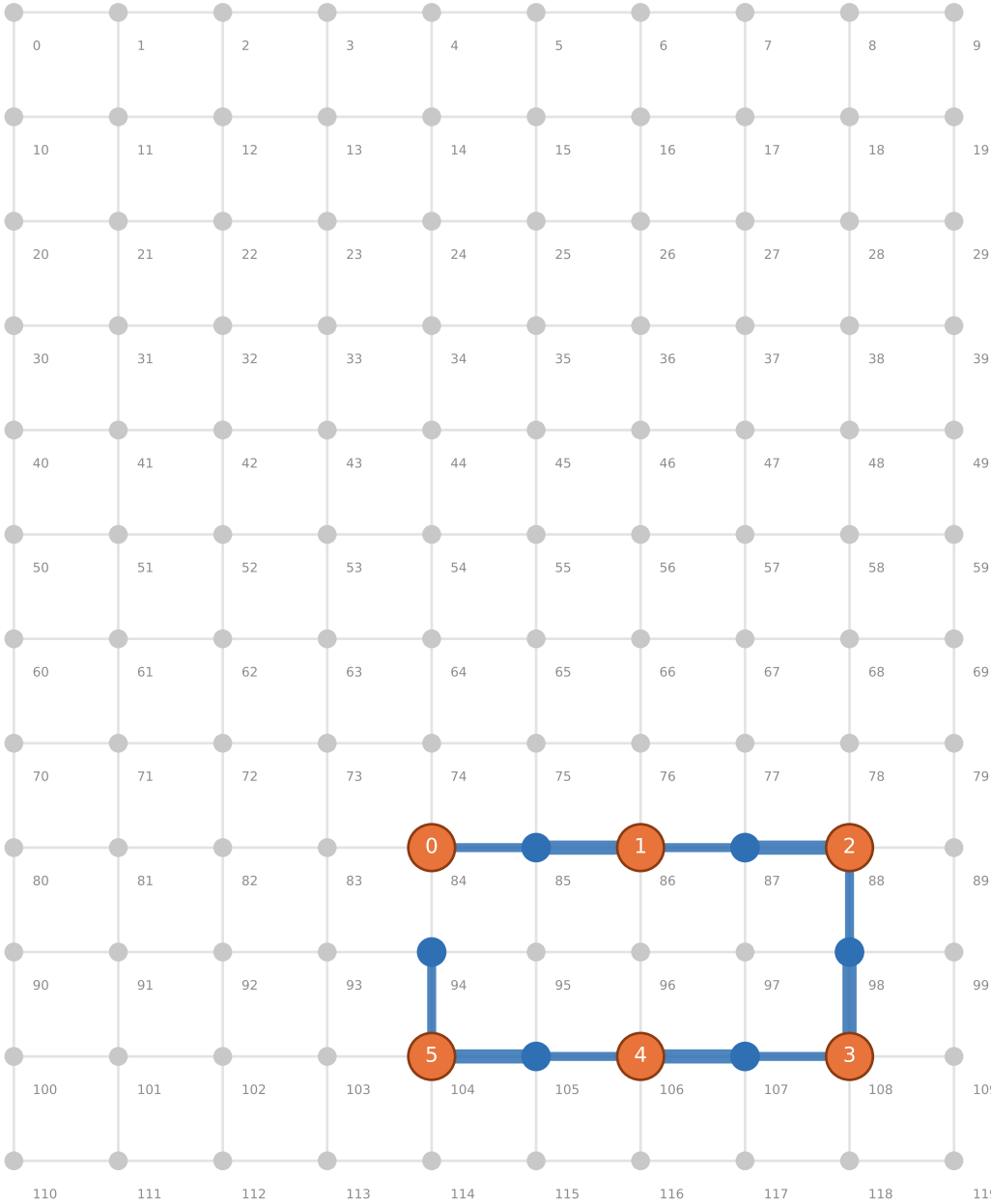
term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers



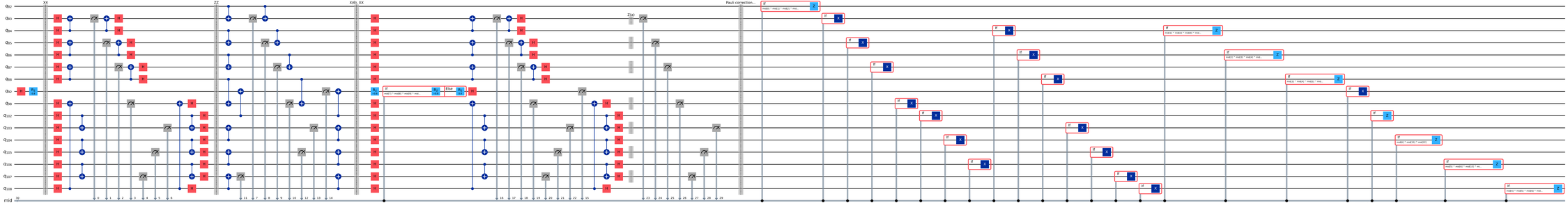
[Z6] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [28, 38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)



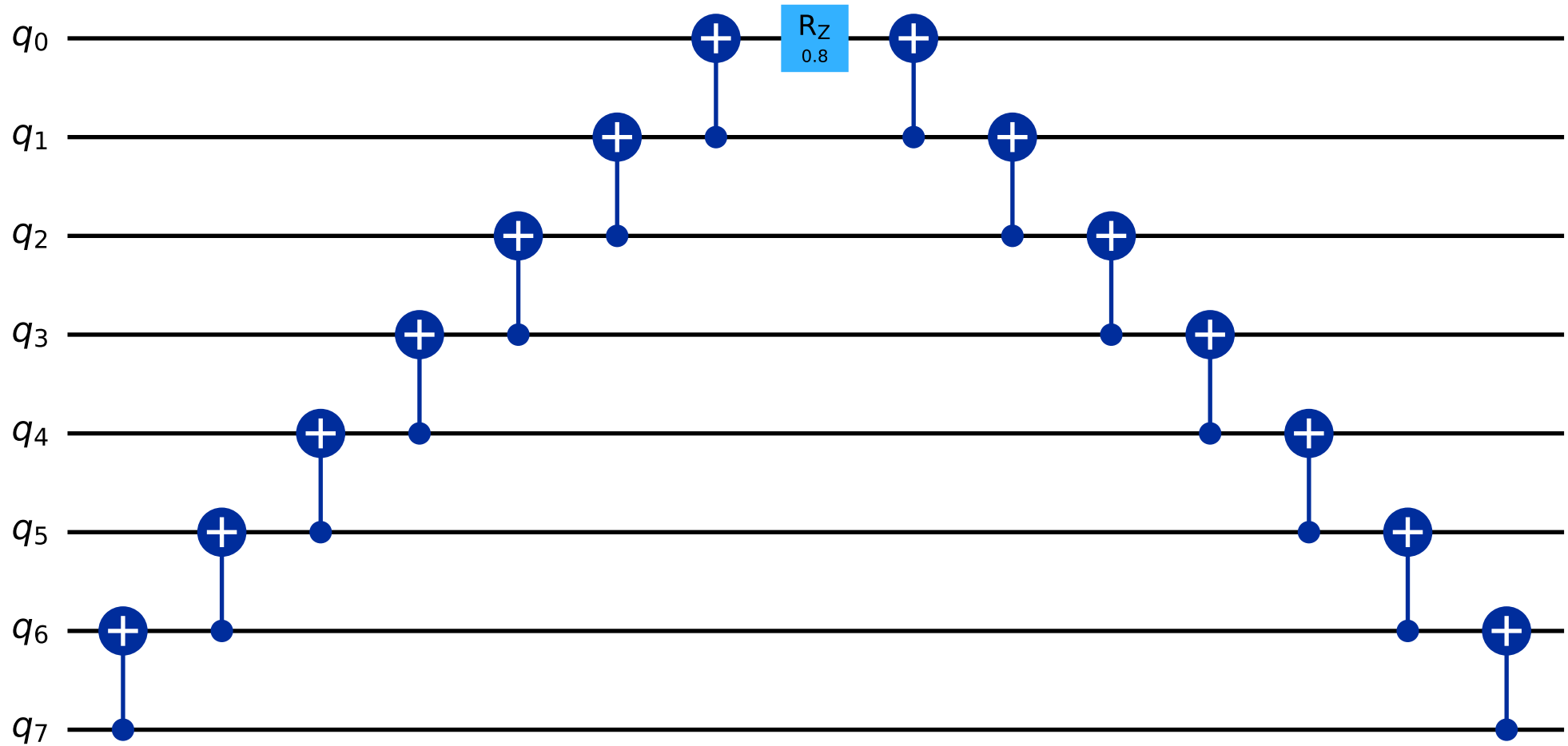
[Z6] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 32



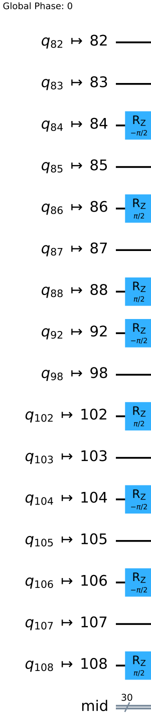
[28] DYNAMIC constant-depth circuit, fault-tolerant ([8]= ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 : [8 logical qubits]
2q depth 6, 30 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]

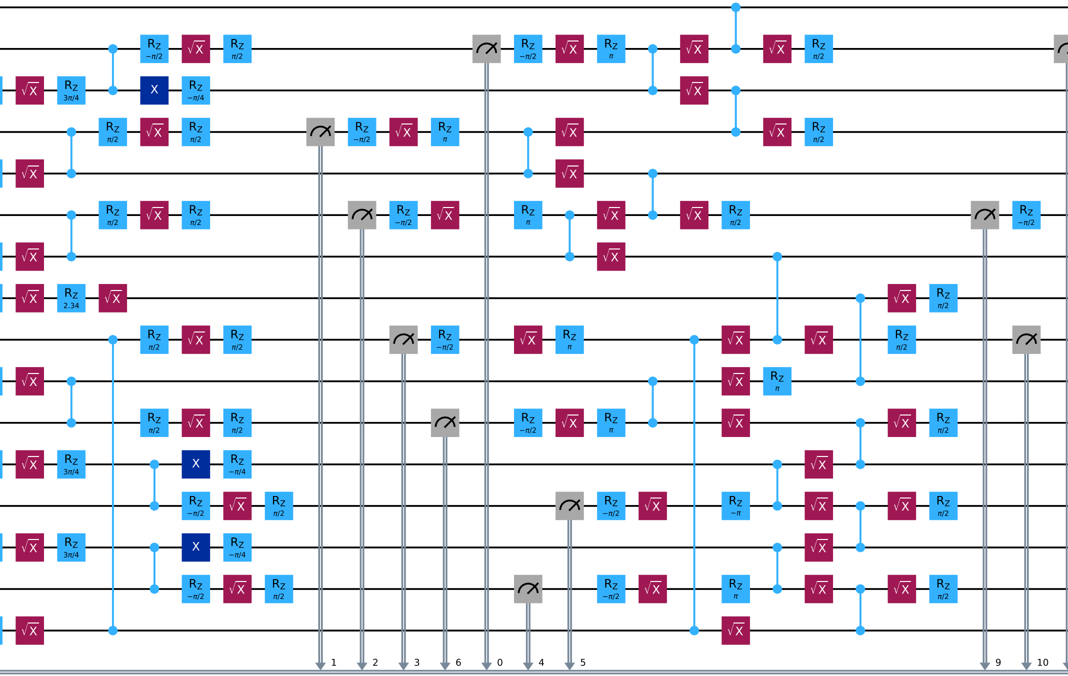


[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14

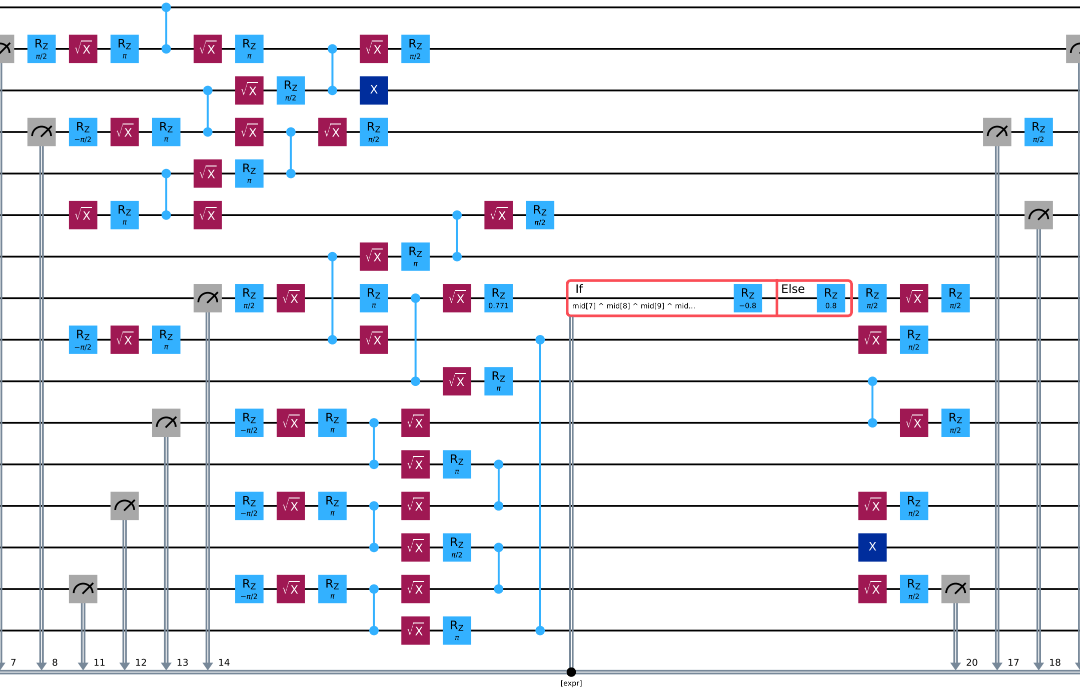


[Z8] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:111 sx:90 cz:44 measure:30 if_else:24 x:7)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 6, total depth 44

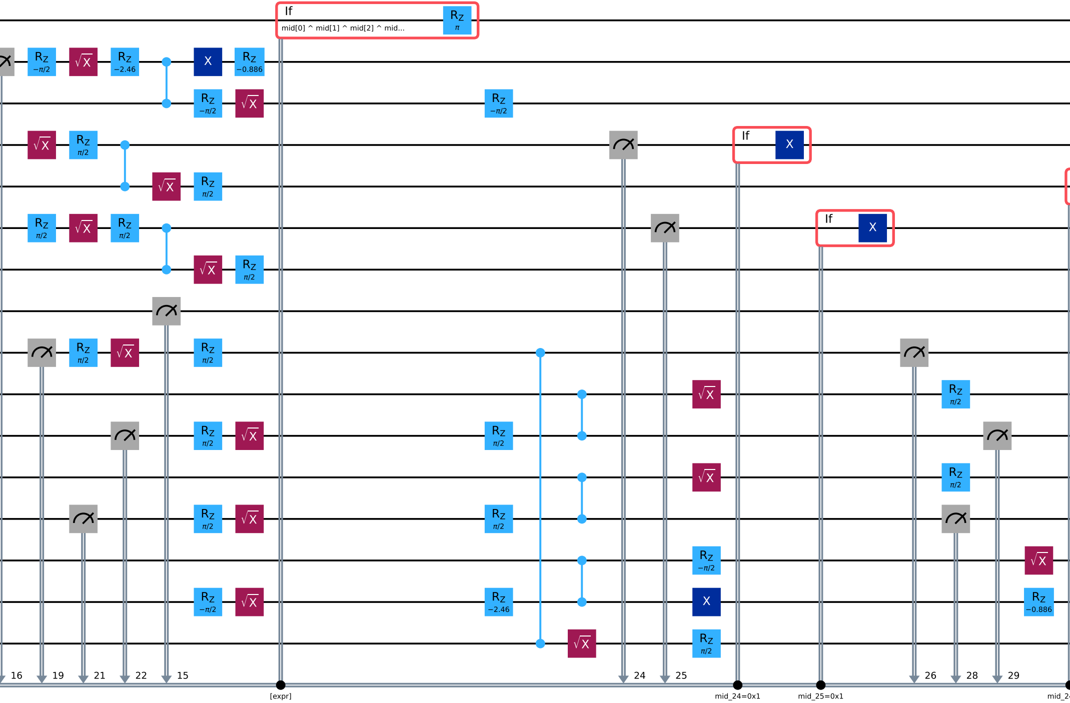

Global Phase: 0
$q_{82} \mapsto 82$
$q_{83} \mapsto 83$
$q_{84} \mapsto 84$
R_Z −π/2
$q_{85} \mapsto 85$
$q_{86} \mapsto 86$
R_Z π/2
$q_{87} \mapsto 87$
$q_{88} \mapsto 88$
R_Z π/2
$q_{92} \mapsto 92$
R_Z −π/2
$q_{98} \mapsto 98$
$q_{102} \mapsto 102$
R_Z π/2
$q_{103} \mapsto 103$
$q_{104} \mapsto 104$
R_Z −π/2
$q_{105} \mapsto 105$
$q_{106} \mapsto 106$
R_Z −π/2
$q_{107} \mapsto 107$
$q_{108} \mapsto 108$
R_Z π/2
mid
30

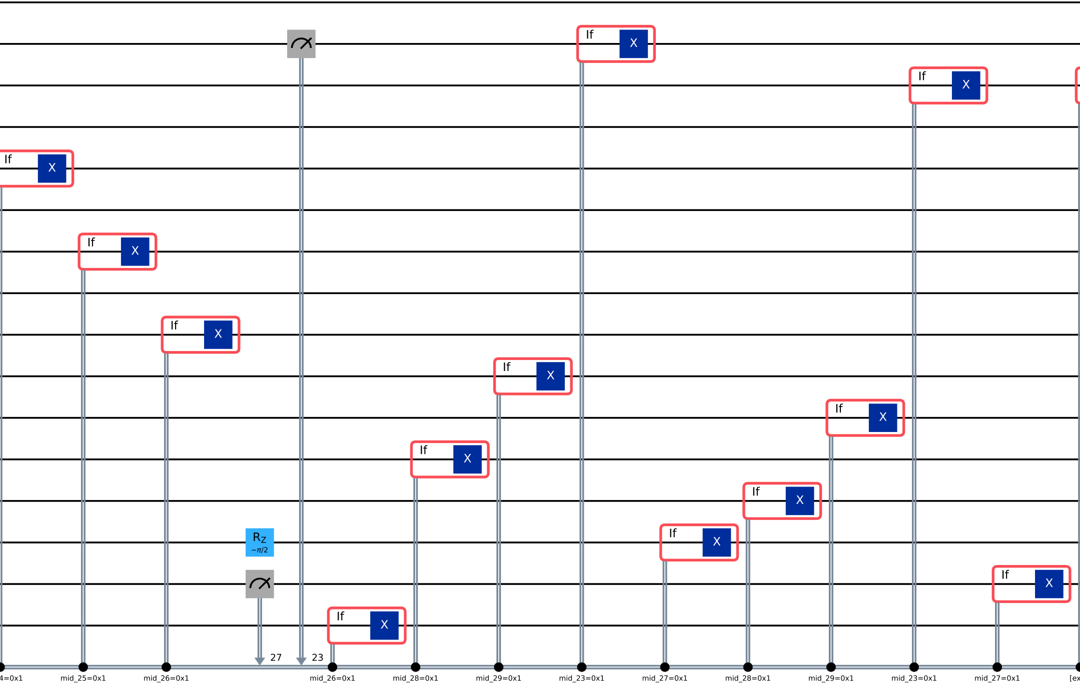


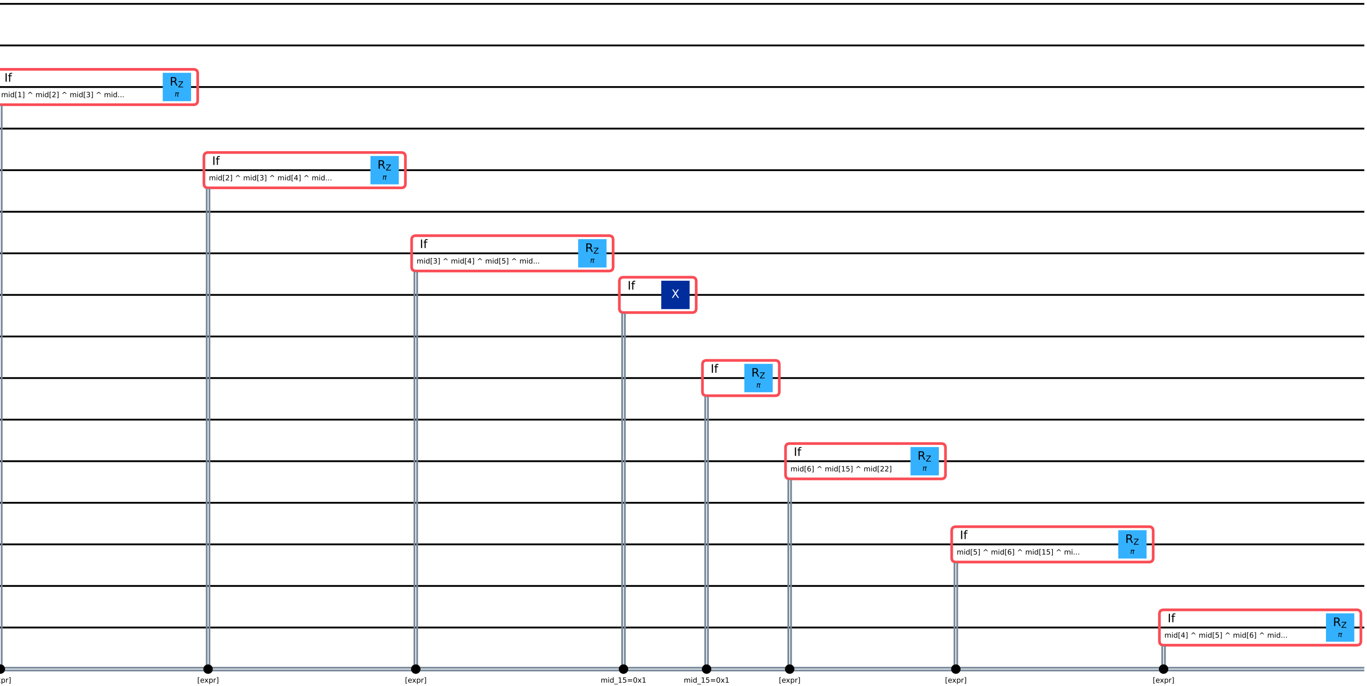
RZ
−π/2
√X
π/2
3π/4
X
−π/4
2.34
π
−π
1
2
3
6
0
4
5
9
10



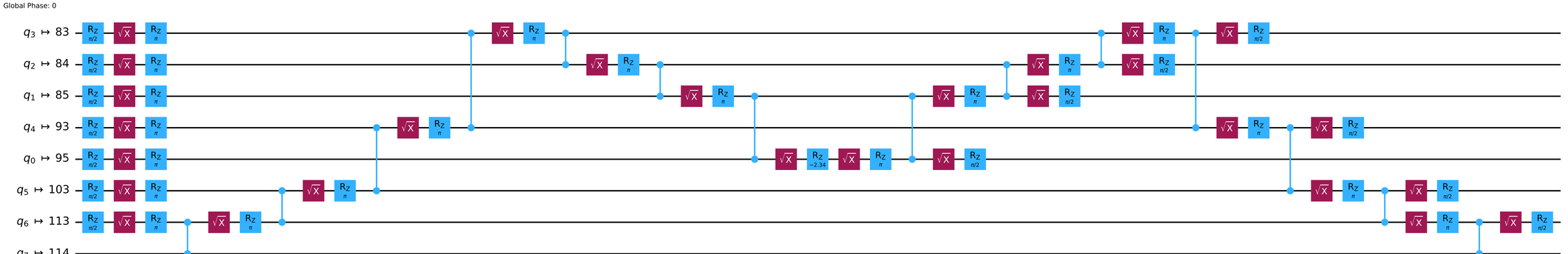
RZ
π/2
√X
RZ
π
RZ
−π/2
X
RZ
0.771
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
RZ
−0.8
Else
RZ
0.8
7
8
11
12
13
14
[expr]
20
17
18


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Rz π
Rz −π/2
√X
Rz −2.46
X
Rz −0.886
Rz −π/2
Rz π/2
If
Rz −2.46
Rz −0.886
16
19
21
22
15
[expr]
24
25
mid_24=0x1
mid_25=0x1
26
28
29
mid_2


If
X
R_Z
$-\pi/2$
27
23
4=0x1
mid_25=0x1
mid_26=0x1
mid_26=0x1
mid_28=0x1
mid_29=0x1
mid_23=0x1
mid_27=0x1
mid_28=0x1
mid_29=0x1
mid_23=0x1
mid_27=0x1
[ex

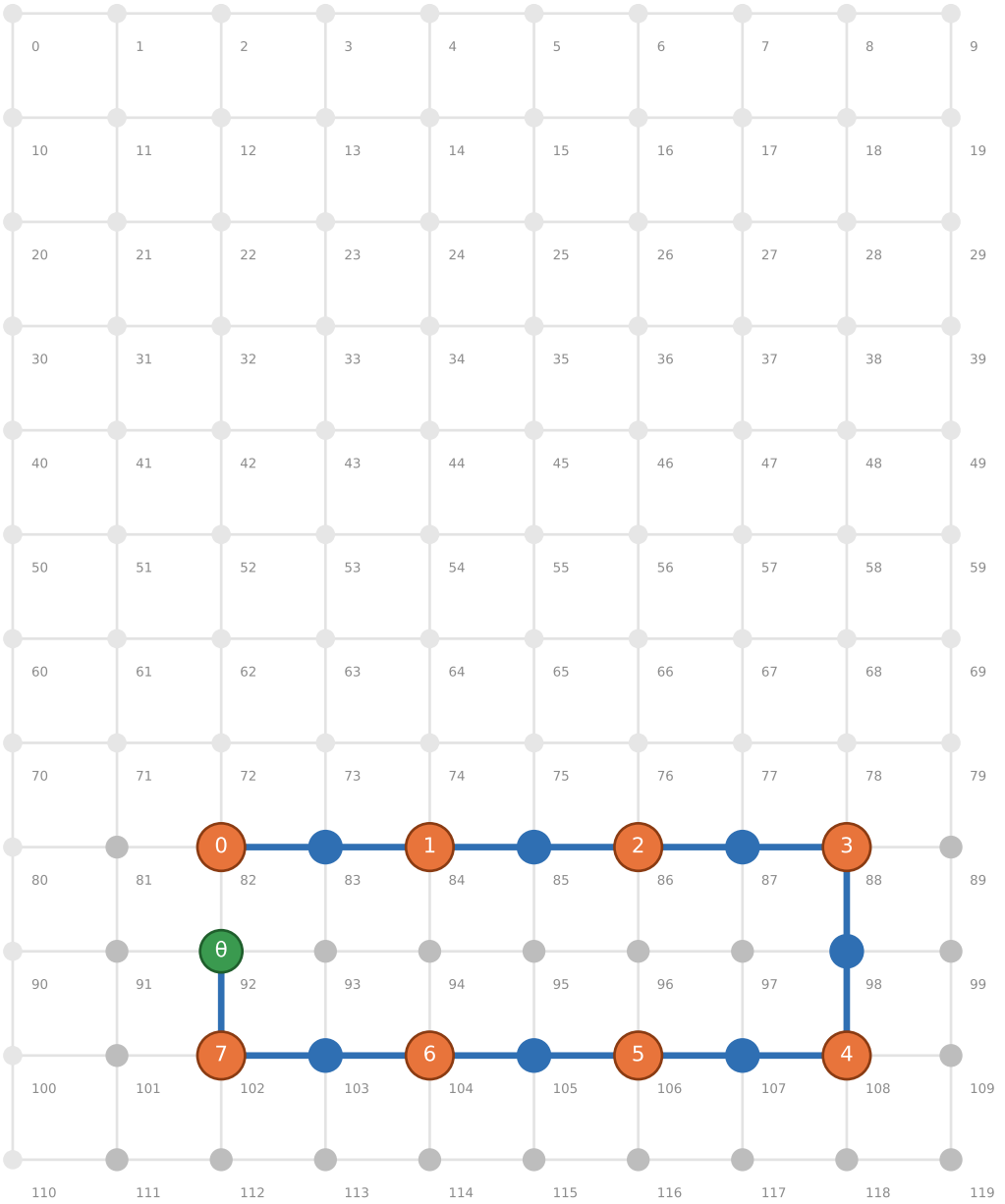

If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
RZ
π
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
RZ
π
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
RZ
π
If
X
If
RZ
π
If
mid[6] ^ mid[15] ^ mid[22]
RZ
π
If
mid[5] ^ mid[6] ^ mid[15] ^ mi...
RZ
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
RZ
π
pr]
[expr]
[expr]
mid_15=0x1
mid_15=0x1
[expr]
[expr]
[expr]

[Z8] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

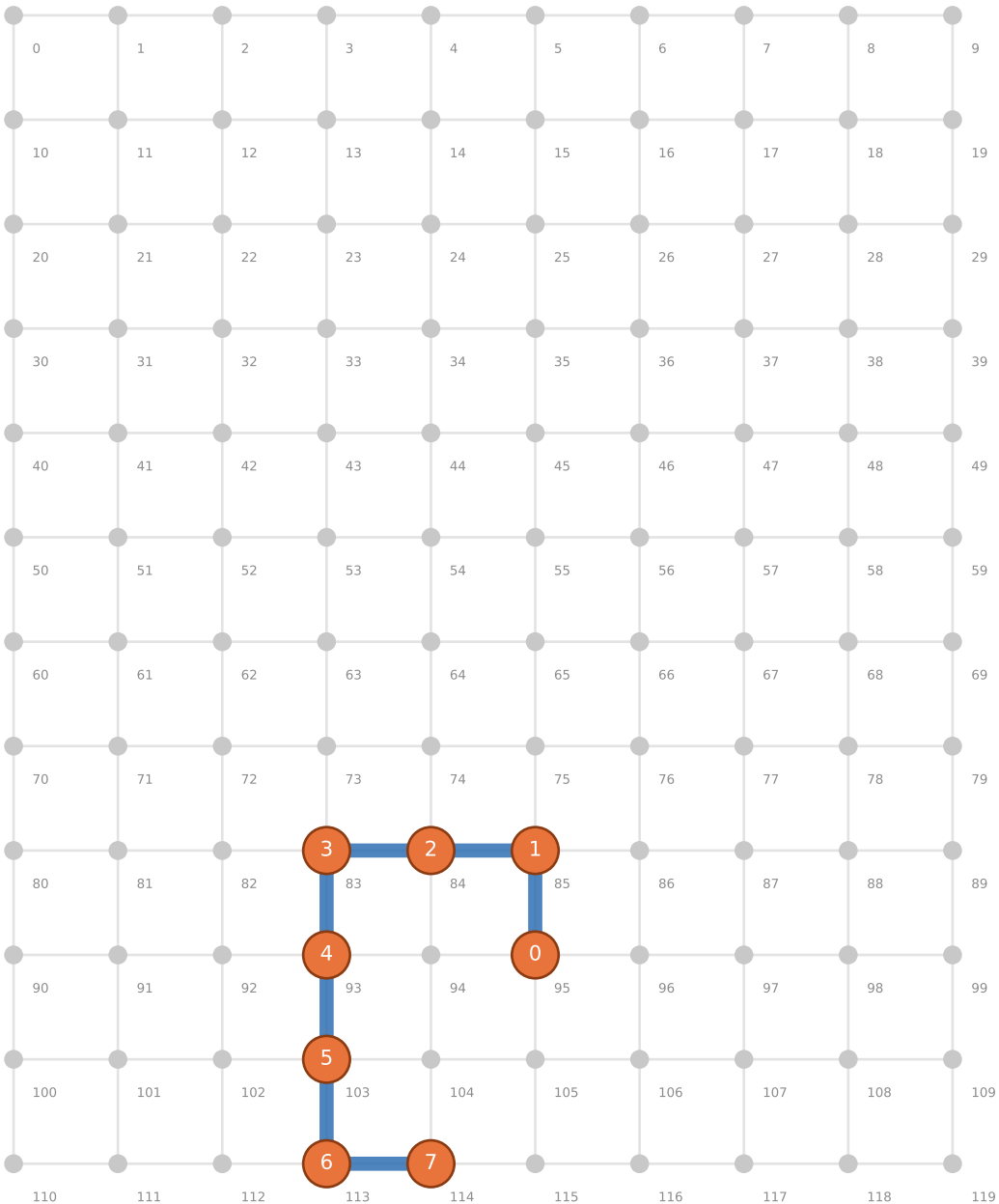


[Z8] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

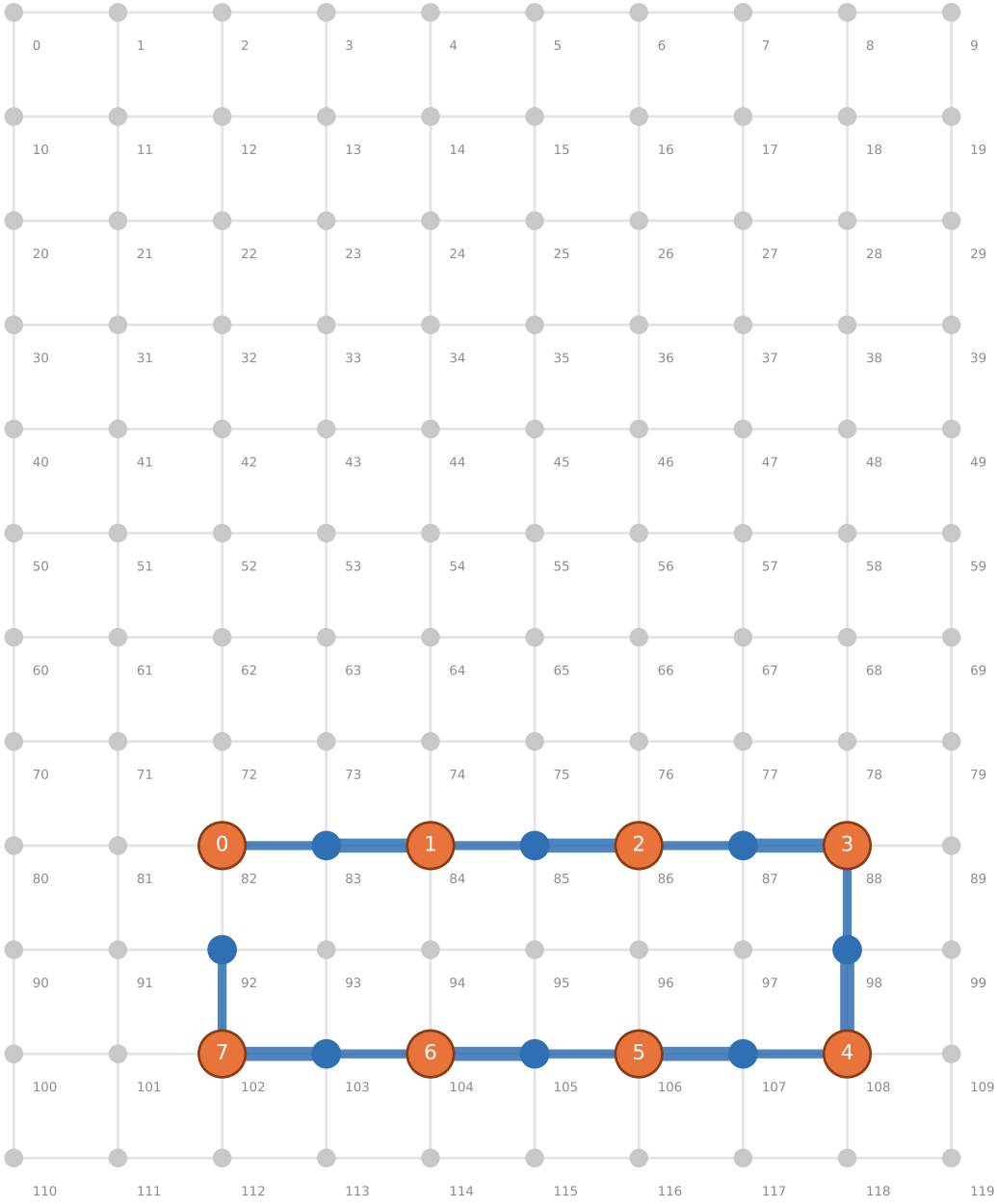
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



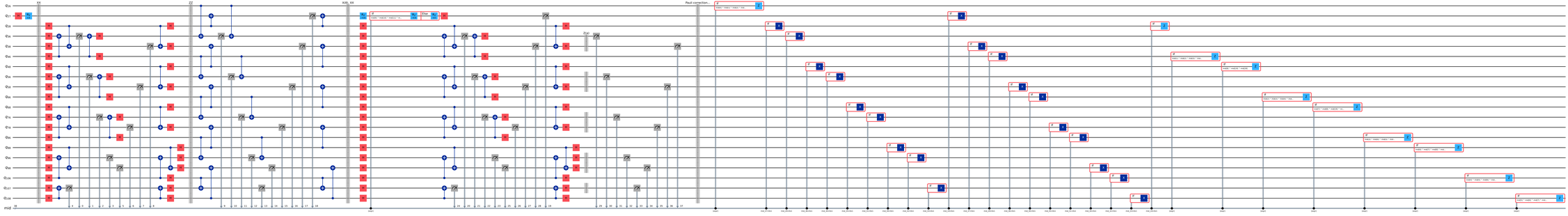
[Z8] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 83, 93, 103, 113, 114] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



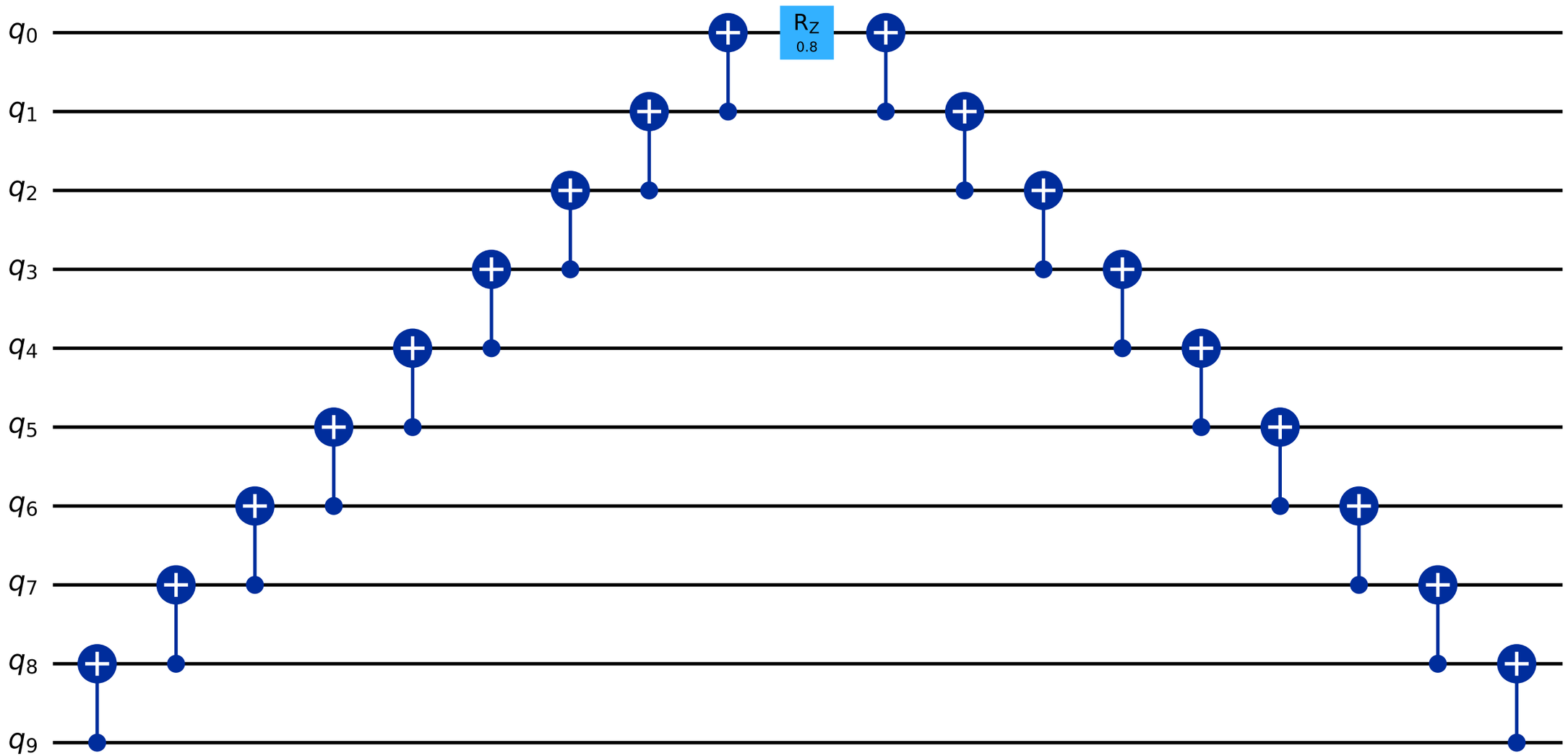
[Z8] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



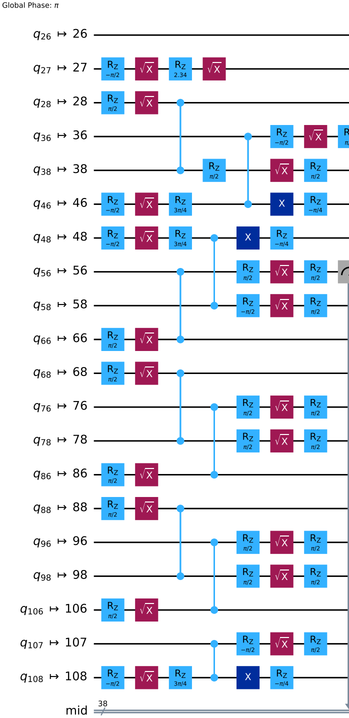
[Z16] DYNAMIC constant-depth circuit, fault-tolerant (8n+ ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=π/4: P = ZQ Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical=physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]

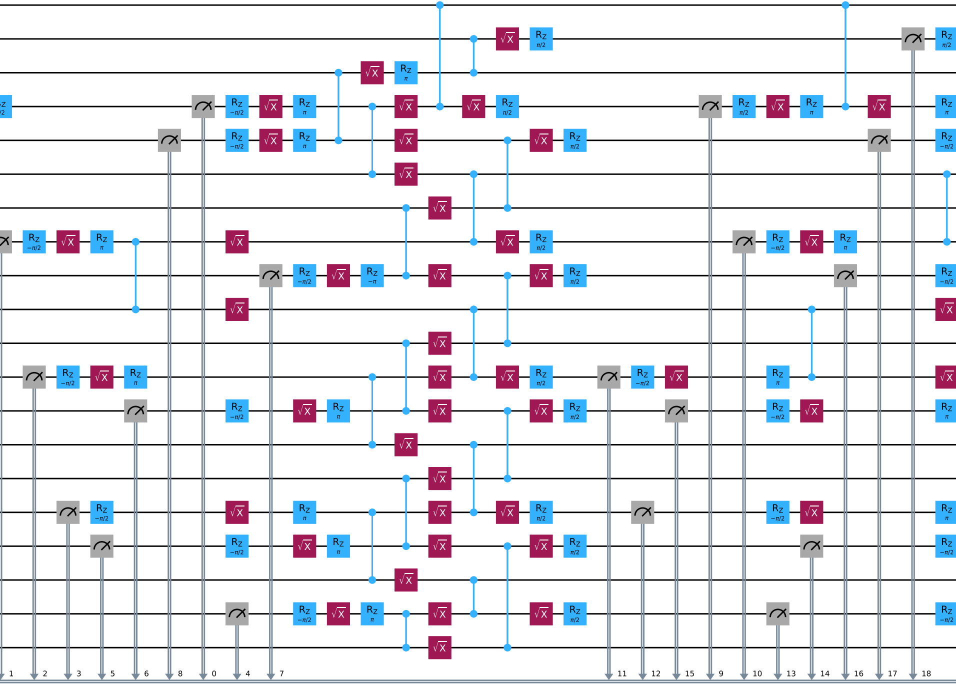


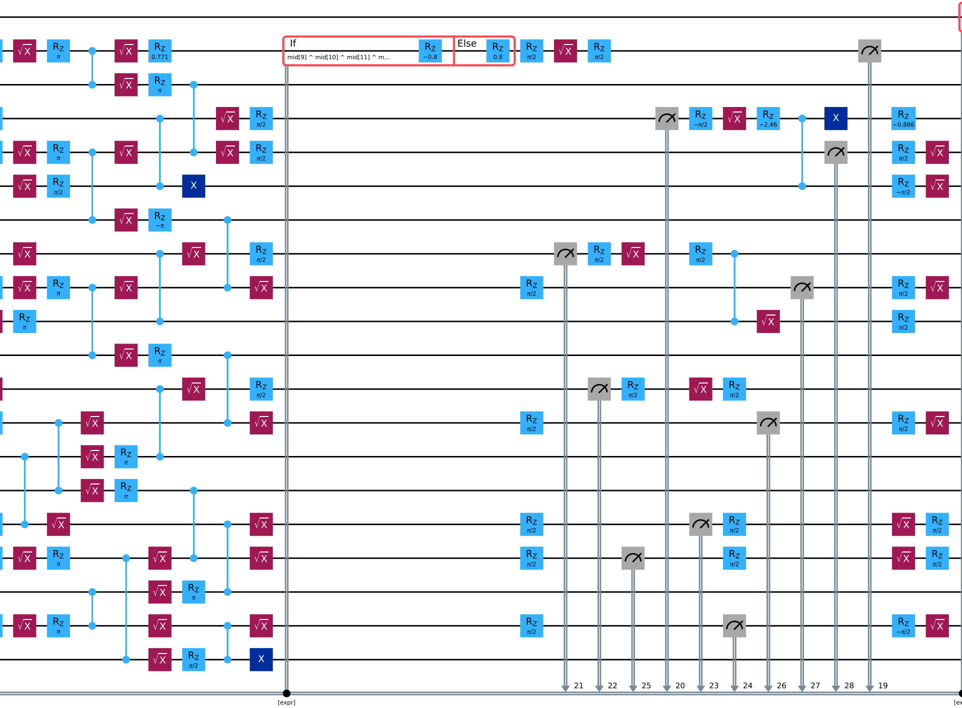
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18

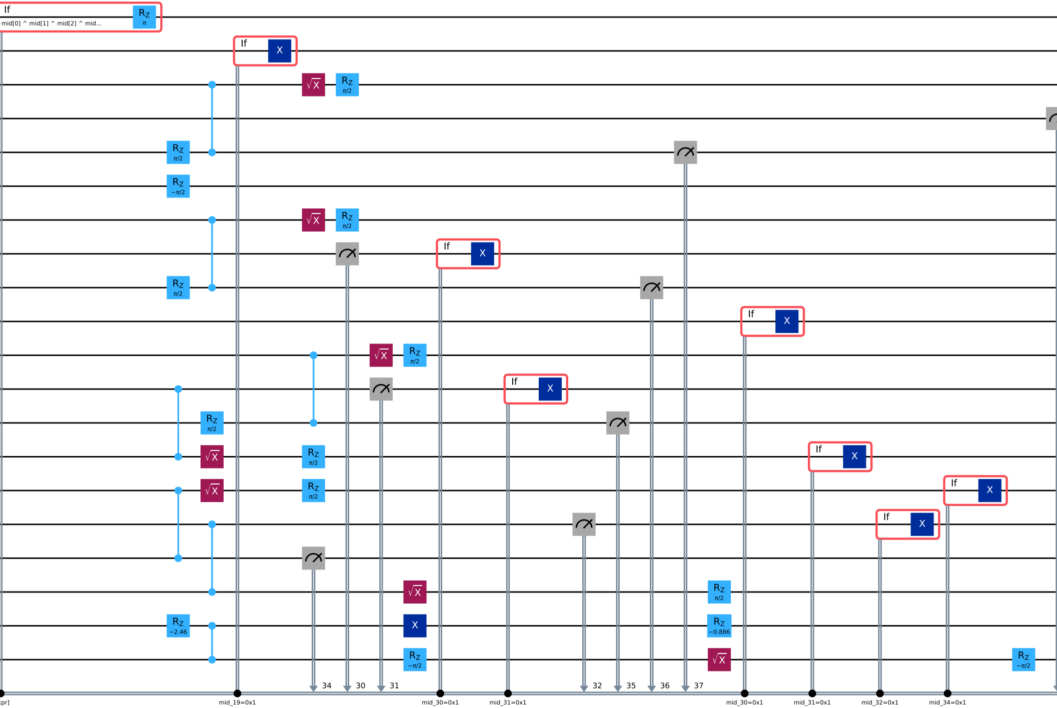


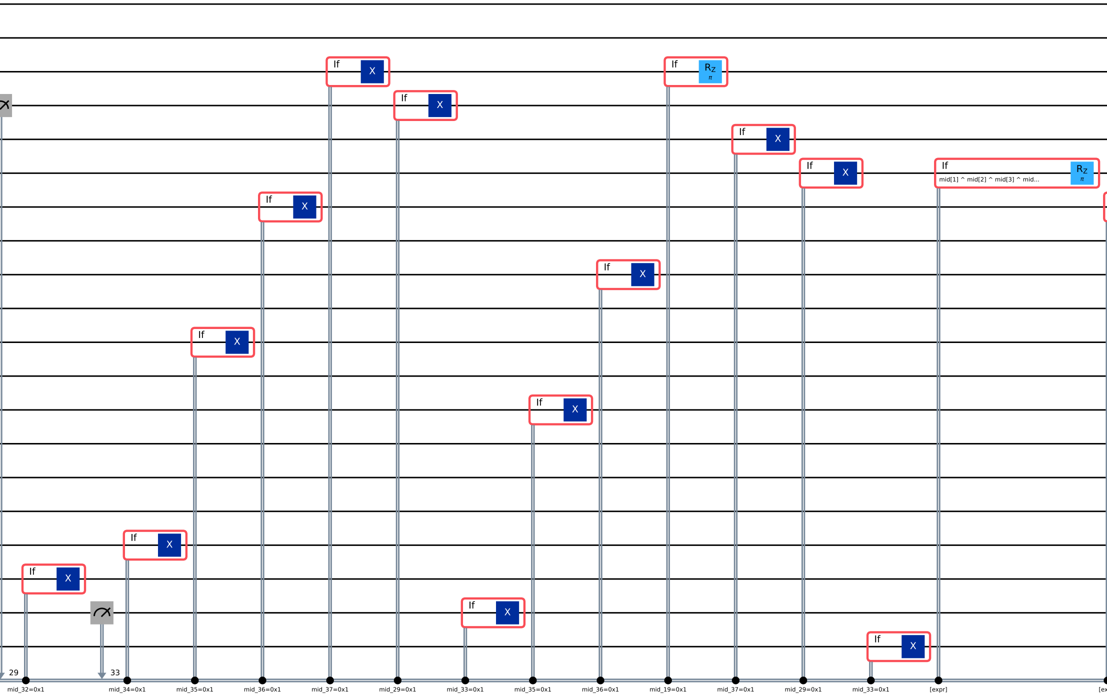
[Z10] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:137 sx:114 cz:56 measure:38 if_else:30 x:7)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 6, total depth 44

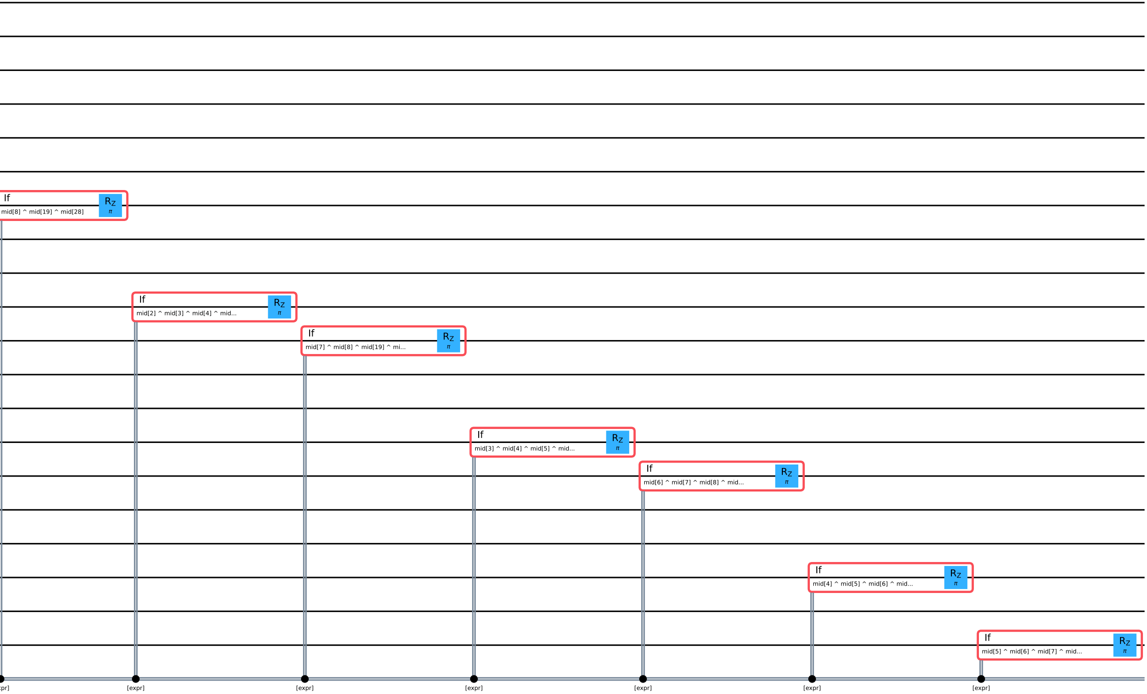

Global Phase: π
$q_{26} \mapsto 26$
$q_{27} \mapsto 27$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z 2.34 $\sqrt{X}$
$q_{28} \mapsto 28$ R_Z $\pi/2$ $\sqrt{X}$
$q_{36} \mapsto 36$ R_Z $-\pi/2$ $\sqrt{X}$
$q_{38} \mapsto 38$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{46} \mapsto 46$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z $3\pi/4$ X R_Z $-\pi/4$
$q_{48} \mapsto 48$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z $3\pi/4$ X R_Z $-\pi/4$
$q_{56} \mapsto 56$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{58} \mapsto 58$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{66} \mapsto 66$ R_Z $\pi/2$ $\sqrt{X}$
$q_{68} \mapsto 68$ R_Z $\pi/2$ $\sqrt{X}$
$q_{76} \mapsto 76$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{78} \mapsto 78$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{86} \mapsto 86$ R_Z $\pi/2$ $\sqrt{X}$
$q_{88} \mapsto 88$ R_Z $\pi/2$ $\sqrt{X}$
$q_{96} \mapsto 96$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{98} \mapsto 98$ R_Z $\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{106} \mapsto 106$ R_Z $\pi/2$ $\sqrt{X}$
$q_{107} \mapsto 107$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z $\pi/2$
$q_{108} \mapsto 108$ R_Z $-\pi/2$ $\sqrt{X}$ R_Z $3\pi/4$ X R_Z $-\pi/4$
mid 38


RZ
√X
π
−π
π/2
−π/2
1
2
3
5
6
8
0
4
7
11
12
15
9
10
13
14
16
17
18

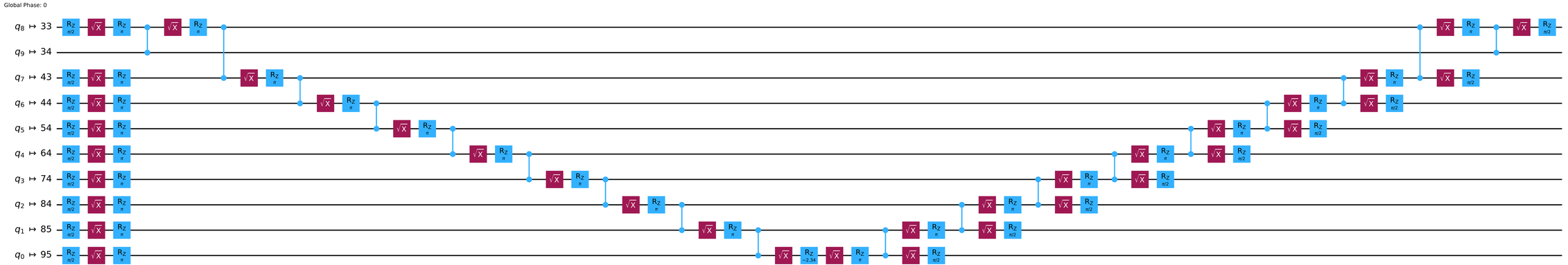

If
mid[9] ^ mid[10] ^ mid[11] ^ m...
Else
RZ −0.8
RZ 0.8
RZ 0.771
RZ π
RZ −π
RZ π/2
RZ −π/2
RZ −2.46
RZ −0.886
√X
X
21
22
25
20
23
24
26
27
28
19
[expr]
[ex


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
RZ
π
If
X
√X
RZ
π/2
RZ
π/2
RZ
−π/2
√X
RZ
π/2
RZ
π/2
If
X
If
X
√X
RZ
π/2
If
X
RZ
π/2
√X
RZ
π/2
√X
RZ
π/2
If
X
If
X
If
X
√X
X
RZ
−π/2
RZ
−2.46
RZ
π/2
RZ
−0.886
√X
RZ
−π/2
34
30
31
32
35
36
37
pr]
mid_19=0x1
mid_30=0x1
mid_31=0x1
mid_30=0x1
mid_31=0x1
mid_32=0x1
mid_34=0x1


29
33
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
R_Z
π
If
X
If
X
If
X
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
R_Z
π
mid_32=0x1
mid_34=0x1
mid_35=0x1
mid_36=0x1
mid_37=0x1
mid_29=0x1
mid_33=0x1
mid_35=0x1
mid_36=0x1
mid_19=0x1
mid_37=0x1
mid_29=0x1
mid_33=0x1
[expr]

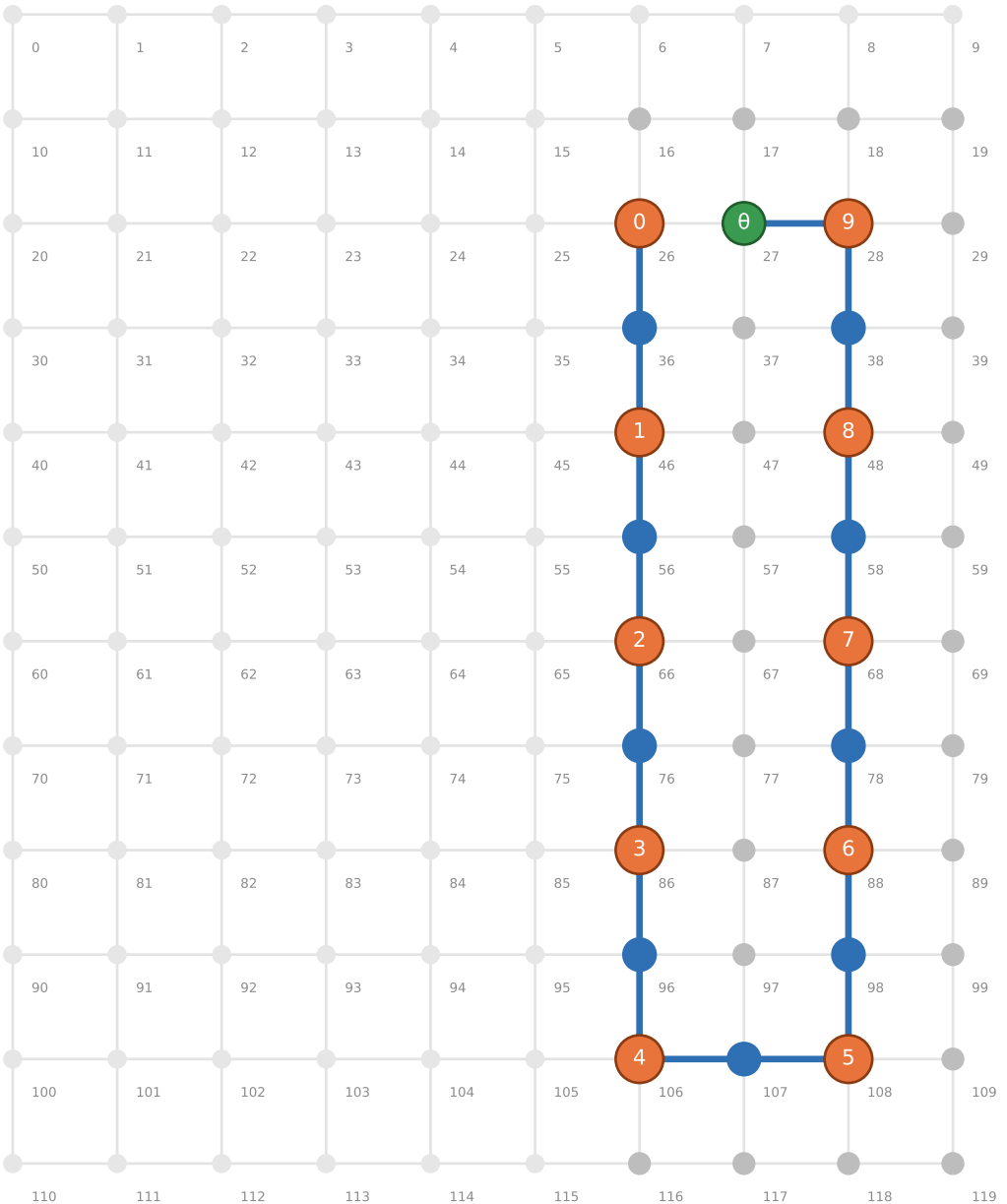

If
mid[8] ^ mid[19] ^ mid[28]
RZ
π
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
RZ
π
If
mid[7] ^ mid[8] ^ mid[19] ^ mi...
RZ
π
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
RZ
π
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
RZ
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
RZ
π
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
RZ
π
pr]
[expr]
[expr]
[expr]
[expr]
[expr]
[expr]

[Z10] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18, total depth 59

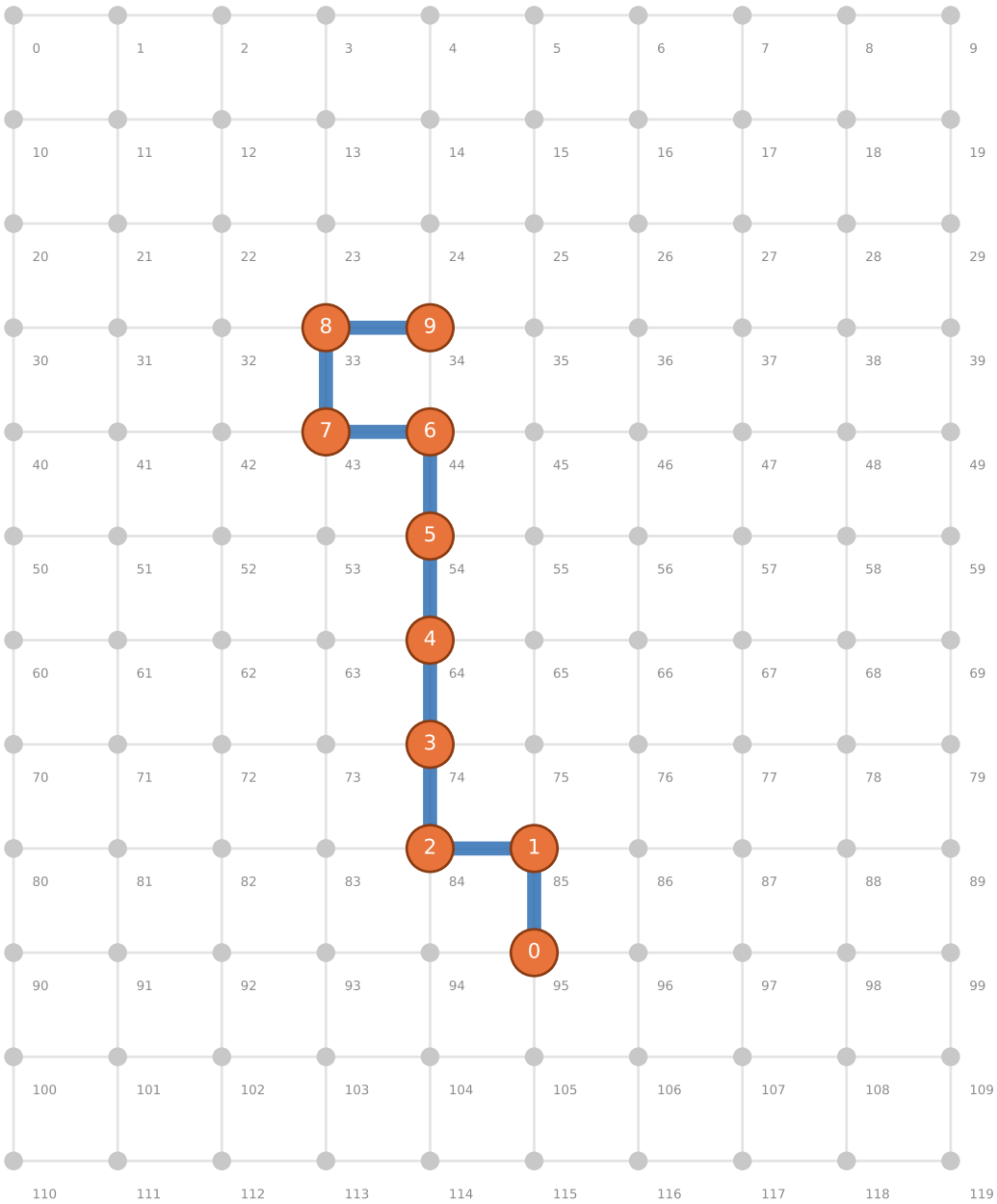


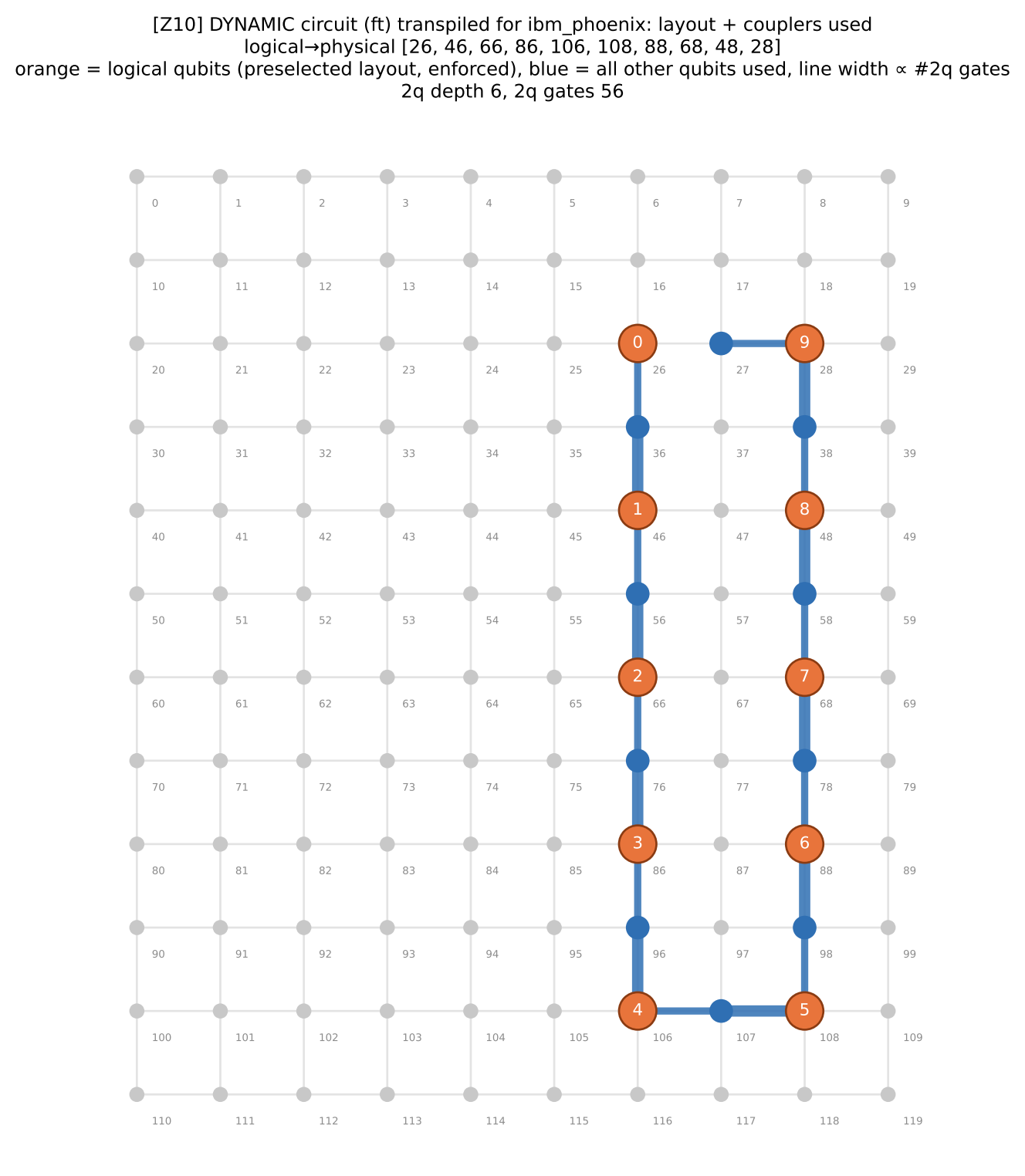
[Z10] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers

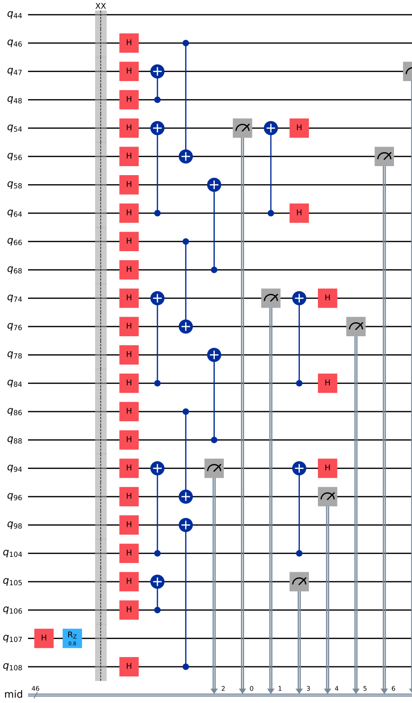


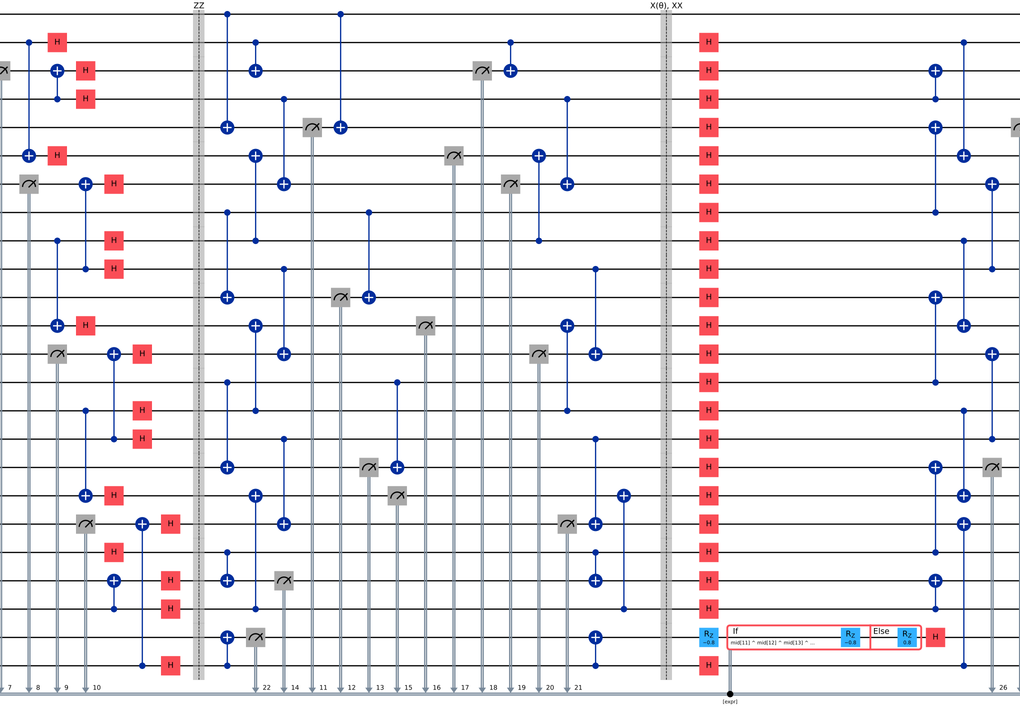
[Z10] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)

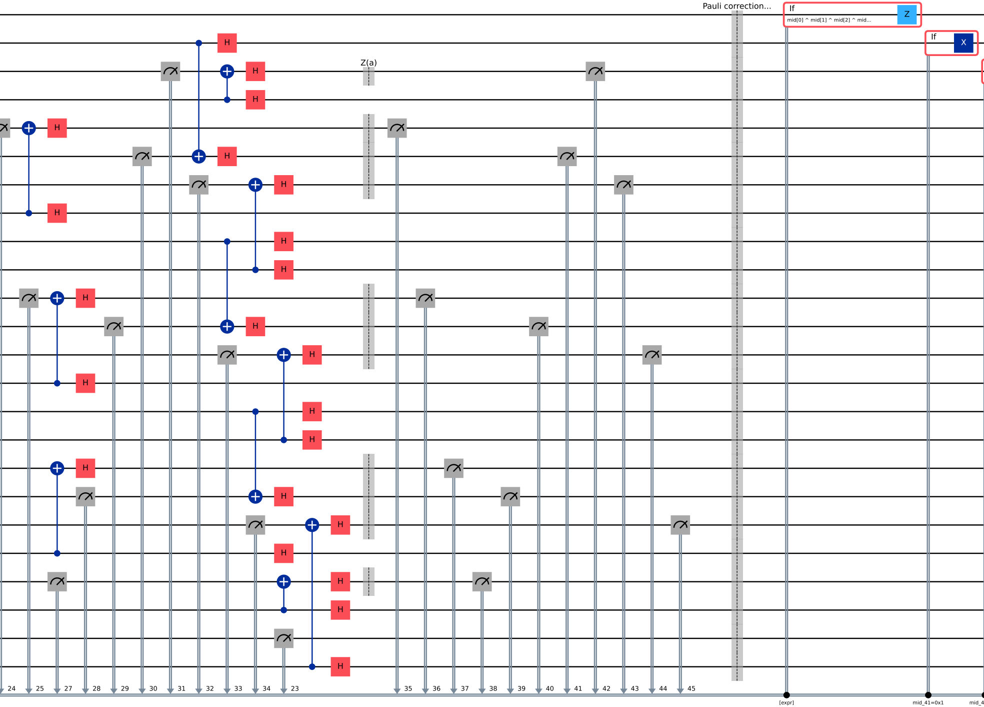


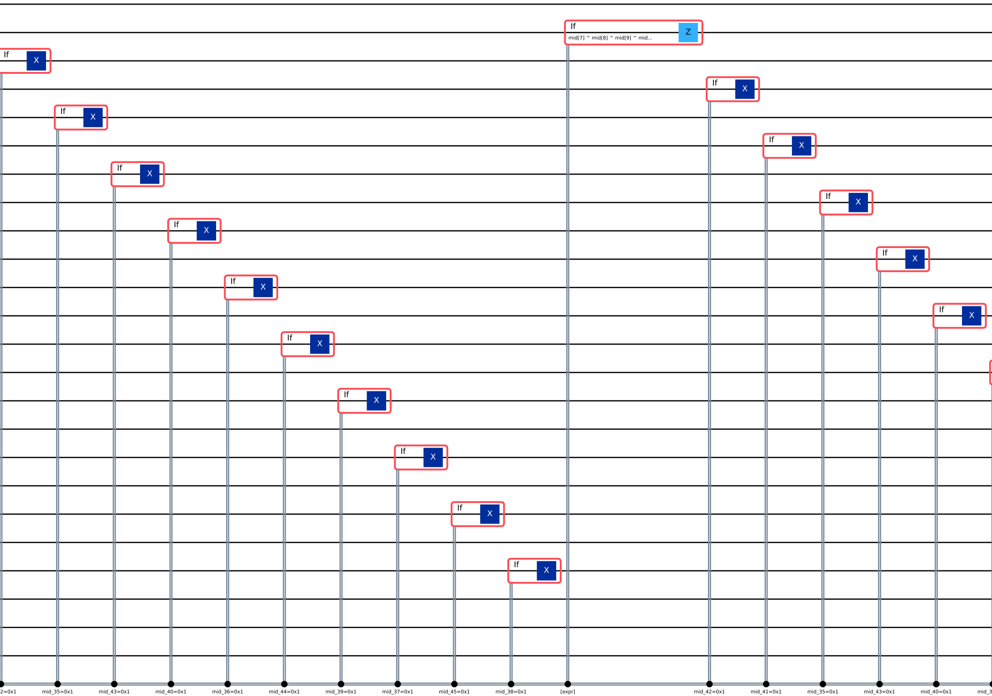

[Z10] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
9
1
8
2
7
3
6
4
5

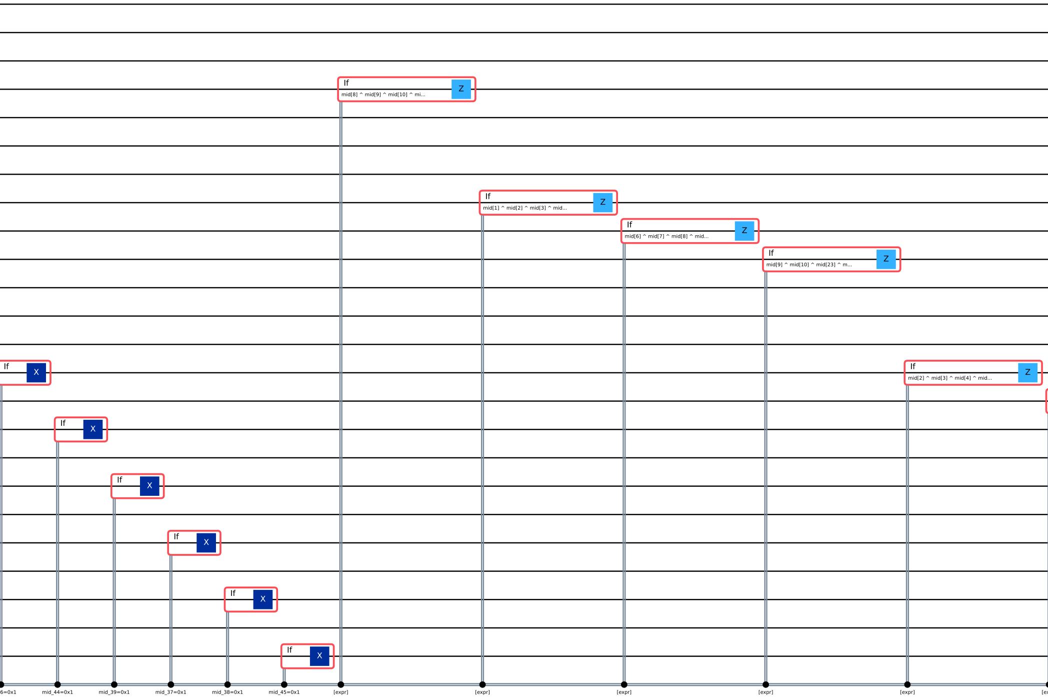
[Z12] DYNAMIC constant-depth circuit, fault-tolerant (|θ> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 6, 46 mid-circuit measurements, logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]

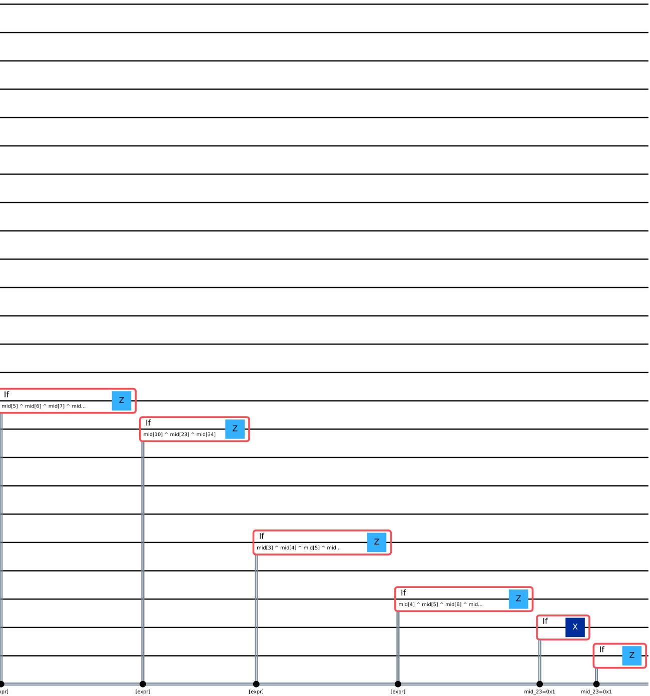

XX
q_{44}
q_{46}
q_{47}
q_{48}
q_{54}
q_{56}
q_{58}
q_{64}
q_{66}
q_{68}
q_{74}
q_{76}
q_{78}
q_{84}
q_{86}
q_{88}
q_{94}
q_{96}
q_{98}
q_{104}
q_{105}
q_{106}
q_{107}
q_{108}
mid
46
H
R_Z
0.8
2
0
1
3
4
5
6


ZZ
X(θ), XX
H
R_Z −0.8
If
mid[11] ^ mid[12] ^ mid[13] ^ ...
R_Z −0.8
Else
R_Z 0.8
H
7
8
9
10
22
14
11
12
13
15
16
17
18
19
20
21
26
[expr]

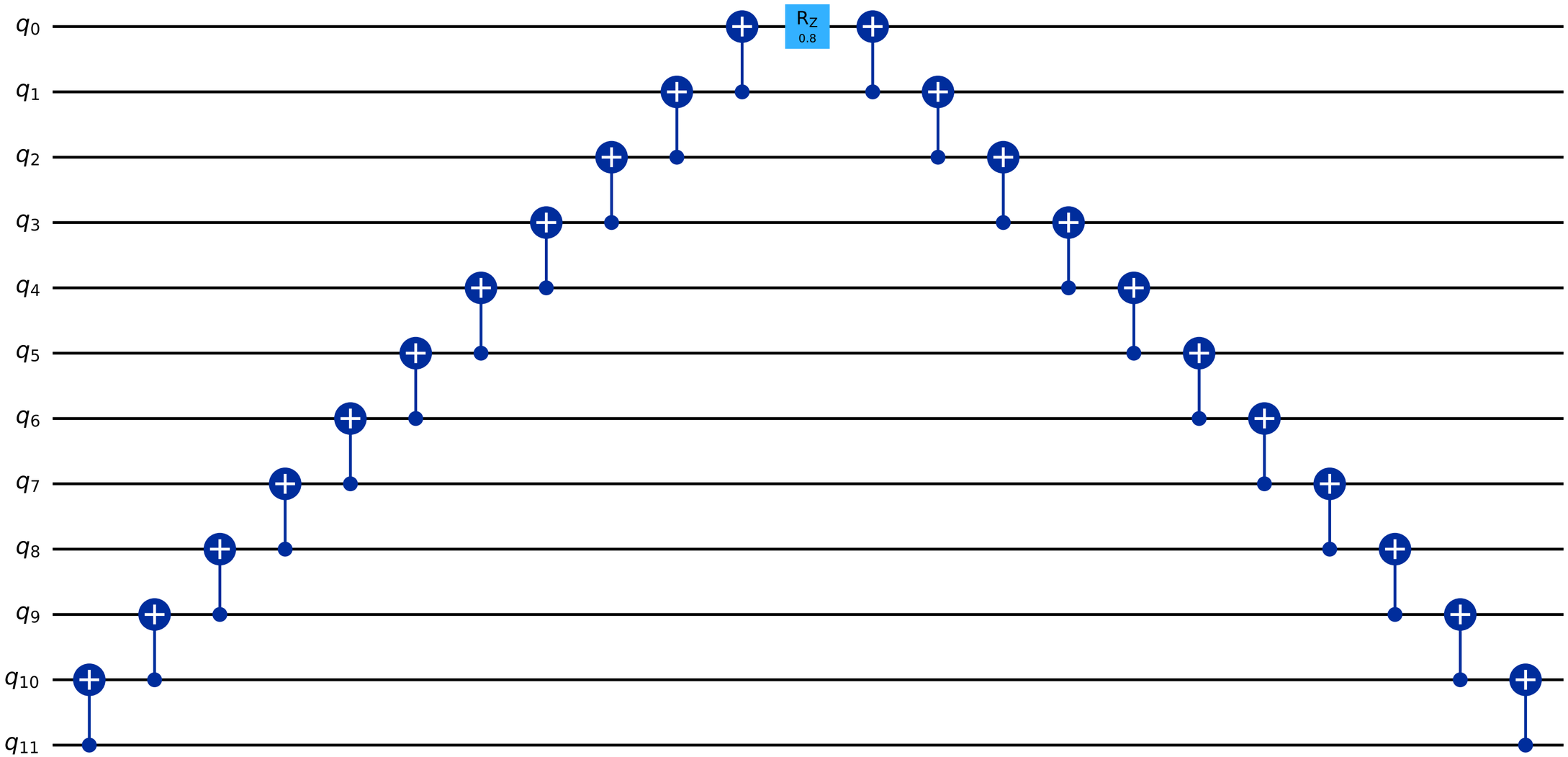

Pauli correction...
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Z
If
X
Z(a)
H
24
25
27
28
29
30
31
32
33
34
23
35
36
37
38
39
40
41
42
43
44
45
[expr]
mid_41=0x1
mid_4


If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
X
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
Z
If
X
If
X
If
X
If
X
If
X
2=0x1
mid_35=0x1
mid_43=0x1
mid_40=0x1
mid_36=0x1
mid_44=0x1
mid_39=0x1
mid_37=0x1
mid_45=0x1
mid_38=0x1
[expr]
mid_42=0x1
mid_41=0x1
mid_35=0x1
mid_43=0x1
mid_40=0x1
mid_3

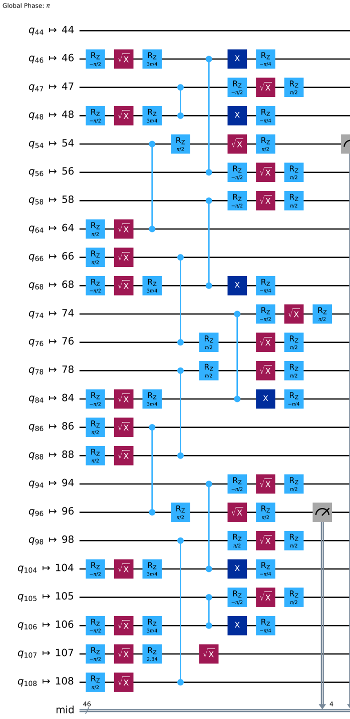

If
X
If
X
If
X
If
X
If
X
If
X
If
mid[8] ^ mid[9] ^ mid[10] ^ mi...
Z
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
Z
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
Z
If
mid[9] ^ mid[10] ^ mid[23] ^ m...
Z
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
Z
6=0x1
mid_44=0x1
mid_39=0x1
mid_37=0x1
mid_38=0x1
mid_45=0x1
[expr]
[expr]
[expr]
[expr]
[expr]

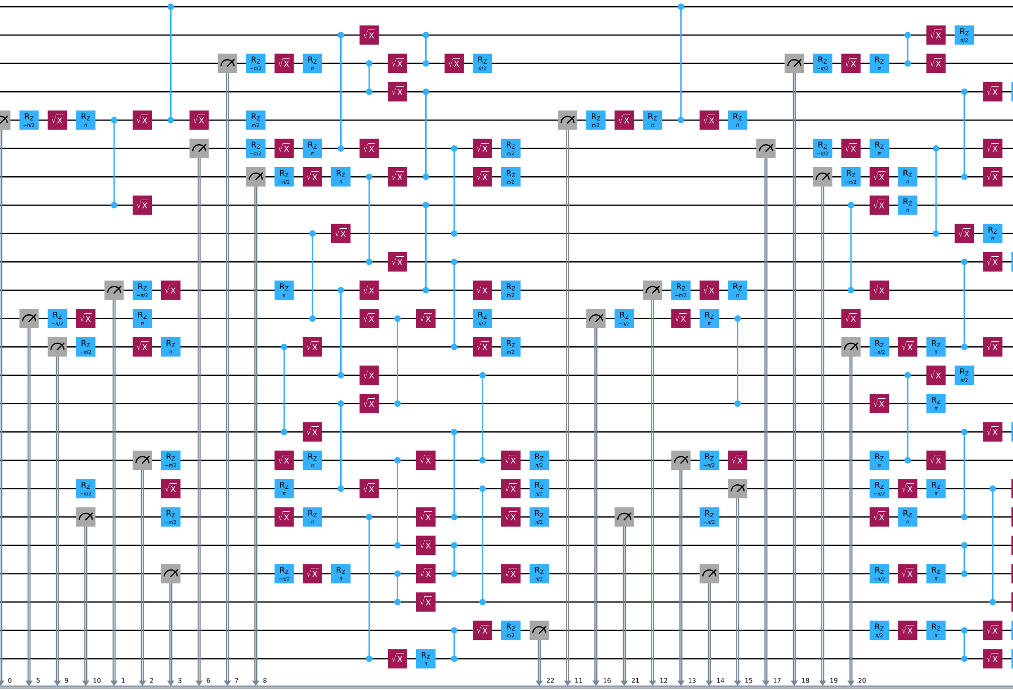

If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
Z
If
mid[10] ^ mid[23] ^ mid[34]
Z
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
Z
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
Z
If
X
If
Z
xpr]
[expr]
[expr]
[expr]
mid_23=0x1
mid_23=0x1

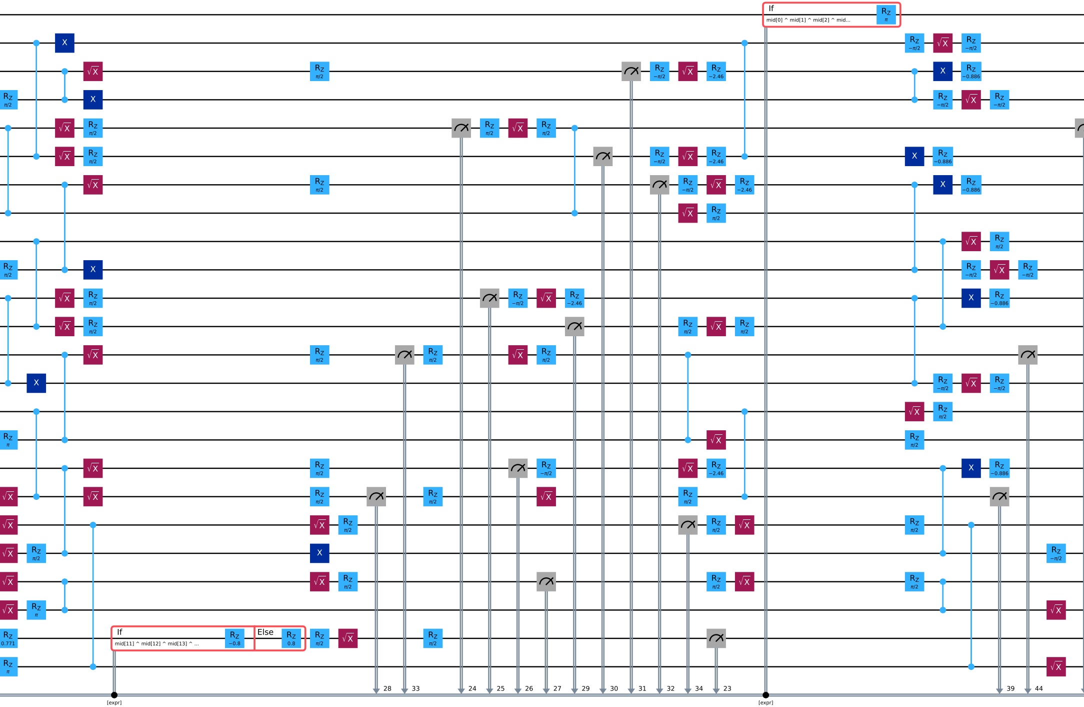
[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22

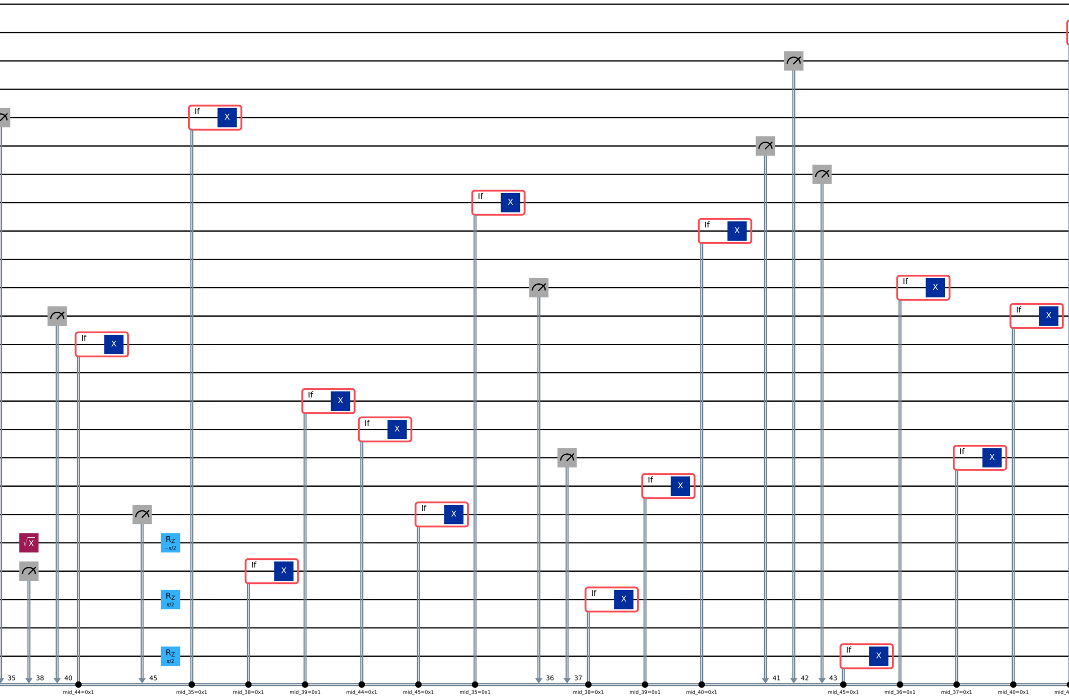


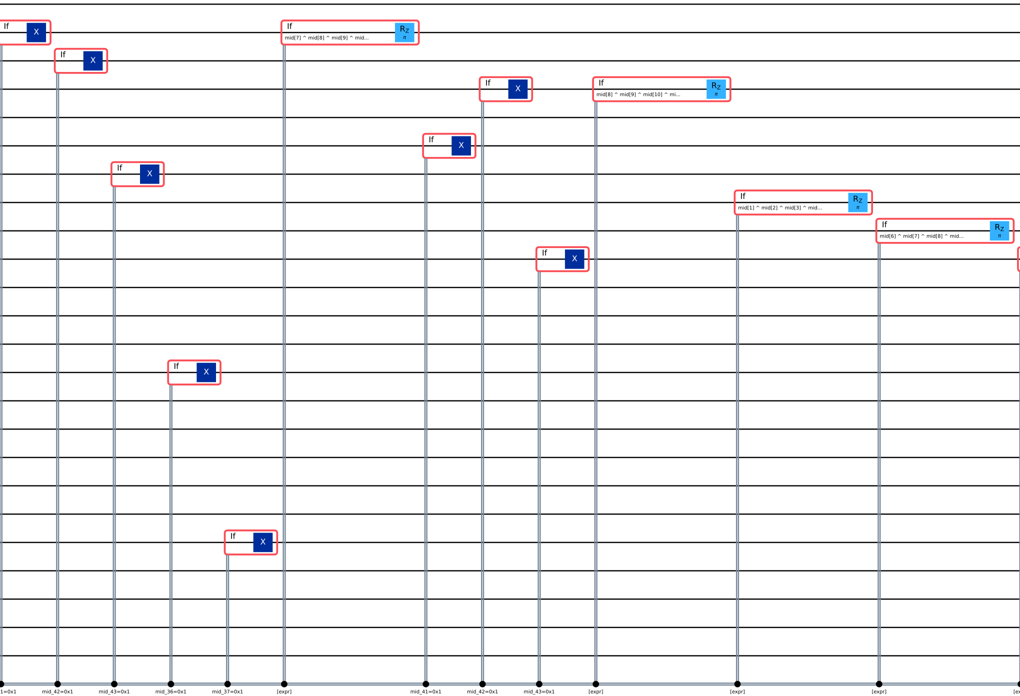
[Z12] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:175 sx:138 cz:68 measure:46 if_else:36 x:16)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 6, total depth 47

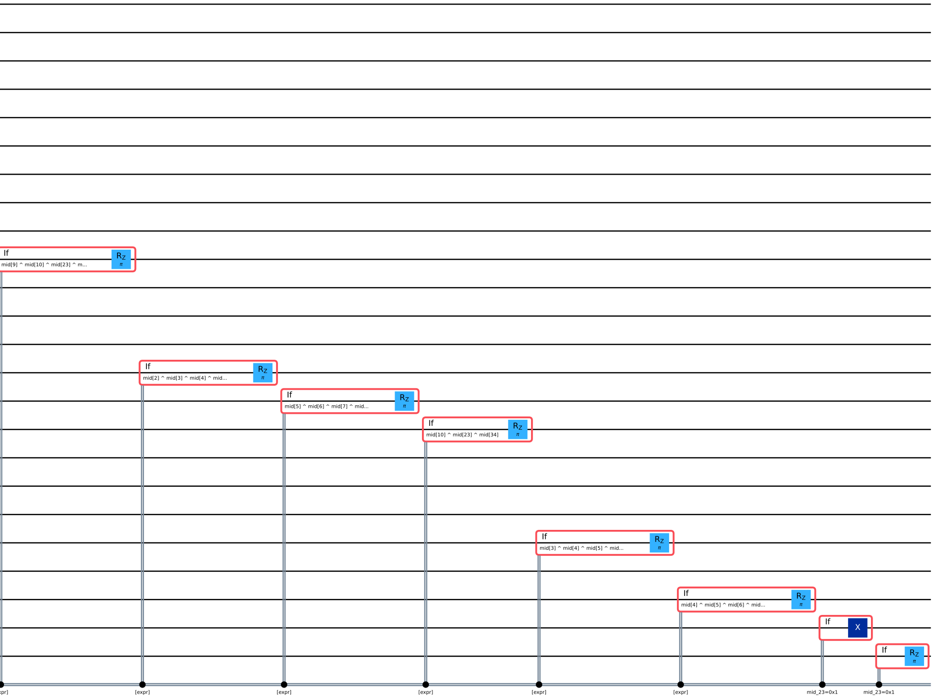

Global Phase: π
q44 ↦ 44
q46 ↦ 46
q47 ↦ 47
q48 ↦ 48
q54 ↦ 54
q56 ↦ 56
q58 ↦ 58
q64 ↦ 64
q66 ↦ 66
q68 ↦ 68
q74 ↦ 74
q76 ↦ 76
q78 ↦ 78
q84 ↦ 84
q86 ↦ 86
q88 ↦ 88
q94 ↦ 94
q96 ↦ 96
q98 ↦ 98
q104 ↦ 104
q105 ↦ 105
q106 ↦ 106
q107 ↦ 107
q108 ↦ 108
mid
46
4
RZ −π/2
RZ π/2
RZ 3π/4
RZ −π/4
RZ 2.34
√X
X


Rz
−π/2
π/2
π
√X
0
5
9
10
1
2
3
6
7
8
22
11
16
21
12
13
14
15
17
18
19
20

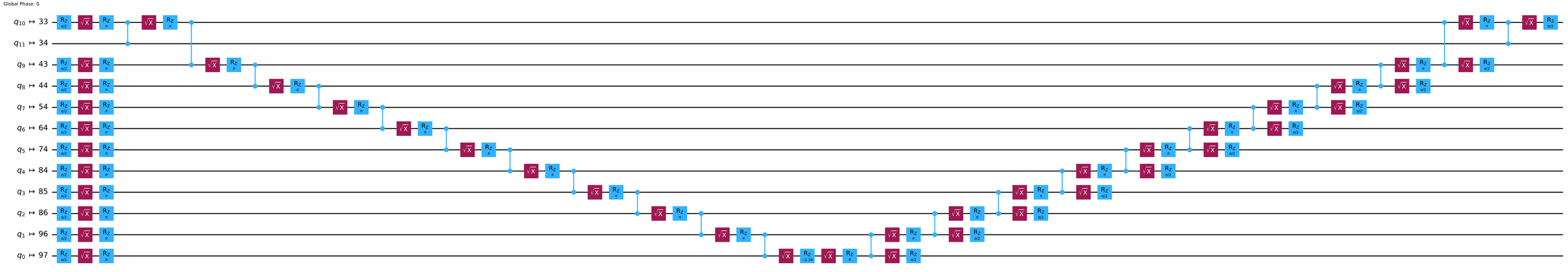

If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
If
mid[11] ^ mid[12] ^ mid[13] ^ ...
Else
Rz
π
−π/2
π/2
−2.46
−0.886
0.771
−0.8
0.8
√X
X
28
33
24
25
26
27
29
30
31
32
34
23
39
44
[expr]
[expr]


√X
35
38
40
mid_44=0x1
If
X
45
R_Z −π/2
R_Z π/2
R_Z π/2
mid_35=0x1
mid_38=0x1
mid_39=0x1
mid_44=0x1
mid_45=0x1
mid_35=0x1
36
37
mid_38=0x1
mid_39=0x1
mid_40=0x1
41
42
43
mid_45=0x1
mid_36=0x1
mid_37=0x1
mid_40=0x1
mid_4


If
X
If
X
If
X
If
X
If
X
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
RZ
π
If
X
If
X
If
X
If
mid[8] ^ mid[9] ^ mid[10] ^ mi...
RZ
π
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
RZ
π
If
mid[6] ^ mid[7] ^ mid[8] ^ mid...
RZ
π
1=0x1
mid_42=0x1
mid_43=0x1
mid_36=0x1
mid_37=0x1
[expr]
mid_41=0x1
mid_42=0x1
mid_43=0x1
[expr]
[expr]
[expr]
[ex

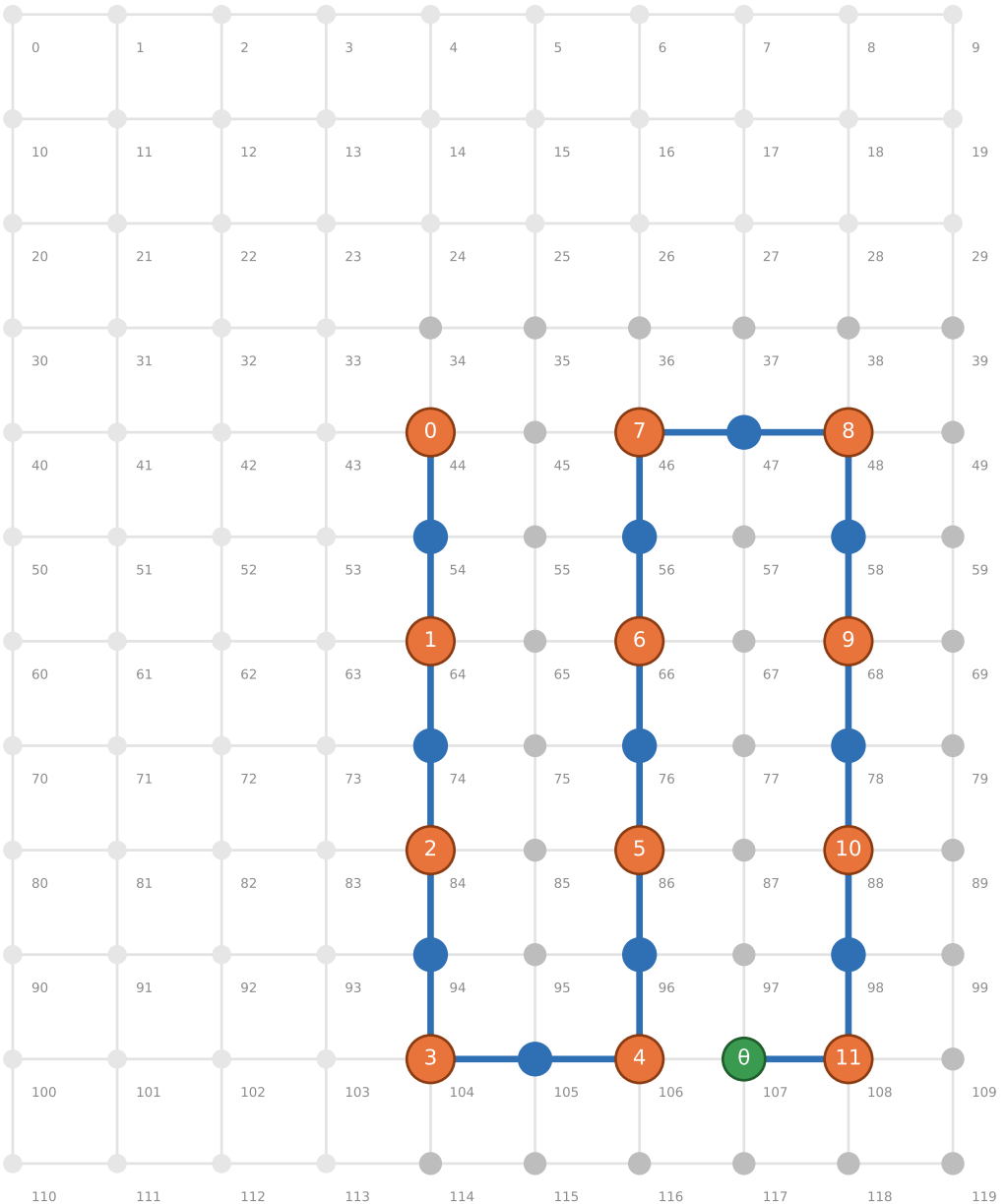

If
R_Z
π
mid[9] ^ mid[10] ^ mid[23] ^ m...
If
R_Z
π
mid[2] ^ mid[3] ^ mid[4] ^ mid...
If
R_Z
π
mid[5] ^ mid[6] ^ mid[7] ^ mid...
If
R_Z
π
mid[10] ^ mid[23] ^ mid[34]
If
R_Z
π
mid[3] ^ mid[4] ^ mid[5] ^ mid...
If
R_Z
π
mid[4] ^ mid[5] ^ mid[6] ^ mid...
If
X
If
R_Z
π
xpr]
[expr]
[expr]
[expr]
[expr]
[expr]
mid_23=0x1
mid_23=0x1

[Z12] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 (12 logical qubits)
2q depth 22, total depth 71

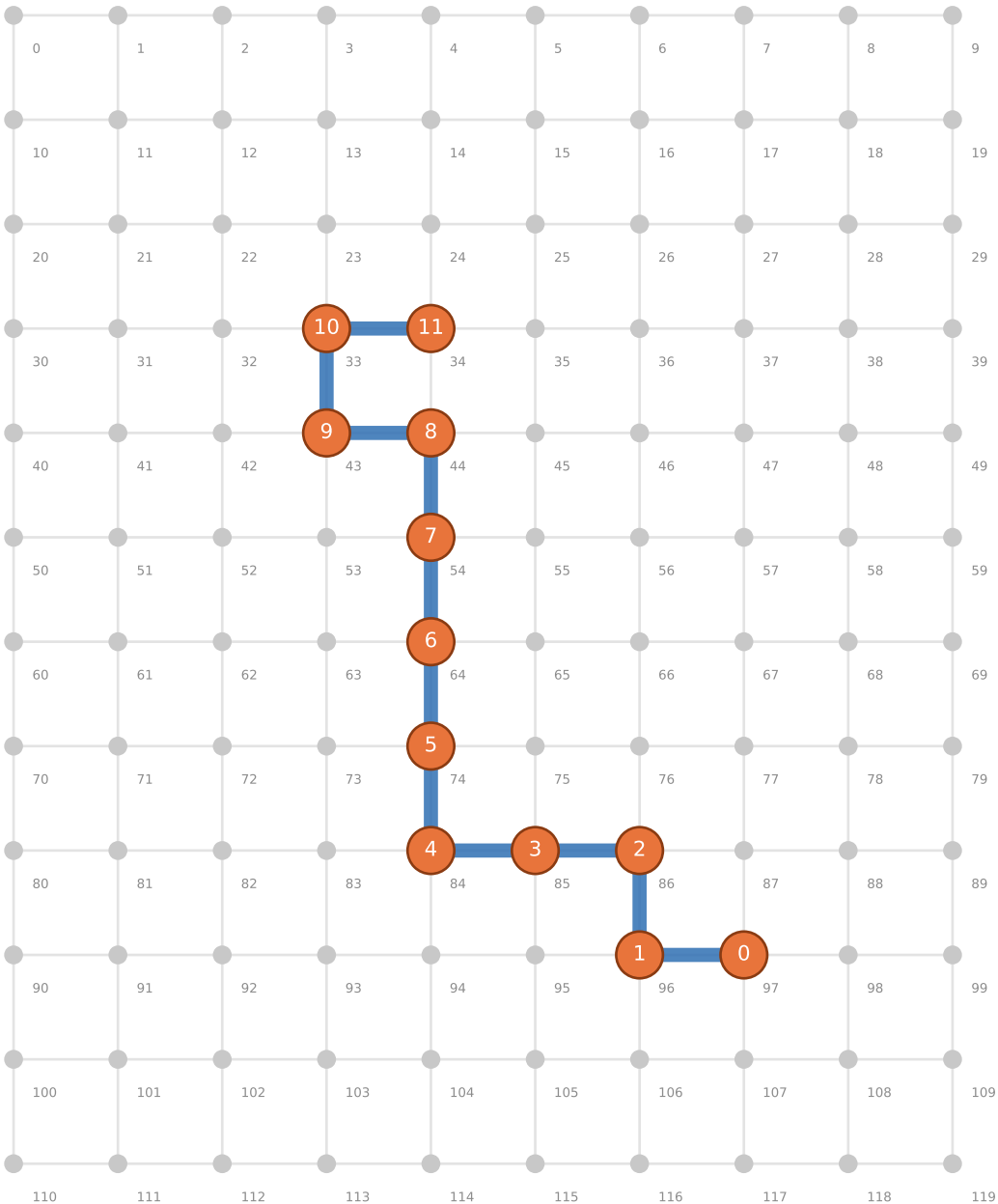


[Z12] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

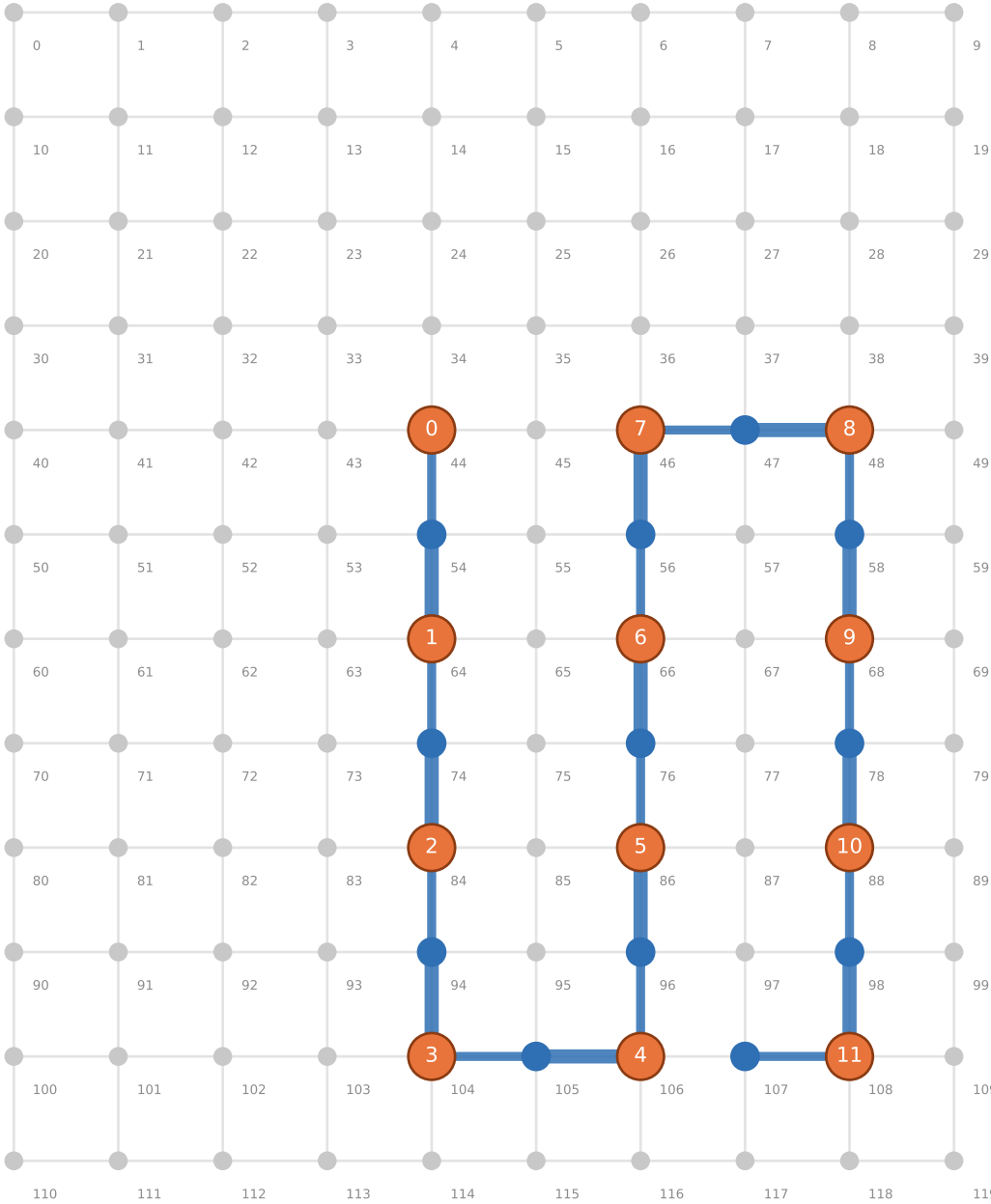
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers



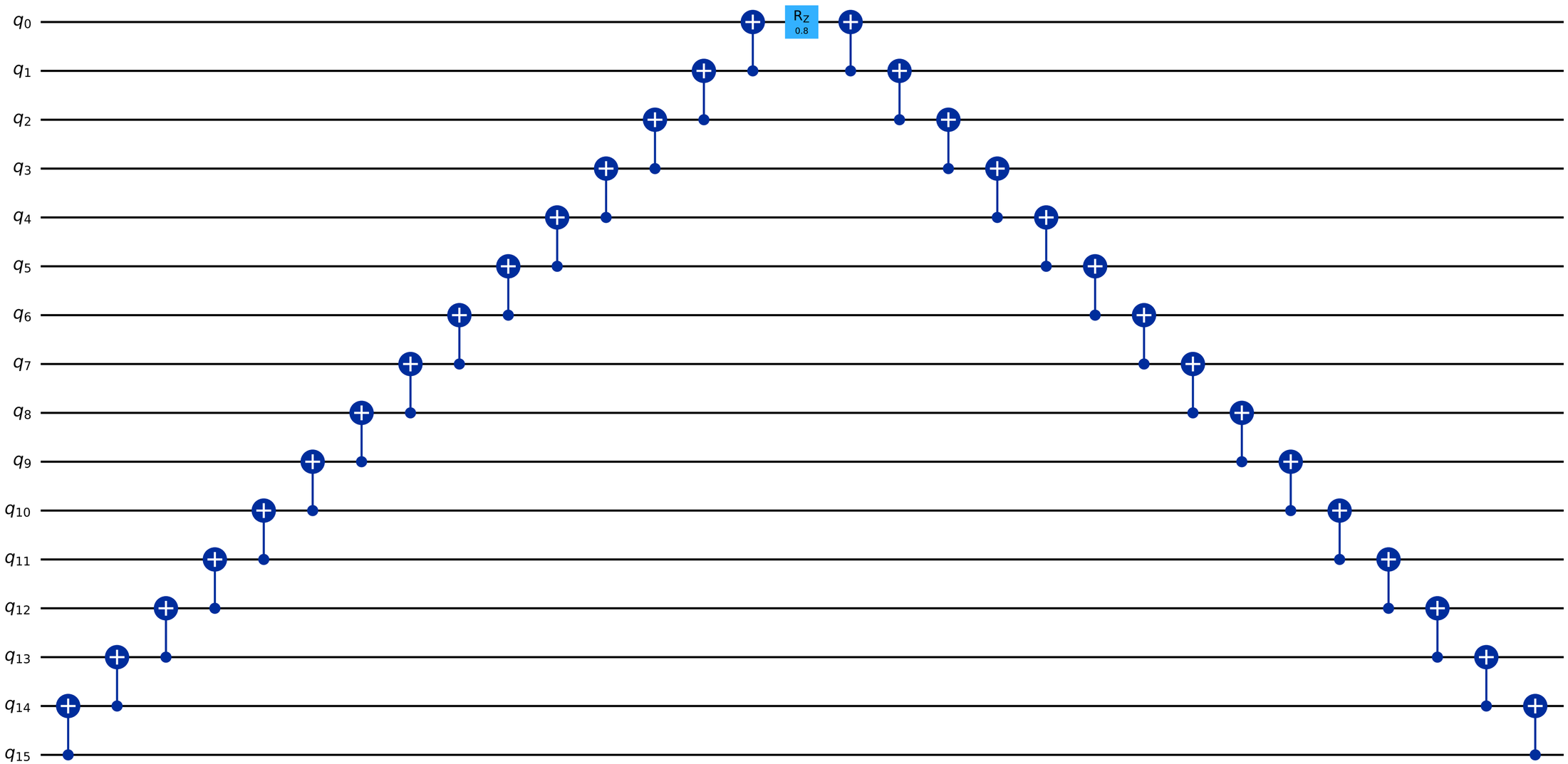
[Z12] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [97, 96, 86, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)



[Z12] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 68

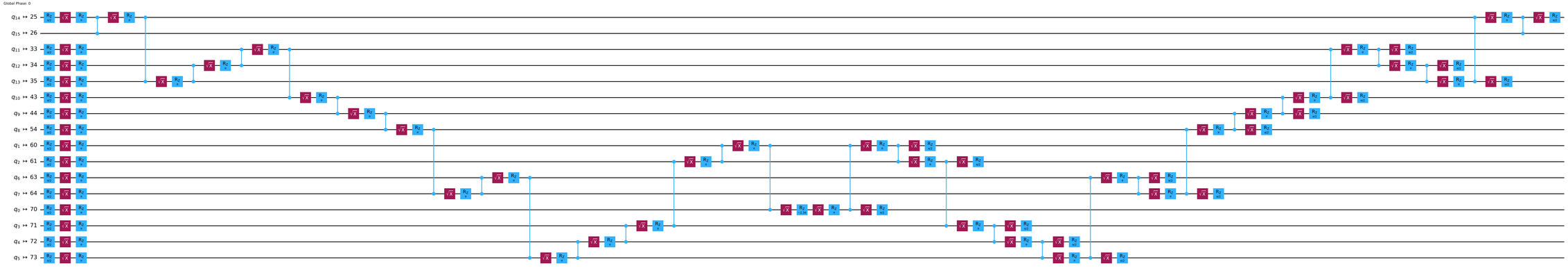


[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30



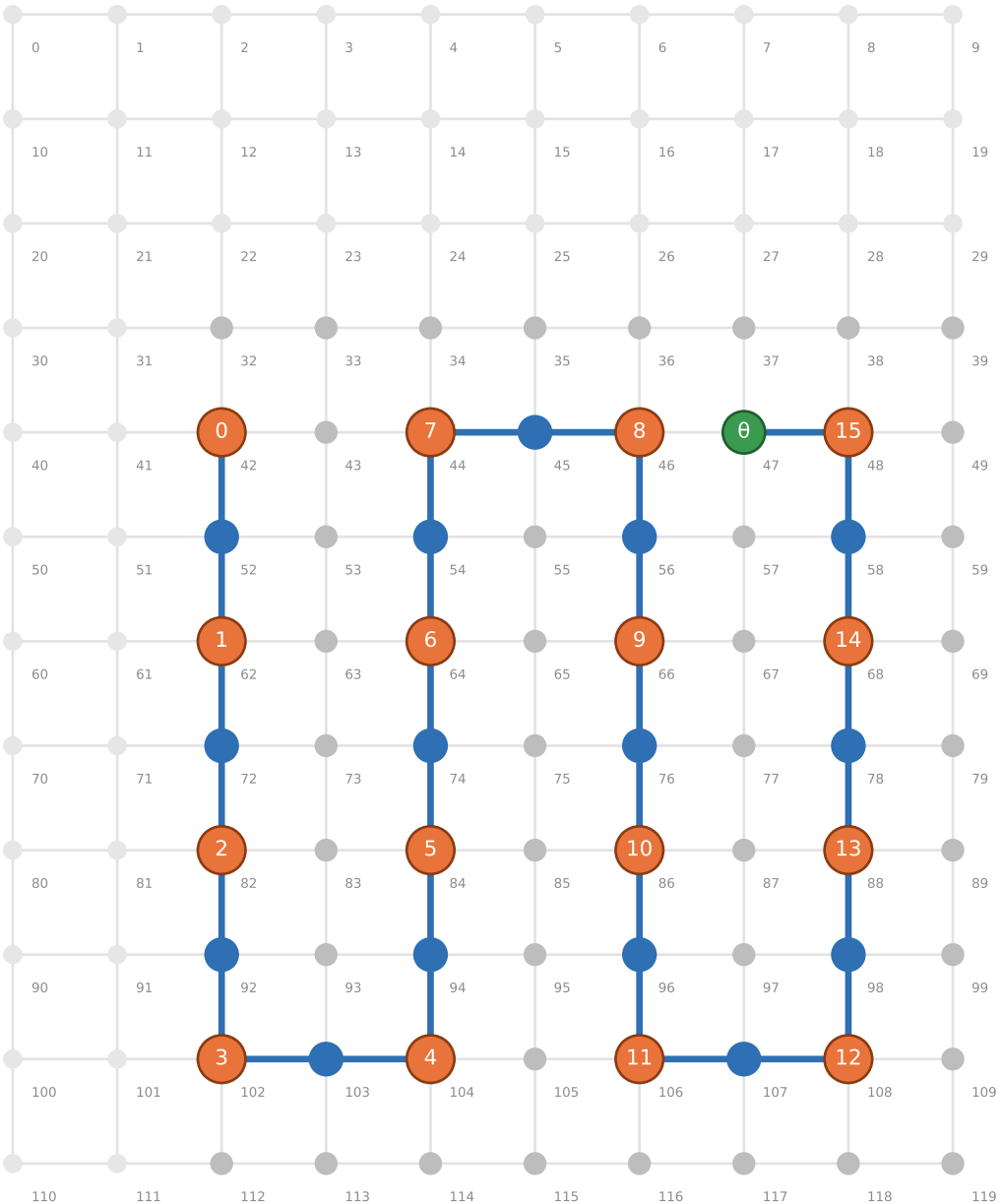
Unreal Frame: 4
Root

[Z16] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

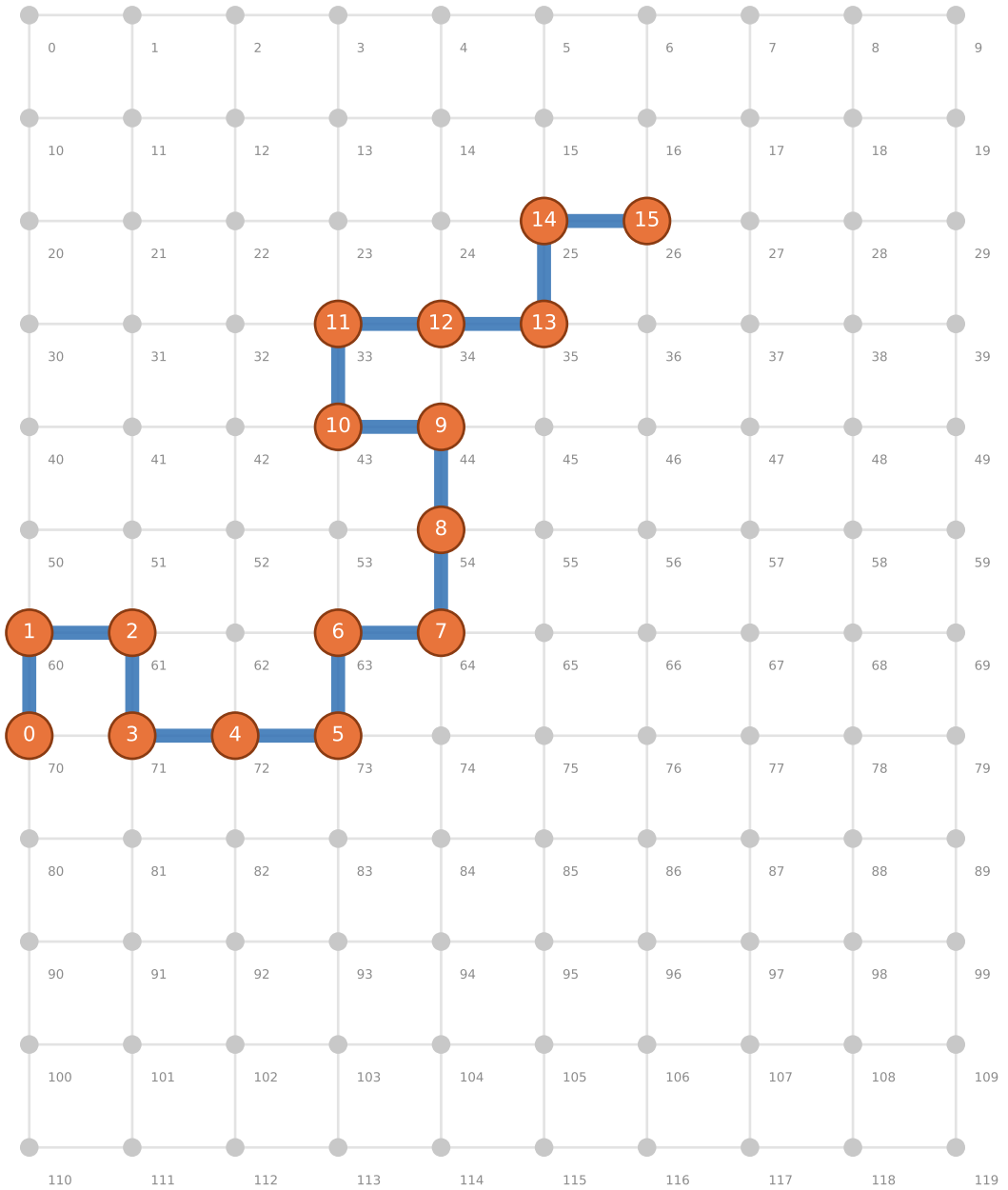


[Z16] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

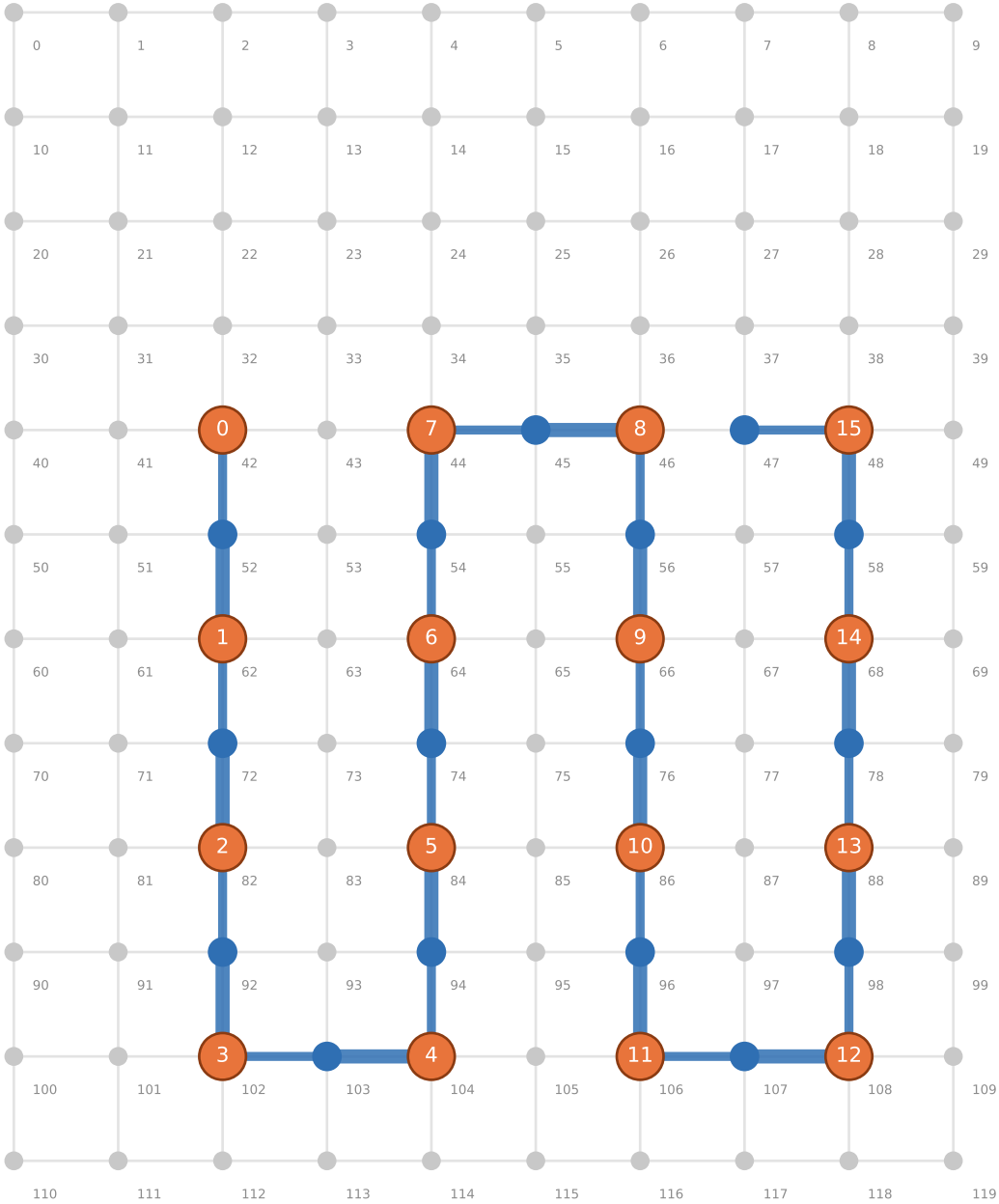
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



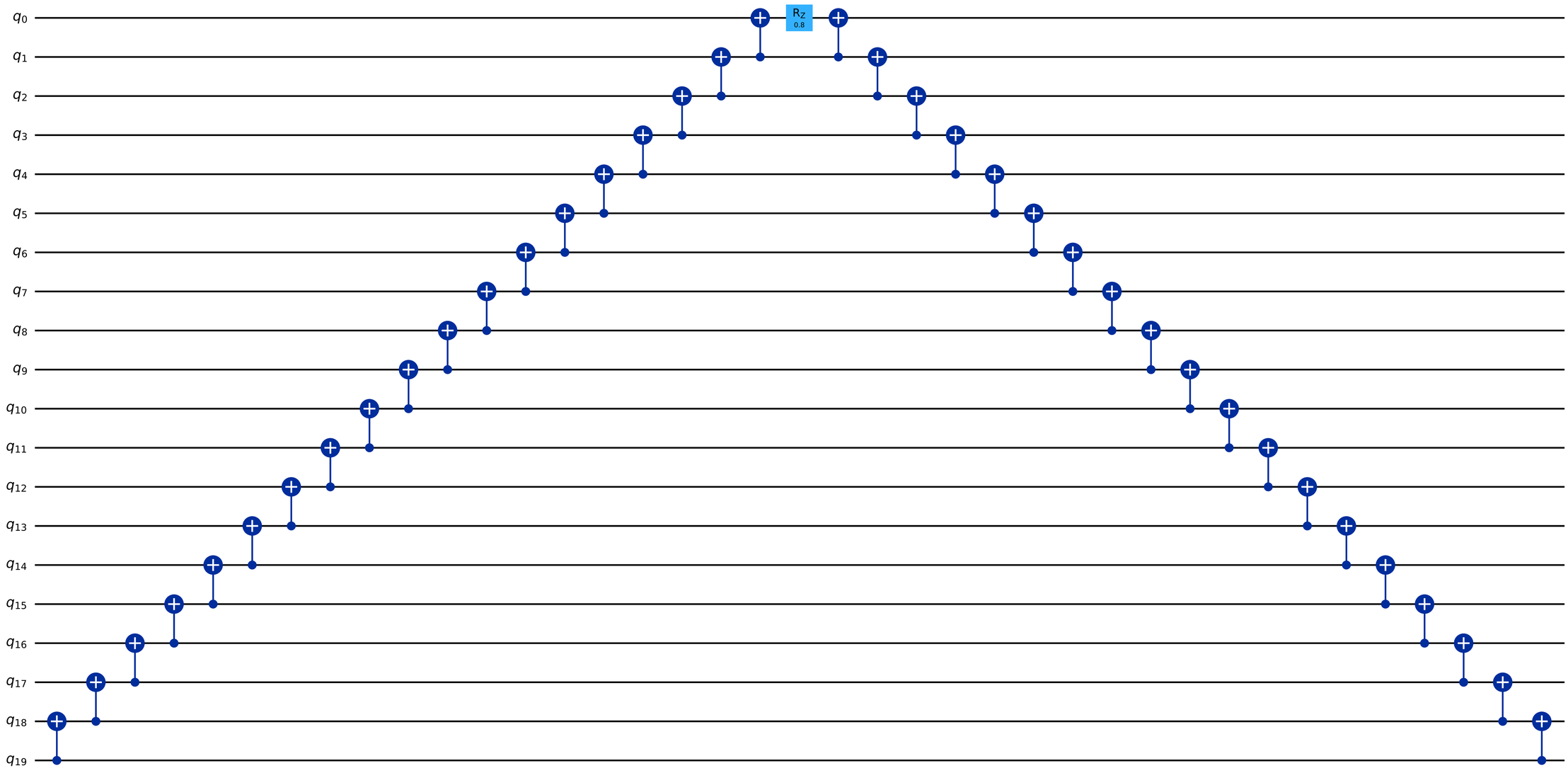
[Z16] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [70, 60, 61, 71, 72, 73, 63, 64, 54, 44, 43, 33, 34, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)



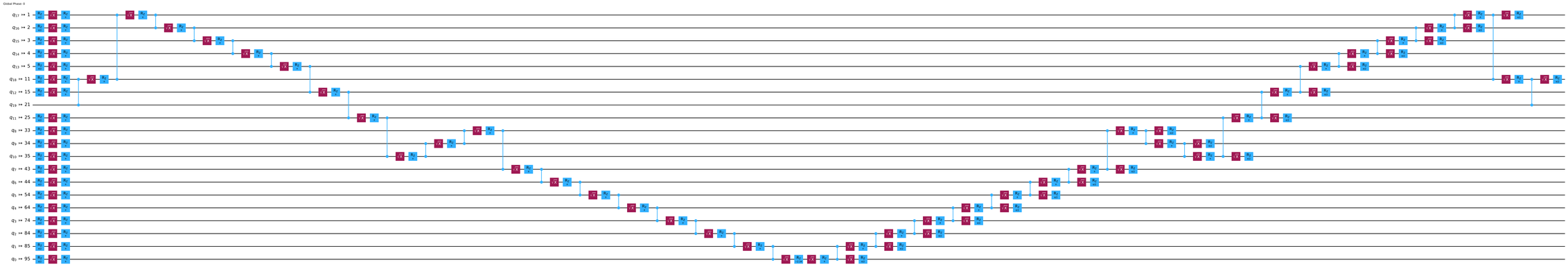
[Z16] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 92



[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

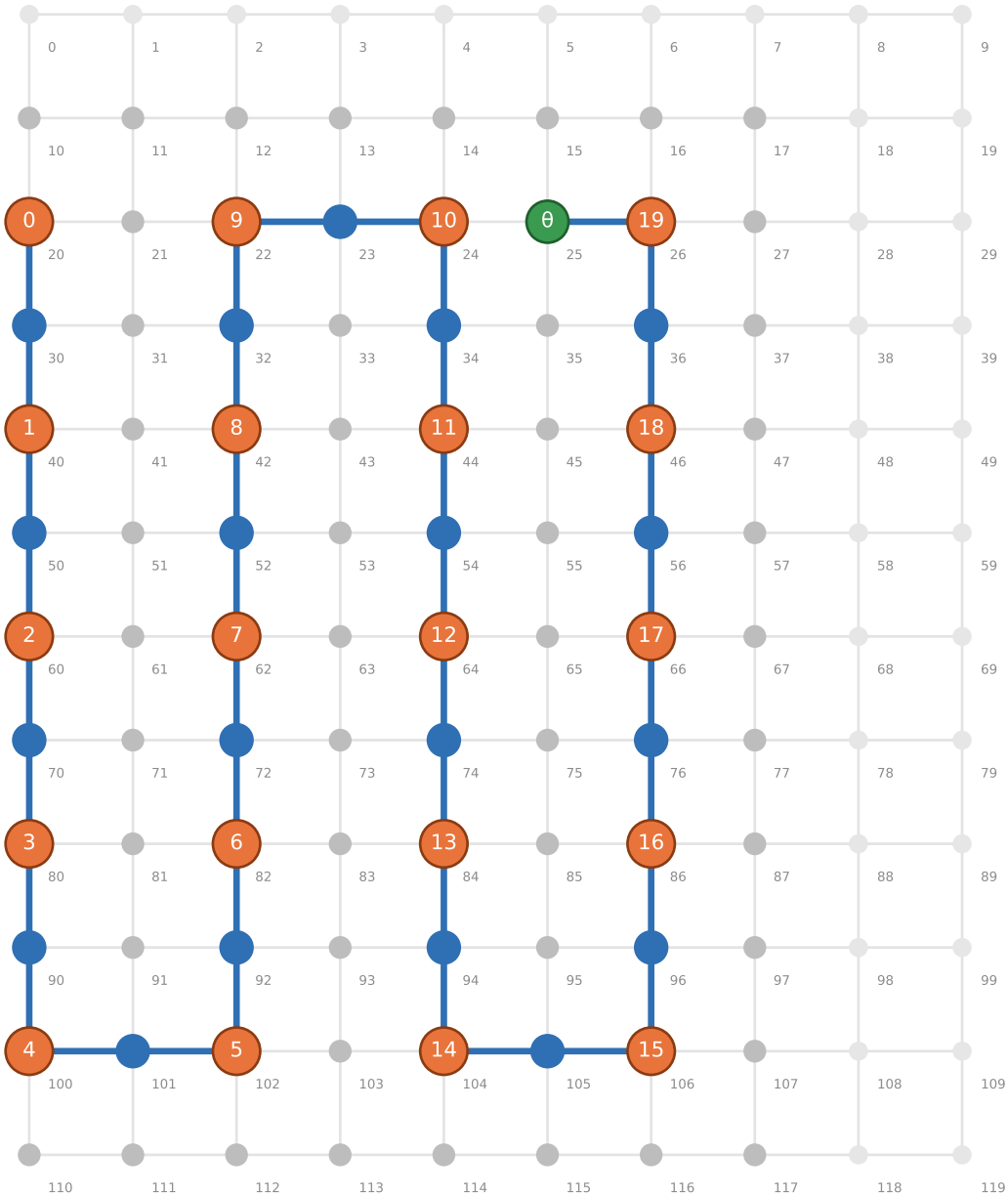


[Z20] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119

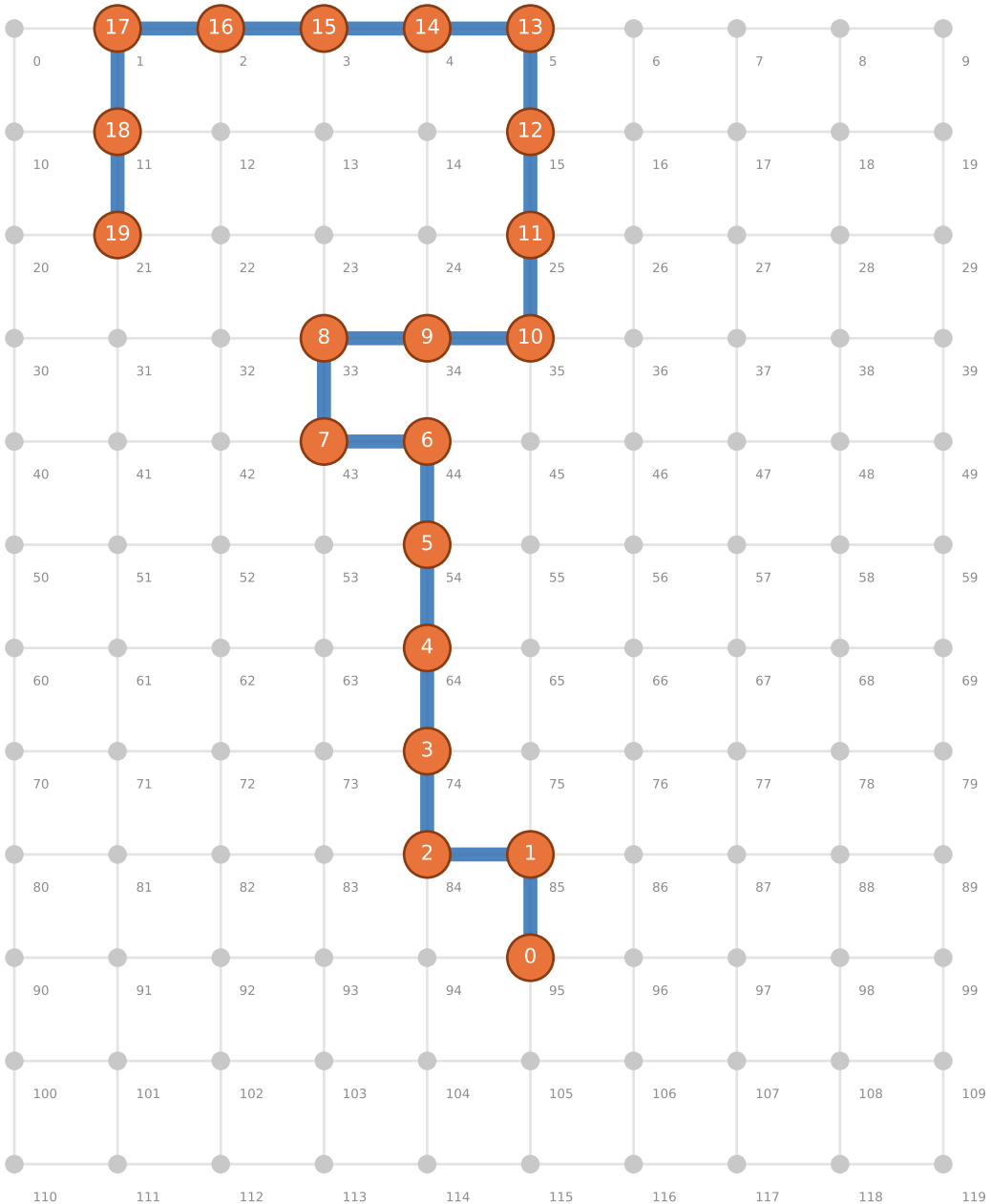


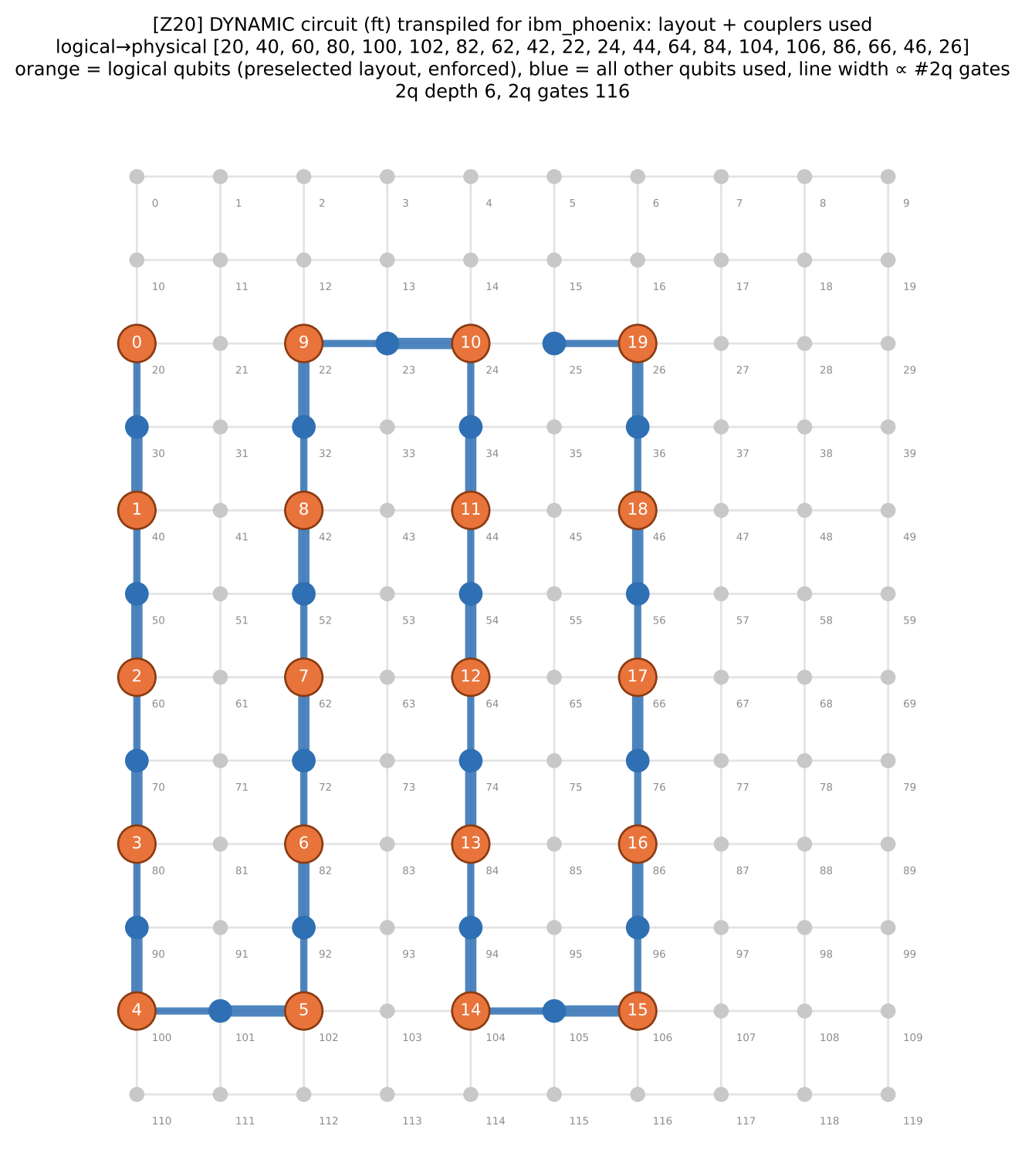
[Z20] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

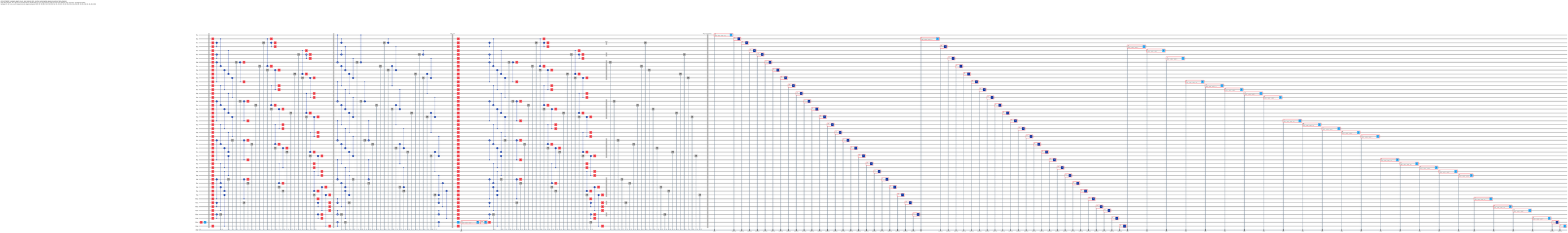
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers



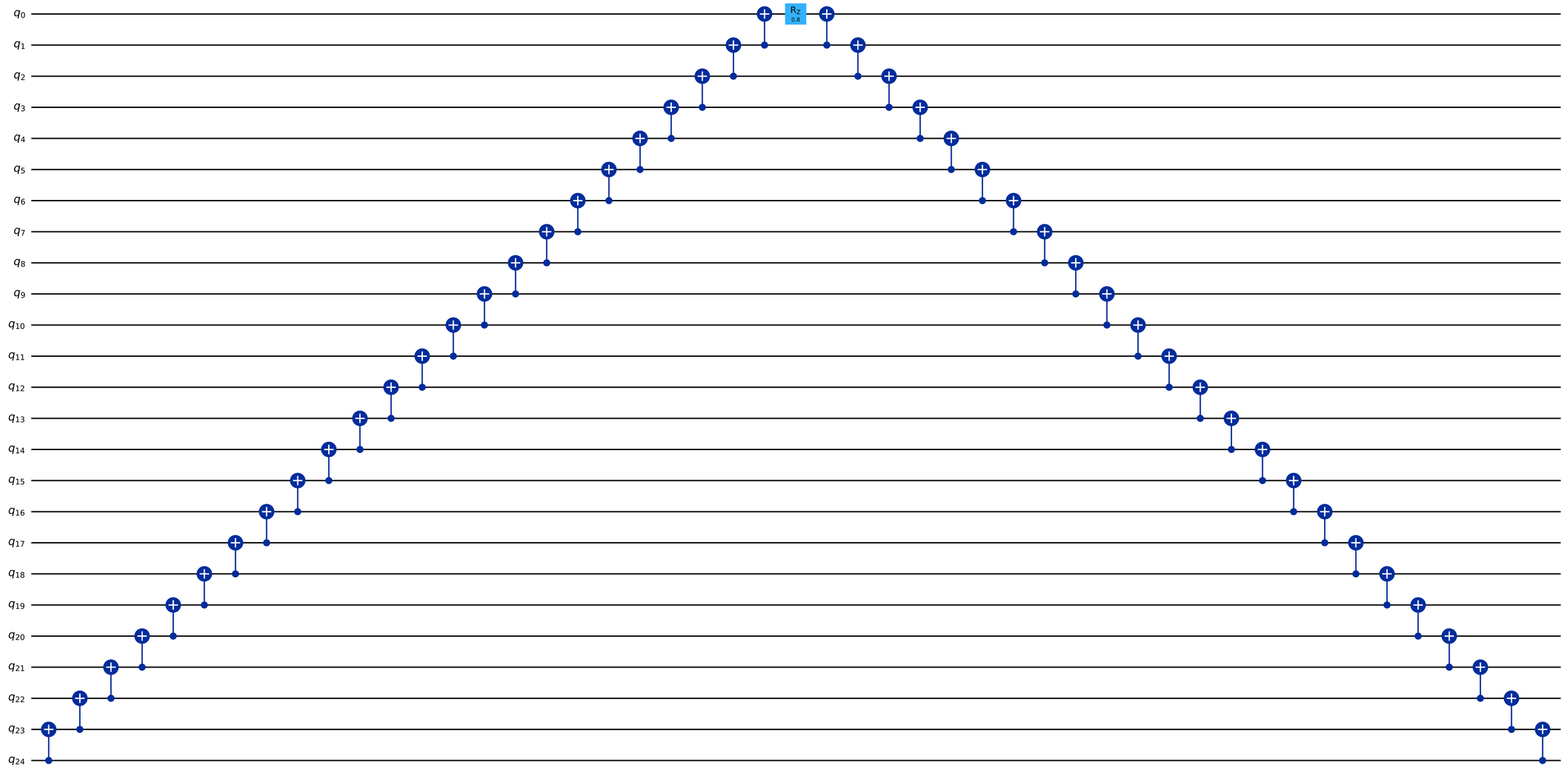
[Z20] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34, 35, 25, 15, 5, 4, 3, 2, 1, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)

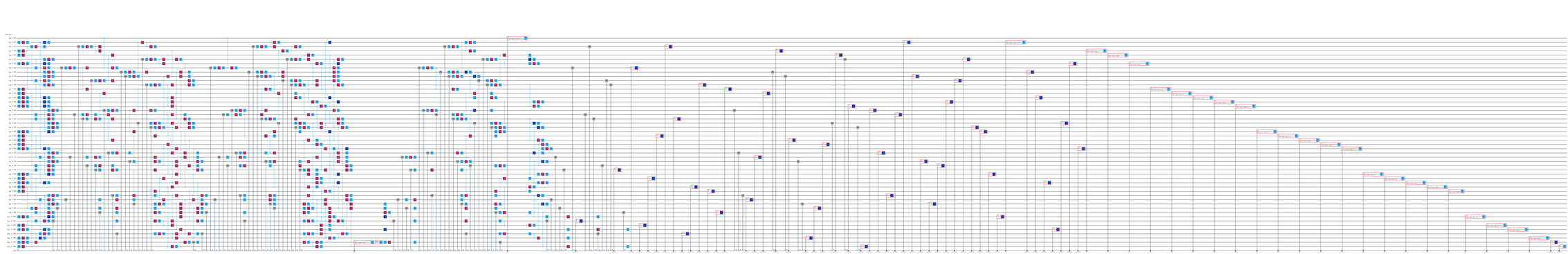



[Z20] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 116
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
0 9 10 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
1 8 11 18
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
2 7 12 17
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
3 6 13 16
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
4 5 14 15
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

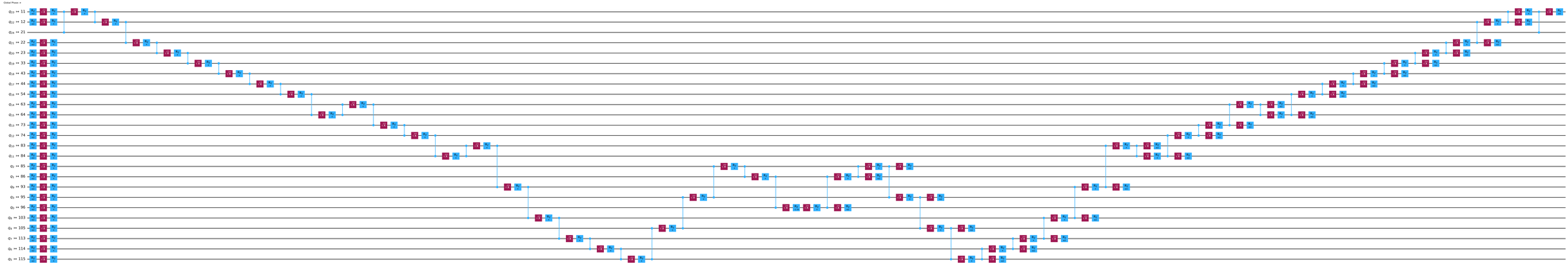


[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48



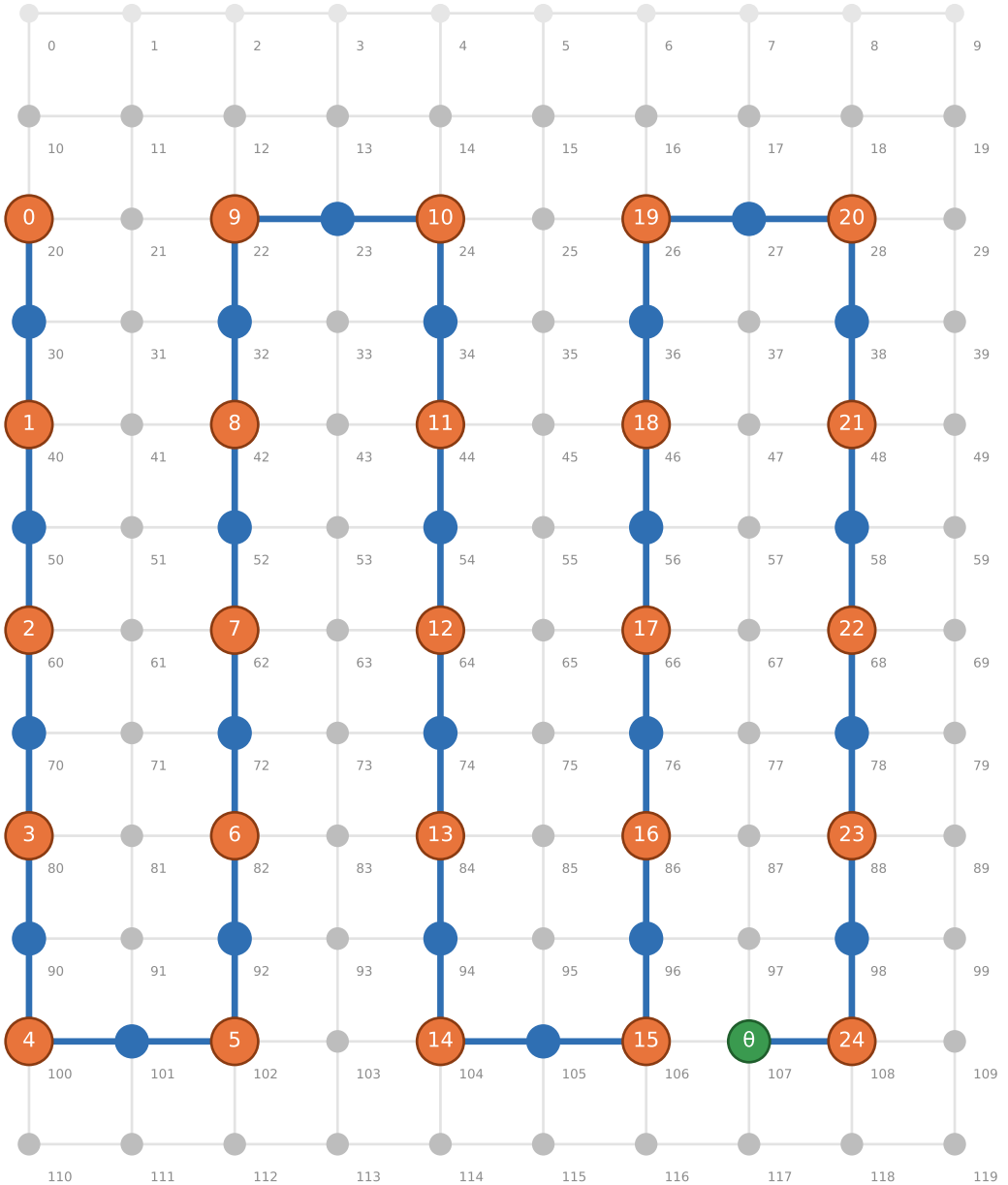


[225] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz/120 sx/96 cz/48)
exp(-i θ P), θ=0.4; P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24 (25 logical qubits)
2q depth 48, total depth 149

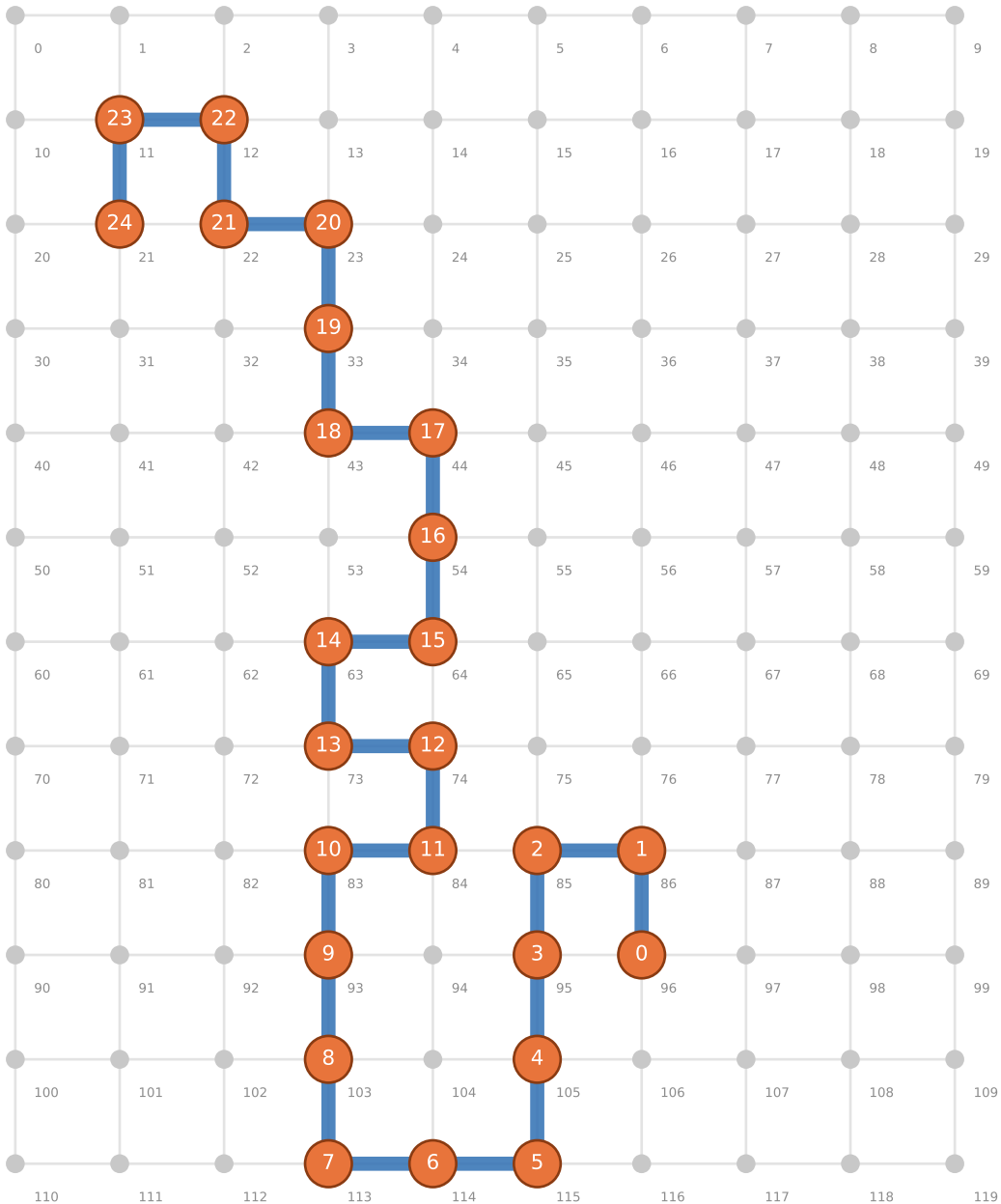


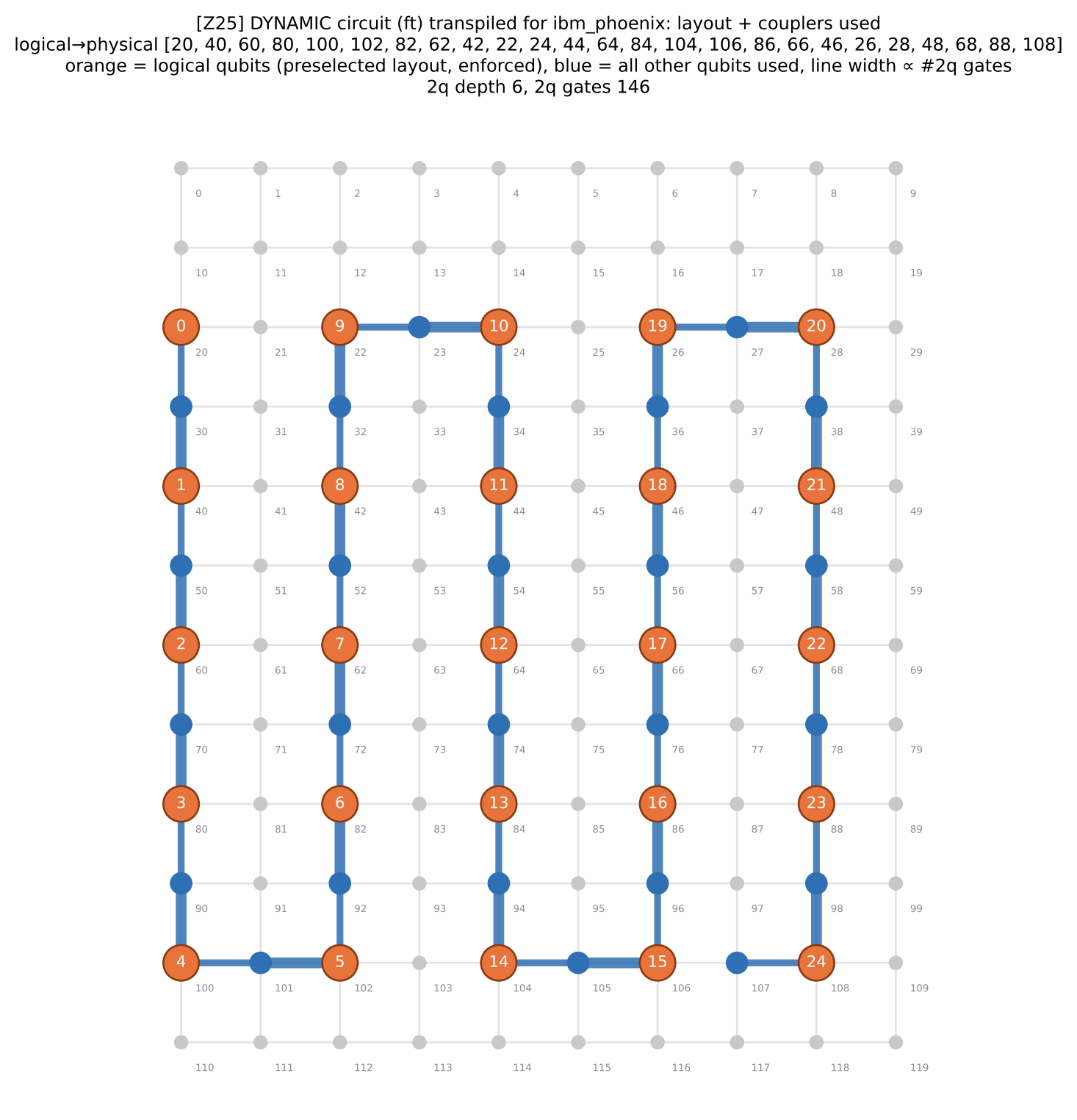
[Z25] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers

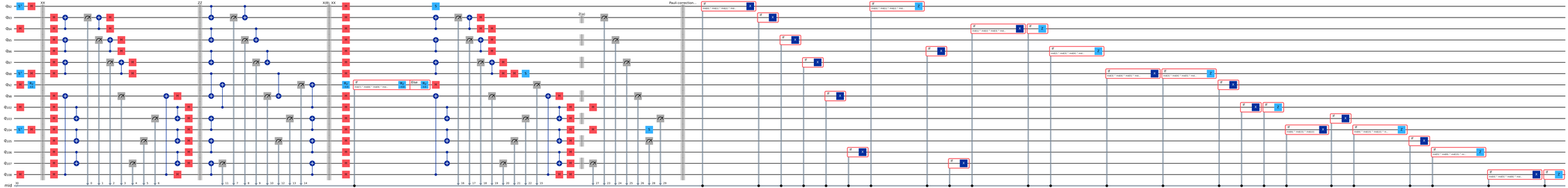


[Z25] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [96, 86, 85, 95, 105, 115, 114, 113, 103, 93, 83, 84, 74, 73, 63, 64, 54, 44, 43, 33, 23, 22, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)

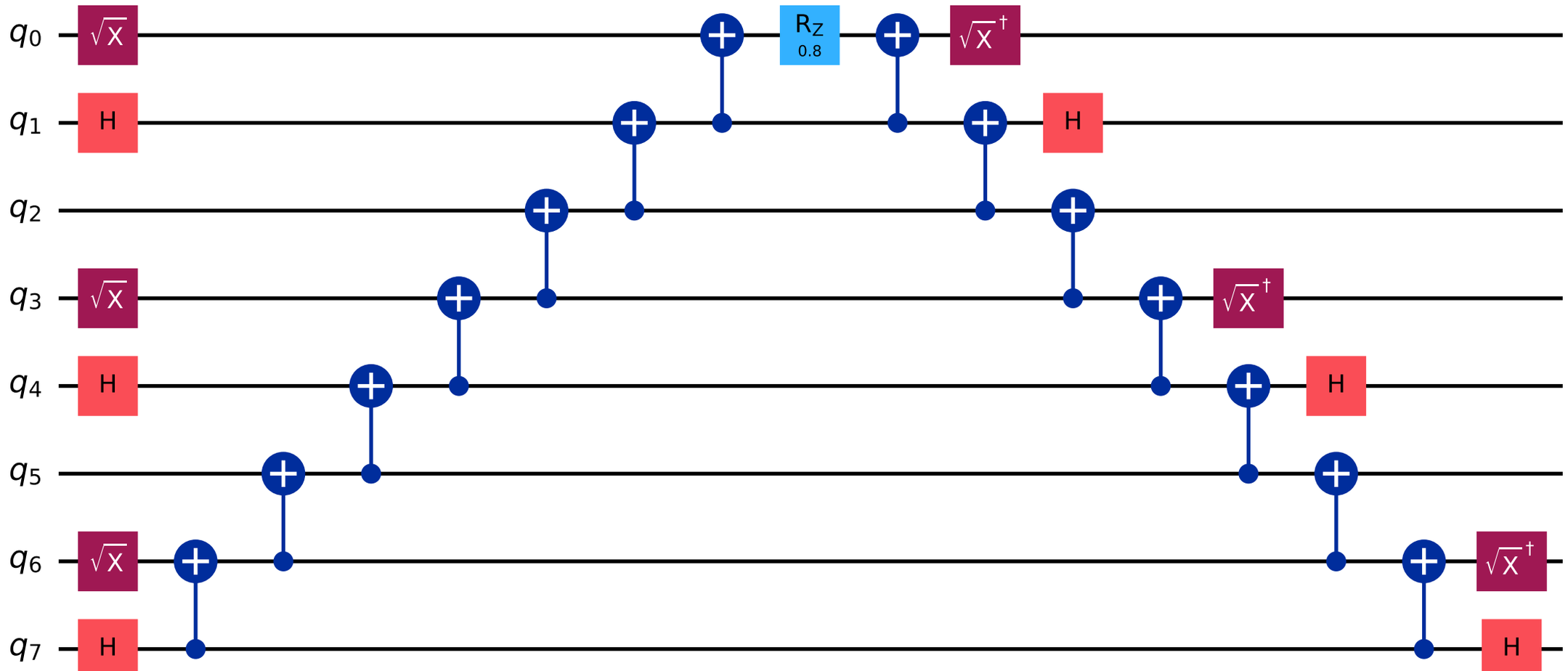



[Z25] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 146
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
0 9 10 19 20
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
1 8 11 18 21
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
2 7 12 17 22
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
3 6 13 16 23
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
4 5 14 15 24
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

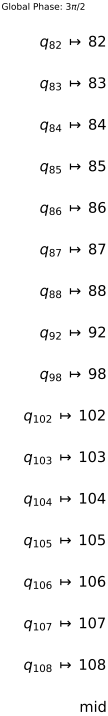
[XYZXYZXY] DYNAMIC constant-depth circuit, fault-tolerant (|0> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P)_{θ=0.4} · P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7 · (8 logical qubits)
2q depth 6, 30 mid-circuit measurements, logical=physical [82, 84, 86, 88, 108, 106, 104, 102]

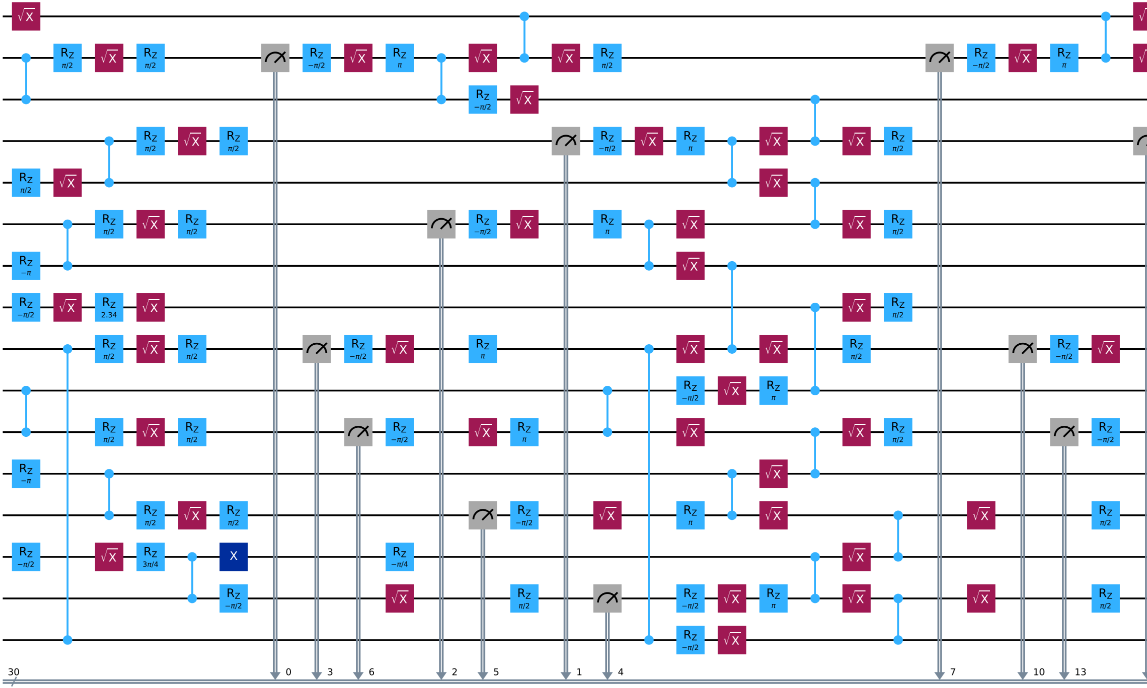


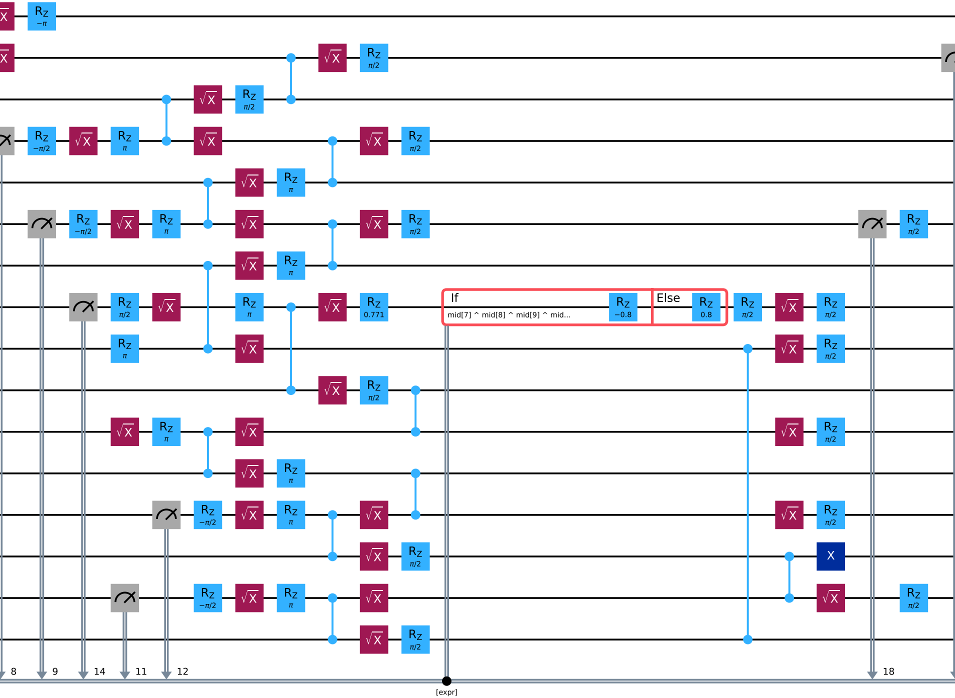
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14



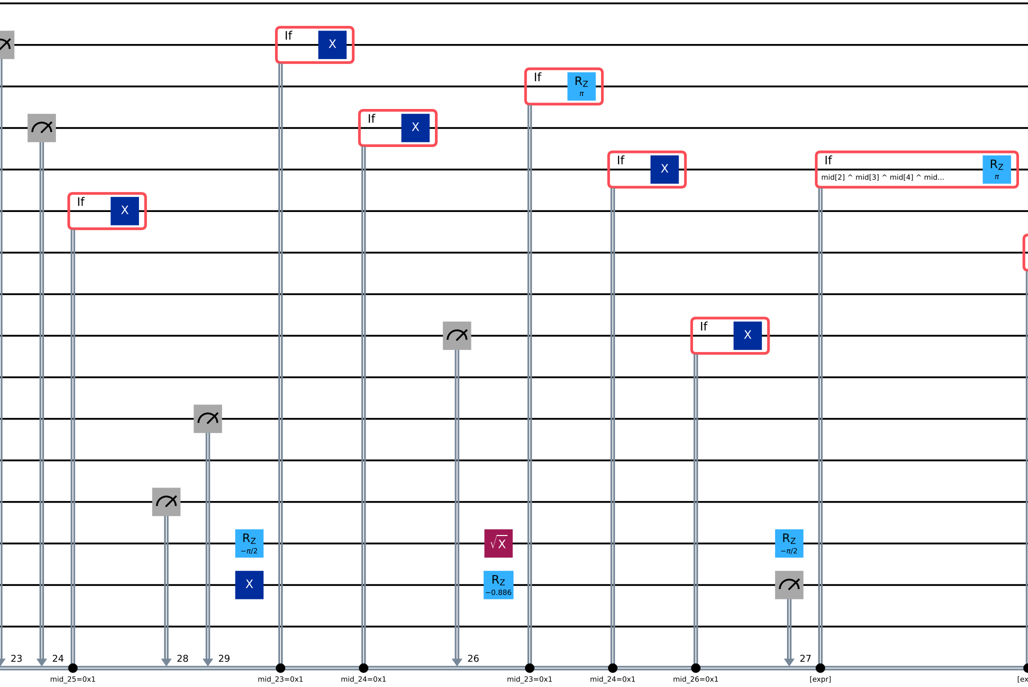
[XYZXYZXY] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:100 sx:82 cz:44 measure:30 if_else:25 x:3)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 6, total depth 43

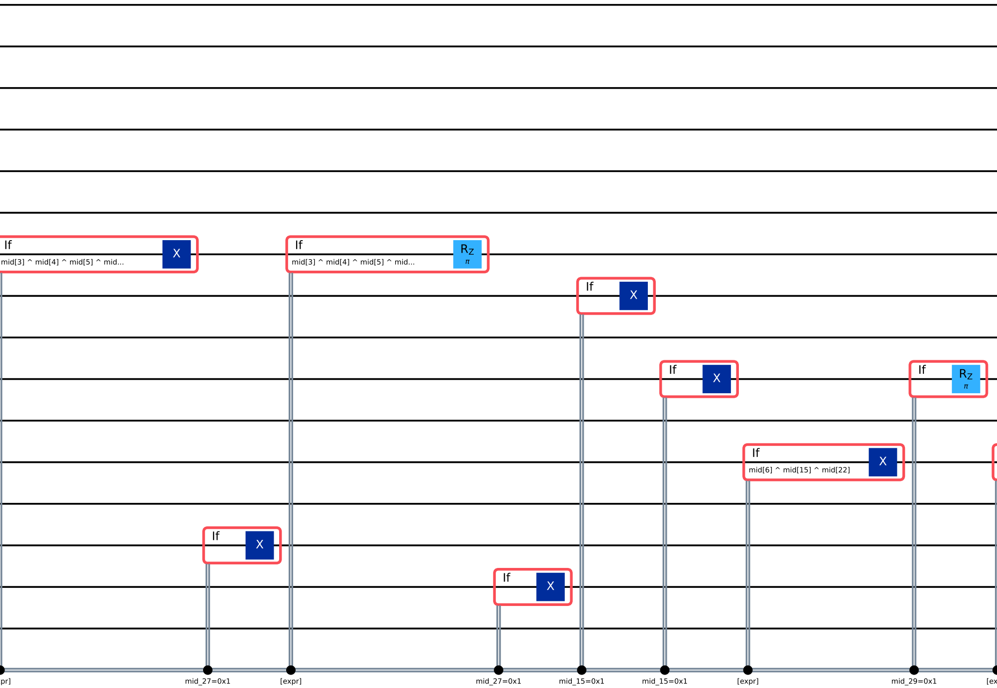

Global Phase: 3π/2
q_{82} ↦ 82
q_{83} ↦ 83
q_{84} ↦ 84
q_{85} ↦ 85
q_{86} ↦ 86
q_{87} ↦ 87
q_{88} ↦ 88
q_{92} ↦ 92
q_{98} ↦ 98
q_{102} ↦ 102
q_{103} ↦ 103
q_{104} ↦ 104
q_{105} ↦ 105
q_{106} ↦ 106
q_{107} ↦ 107
q_{108} ↦ 108
mid

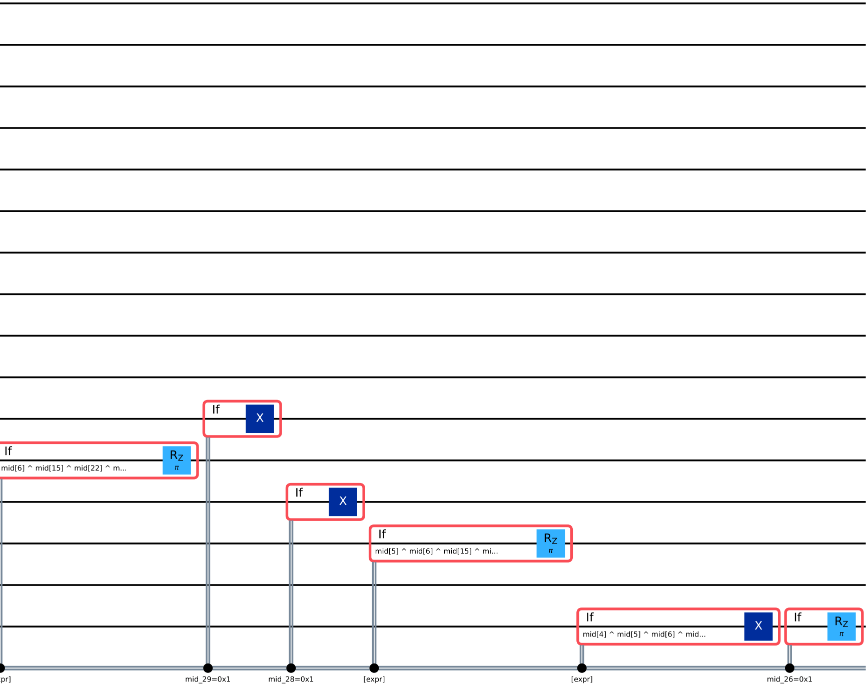

√X
R_Z
π/2
−π/2
π
−π
2.34
3π/4
−π/4
X
30
0
3
6
2
5
1
4
7
10
13


√X
RZ
−π
π/2
−π/2
π
0.771
If
mid[7] ^ mid[8] ^ mid[9] ^ mid...
−0.8
Else
0.8
X
8
9
14
11
12
[expr]
18

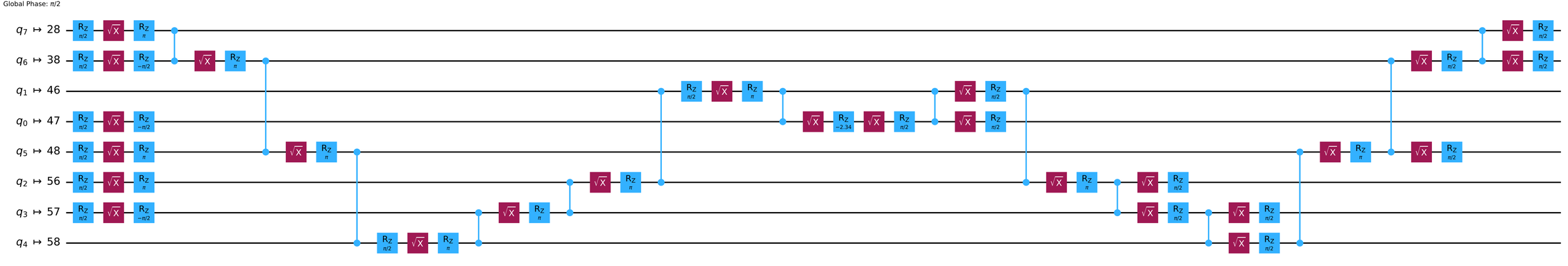
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
X
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
R_Z
π
If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
X
R_Z
$\pi/2$
$\sqrt{X}$
R_Z
$-\pi/2$
R_Z
-2.46
16
17
21
22
15
19
20
25
[expr]
[expr]
[expr]


23
24
If
X
mid_25=0x1
28
29
RZ
−π/2
X
If
X
mid_23=0x1
If
X
mid_24=0x1
26
√X
RZ
−0.886
If
RZ
π
mid_23=0x1
If
X
mid_24=0x1
If
X
mid_26=0x1
RZ
−π/2
27
If
mid[2] ^ mid[3] ^ mid[4] ^ mid...
RZ
π
[expr]


If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
X
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
R_Z
π
If
X
If
X
If
R_Z
π
If
mid[6] ^ mid[15] ^ mid[22]
X
If
X
If
X
pr]
mid_27=0x1
[expr]
mid_27=0x1
mid_15=0x1
mid_15=0x1
[expr]
mid_29=0x1
[ex

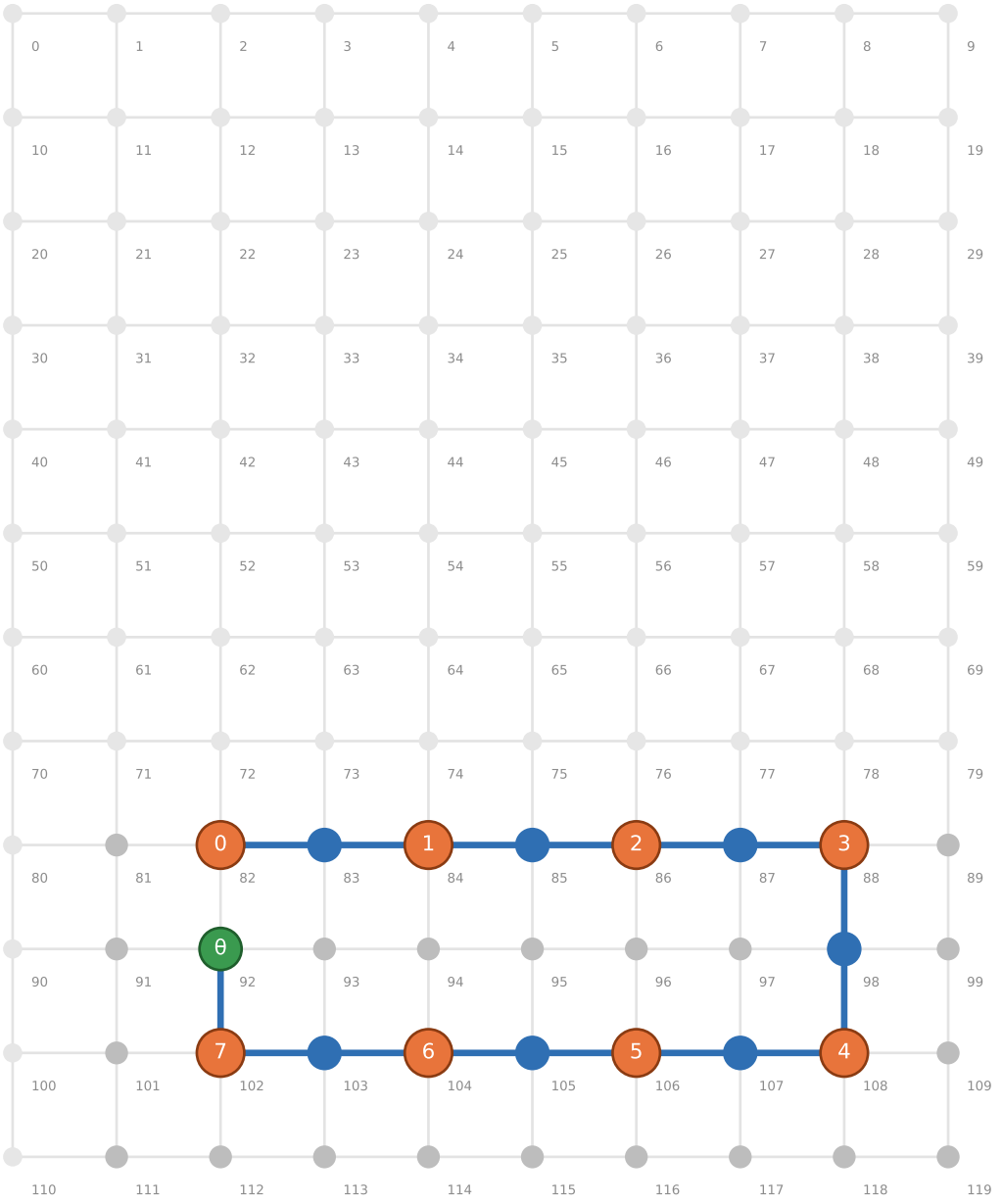

If
X
If
mid[6] ^ mid[15] ^ mid[22] ^ m...
R_Z
π
If
X
If
mid[5] ^ mid[6] ^ mid[15] ^ mi...
R_Z
π
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
X
If
R_Z
π
pr]
mid_29=0x1
mid_28=0x1
[expr]
[expr]
mid_26=0x1

[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:34 sx:26 cz:14)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 49

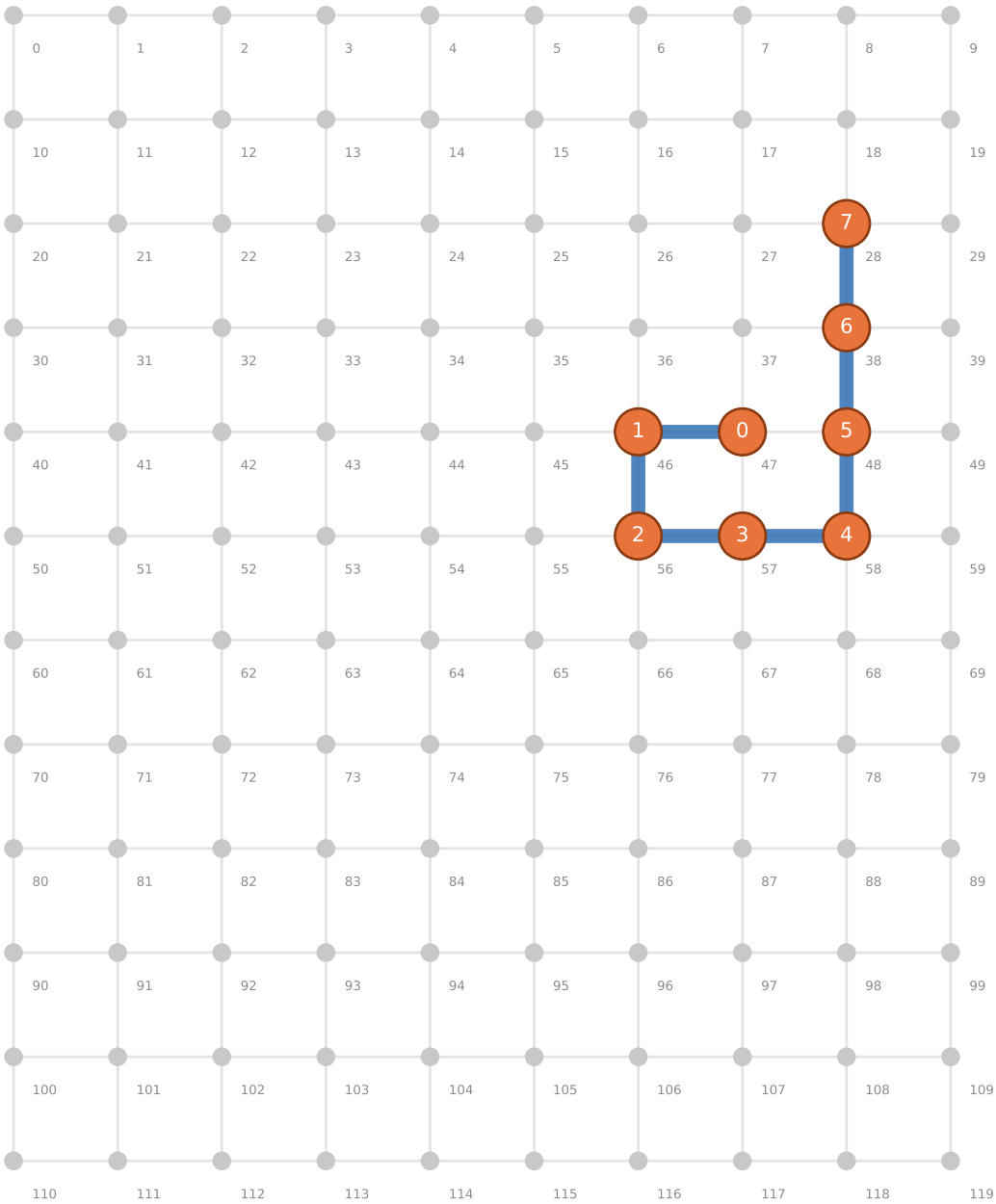


[XYZXYZXY] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

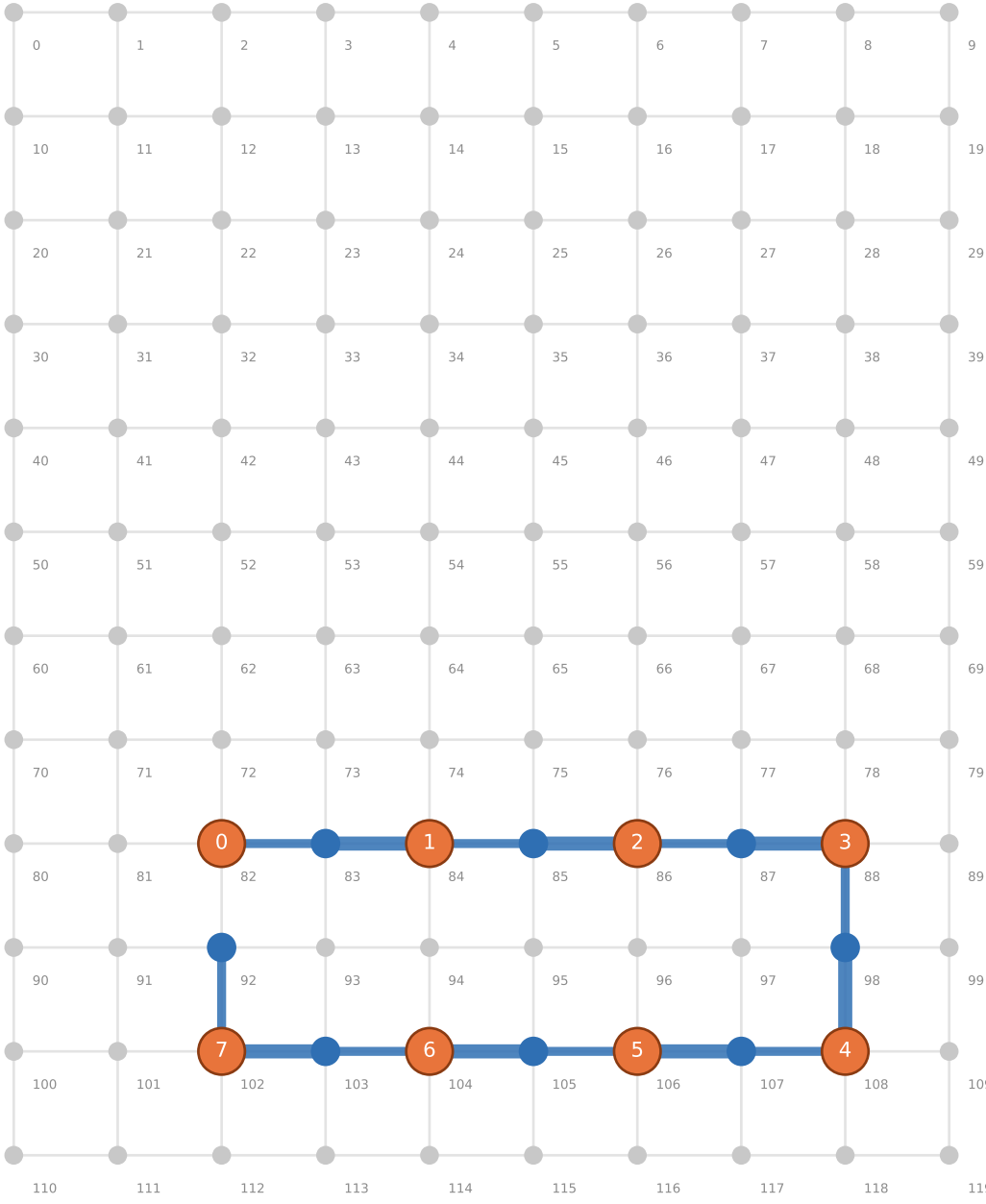
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers



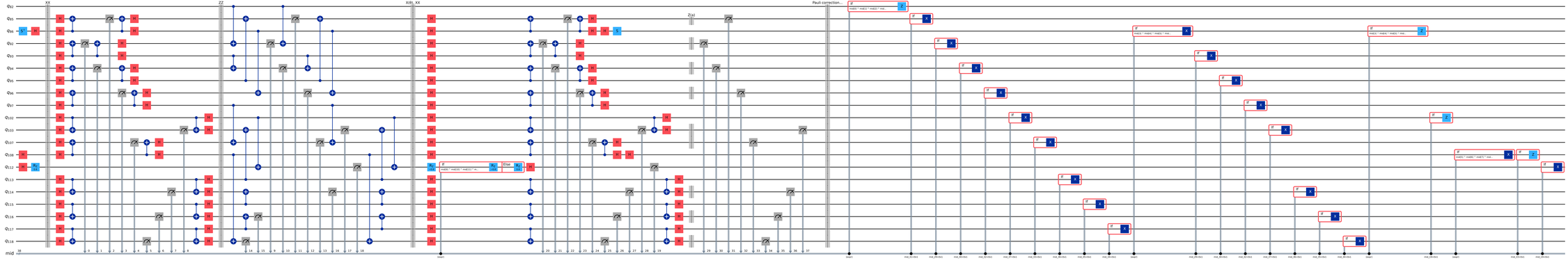
[XYZXYZXY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [47, 46, 56, 57, 58, 48, 38, 28] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



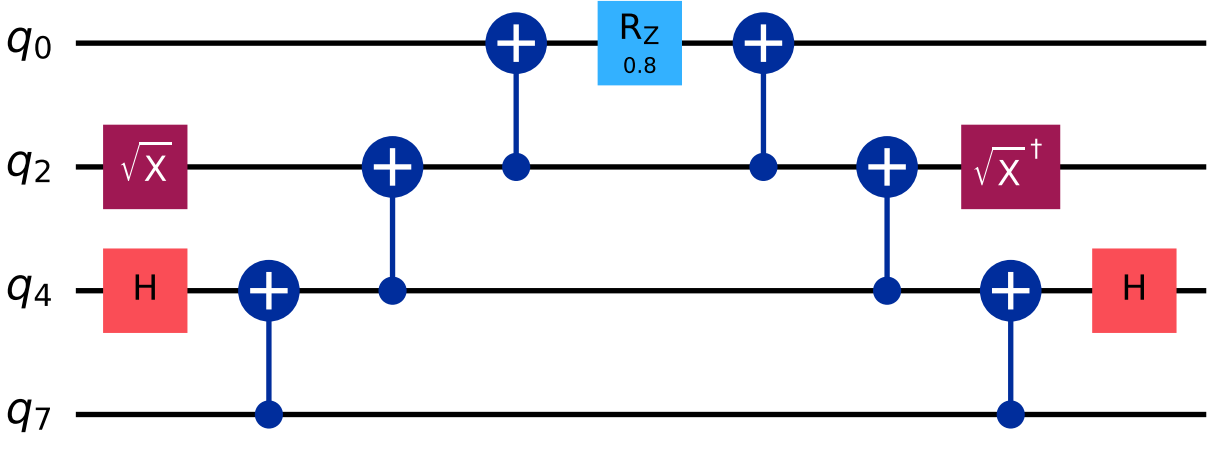
[XYZXYZXY] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 44



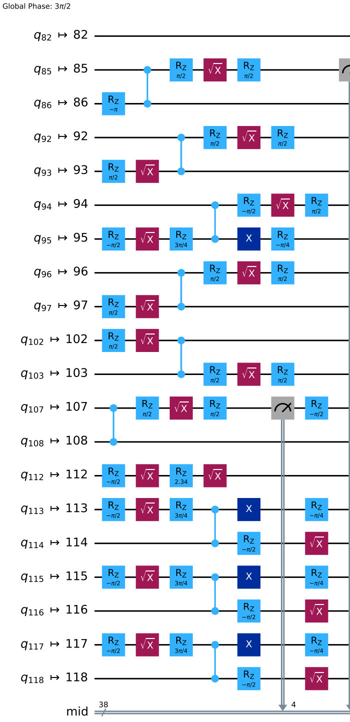
[ZIXIYIZ_gapped] DYNAMIC constant-depth circuit, fault-tolerant (|0> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P)_θ=0.4: P = ZQ Y2 X4 Z7 (8 logical qubits)
2q depth 6, 38 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]

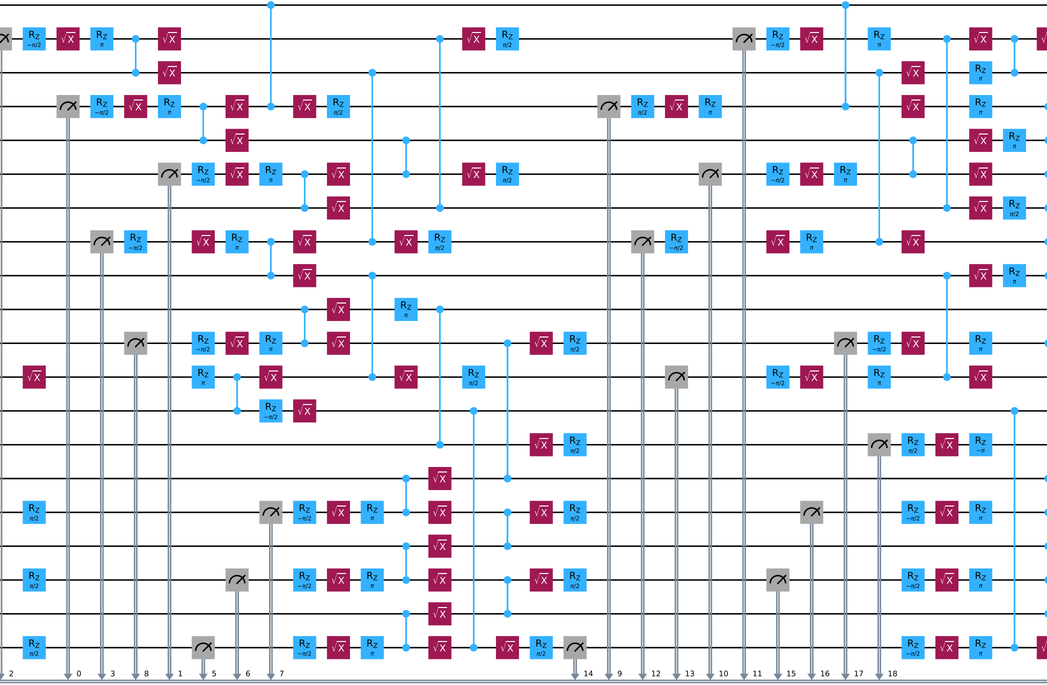


[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6

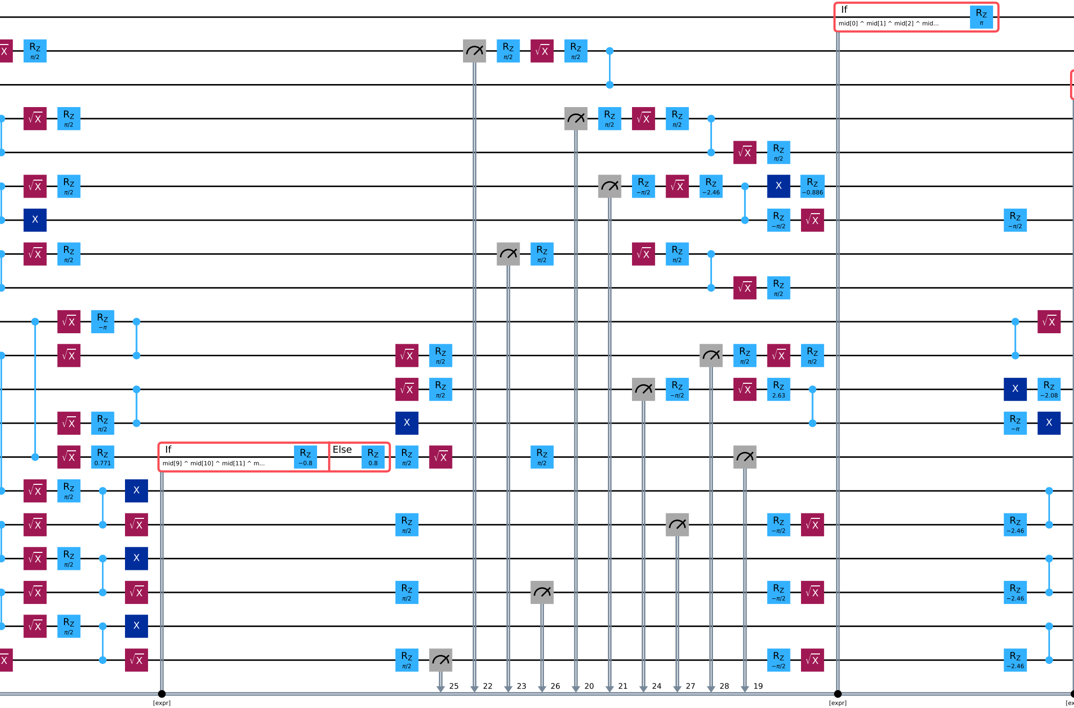


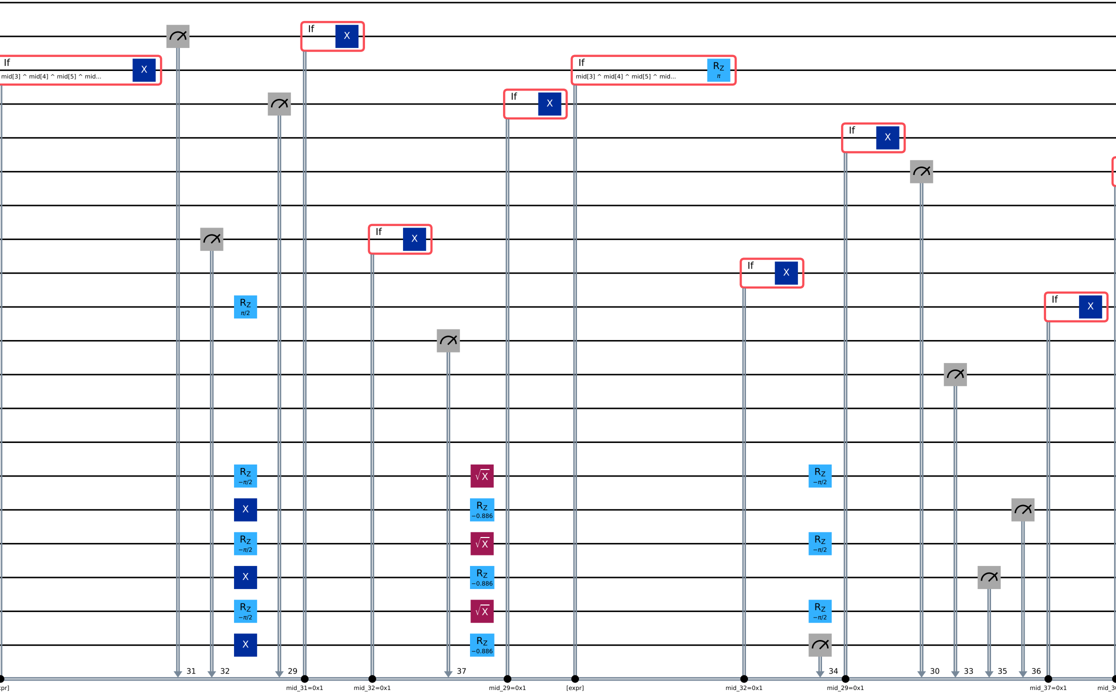
[ZIIXIYIZ_gapped] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:143 sx:110 cz:56 measure:38 if_else:24 x:15)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6, total depth 37

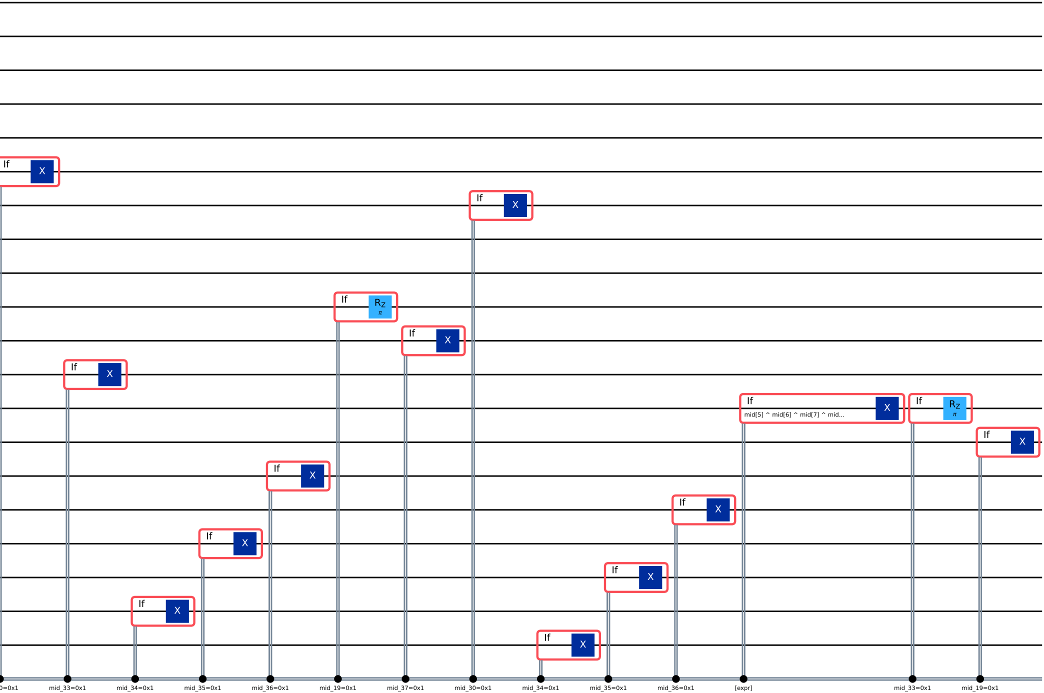

Global Phase: 3π/2
q_{82} ↦ 82
q_{85} ↦ 85
q_{86} ↦ 86
q_{92} ↦ 92
q_{93} ↦ 93
q_{94} ↦ 94
q_{95} ↦ 95
q_{96} ↦ 96
q_{97} ↦ 97
q_{102} ↦ 102
q_{103} ↦ 103
q_{107} ↦ 107
q_{108} ↦ 108
q_{112} ↦ 112
q_{113} ↦ 113
q_{114} ↦ 114
q_{115} ↦ 115
q_{116} ↦ 116
q_{117} ↦ 117
q_{118} ↦ 118
mid
38
4
Rz −π
Rz π/2
Rz −π/2
Rz 3π/4
Rz −π/4
Rz 2.34
√X
X



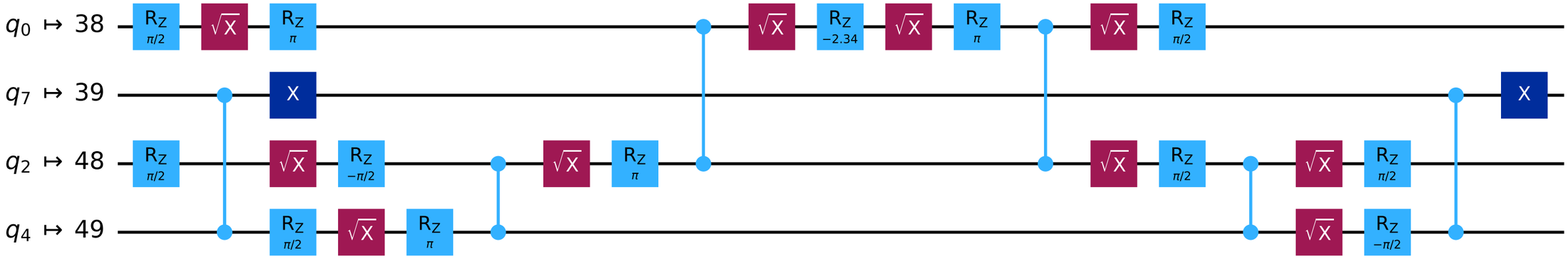
RZ
−π/2
π/2
π
−π
√X
2
0
3
8
1
5
6
7
14
9
12
13
10
11
15
16
17
18


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
Rz π
If
mid[9] ^ mid[10] ^ mid[11] ^ m...
Rz −0.8
Else
Rz 0.8
Rz π/2
Rz −π/2
Rz −π
Rz 0.771
Rz −2.46
Rz −0.886
Rz 2.63
Rz −2.08
√X
X
25
22
23
26
20
21
24
27
28
19
[expr]
[expr]


If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
X
If
X
If
mid[3] ^ mid[4] ^ mid[5] ^ mid...
R_Z
π
If
X
If
X
If
X
If
X
If
X
R_Z
$\pi/2$
R_Z
$-\pi/2$
X
R_Z
$-\pi/2$
X
R_Z
$-\pi/2$
X
$\sqrt{X}$
R_Z
-0.886
$\sqrt{X}$
R_Z
-0.886
$\sqrt{X}$
R_Z
-0.886
R_Z
$-\pi/2$
R_Z
$-\pi/2$
R_Z
$-\pi/2$
31
32
29
37
34
30
33
35
36
pr]
mid_31=0x1
mid_32=0x1
mid_29=0x1
[expr]
mid_32=0x1
mid_29=0x1
mid_37=0x1
mid_3

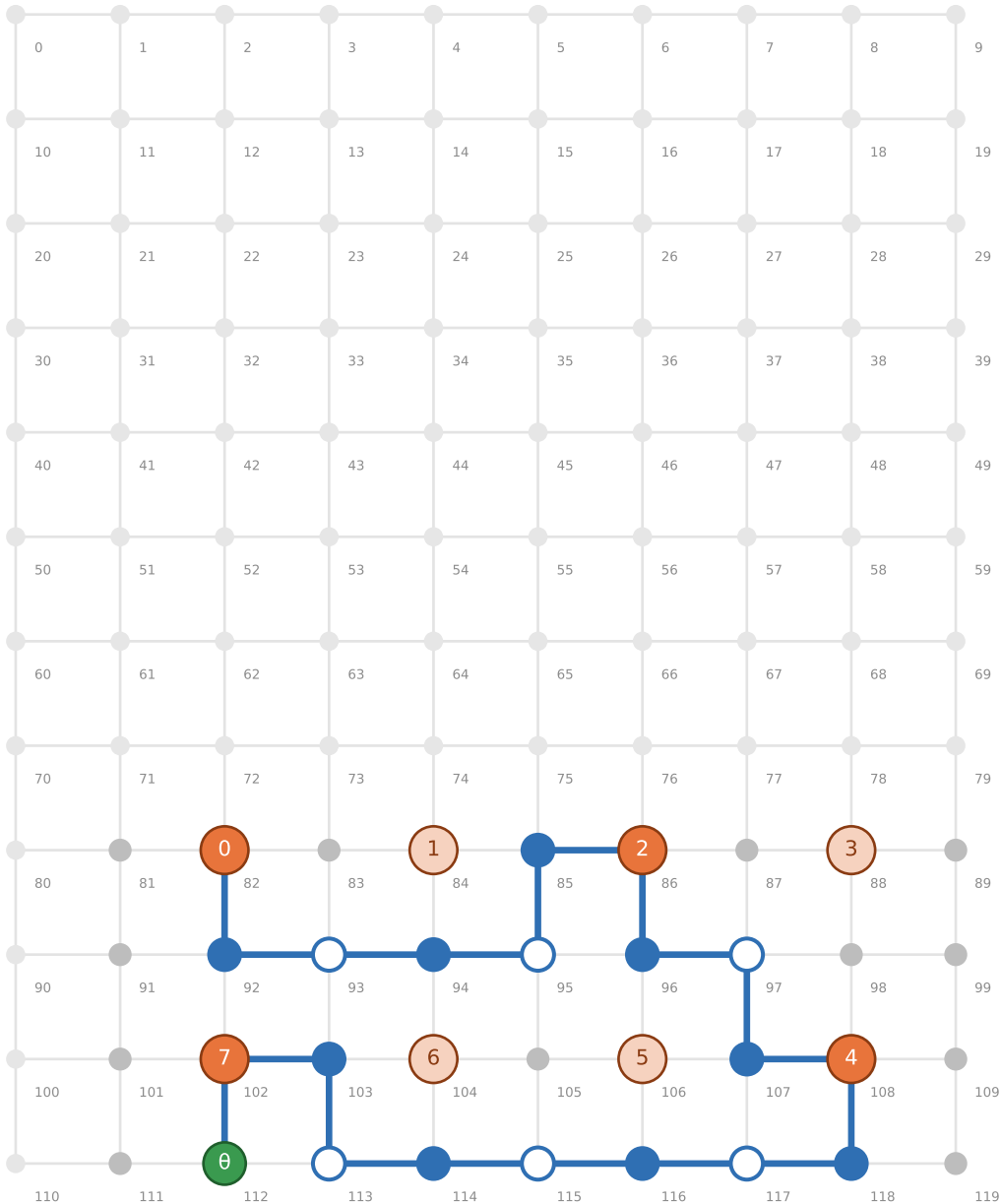

If
X
If
X
If
X
If
X
If
X
If
R_Z
π
If
X
If
X
If
X
If
X
If
X
If
mid[5] ^ mid[6] ^ mid[7] ^ mid...
X
If
R_Z
π
If
X
0=0x1
mid_33=0x1
mid_34=0x1
mid_35=0x1
mid_36=0x1
mid_19=0x1
mid_37=0x1
mid_30=0x1
mid_34=0x1
mid_35=0x1
mid_36=0x1
[expr]
mid_33=0x1
mid_19=0x1

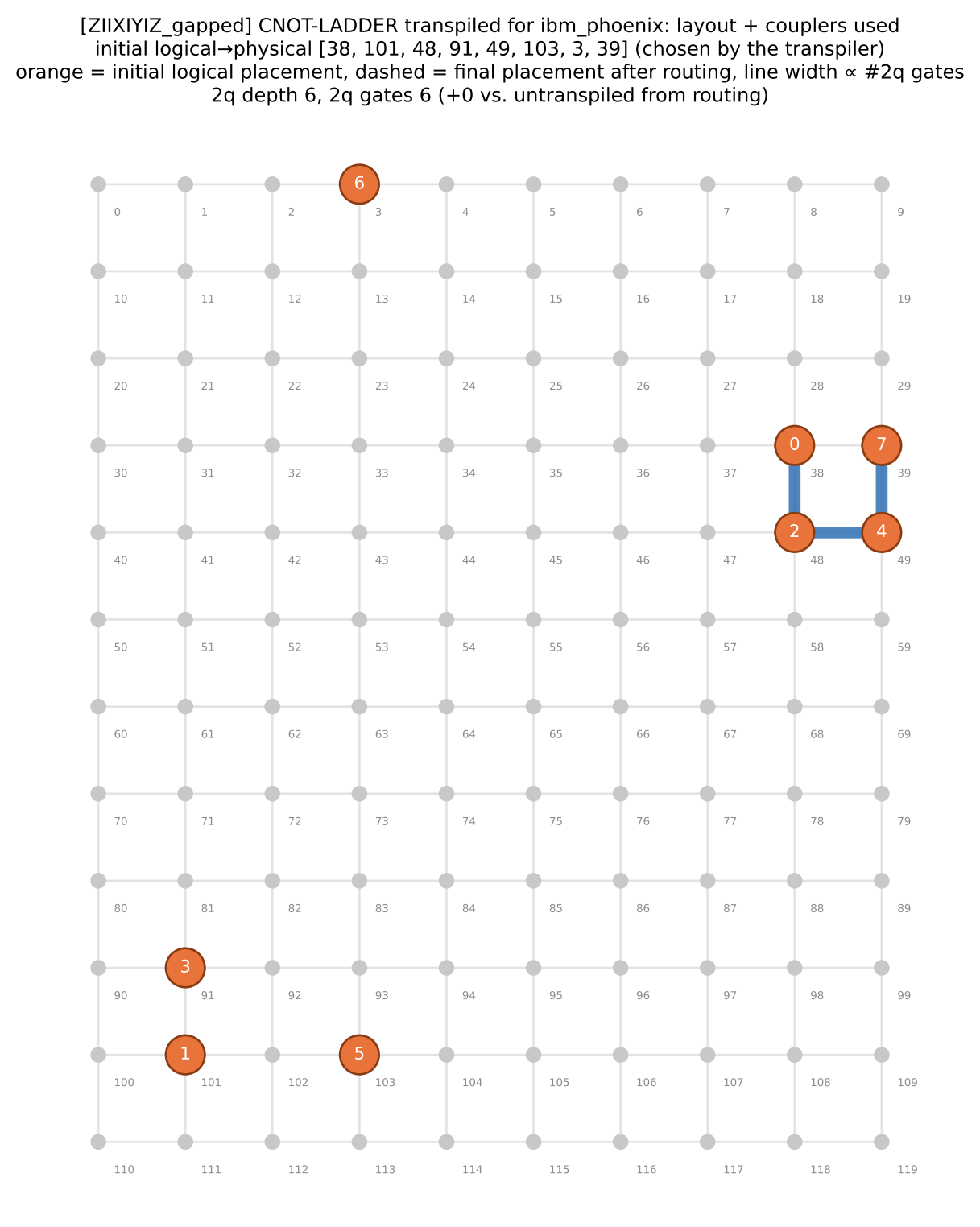
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7 (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



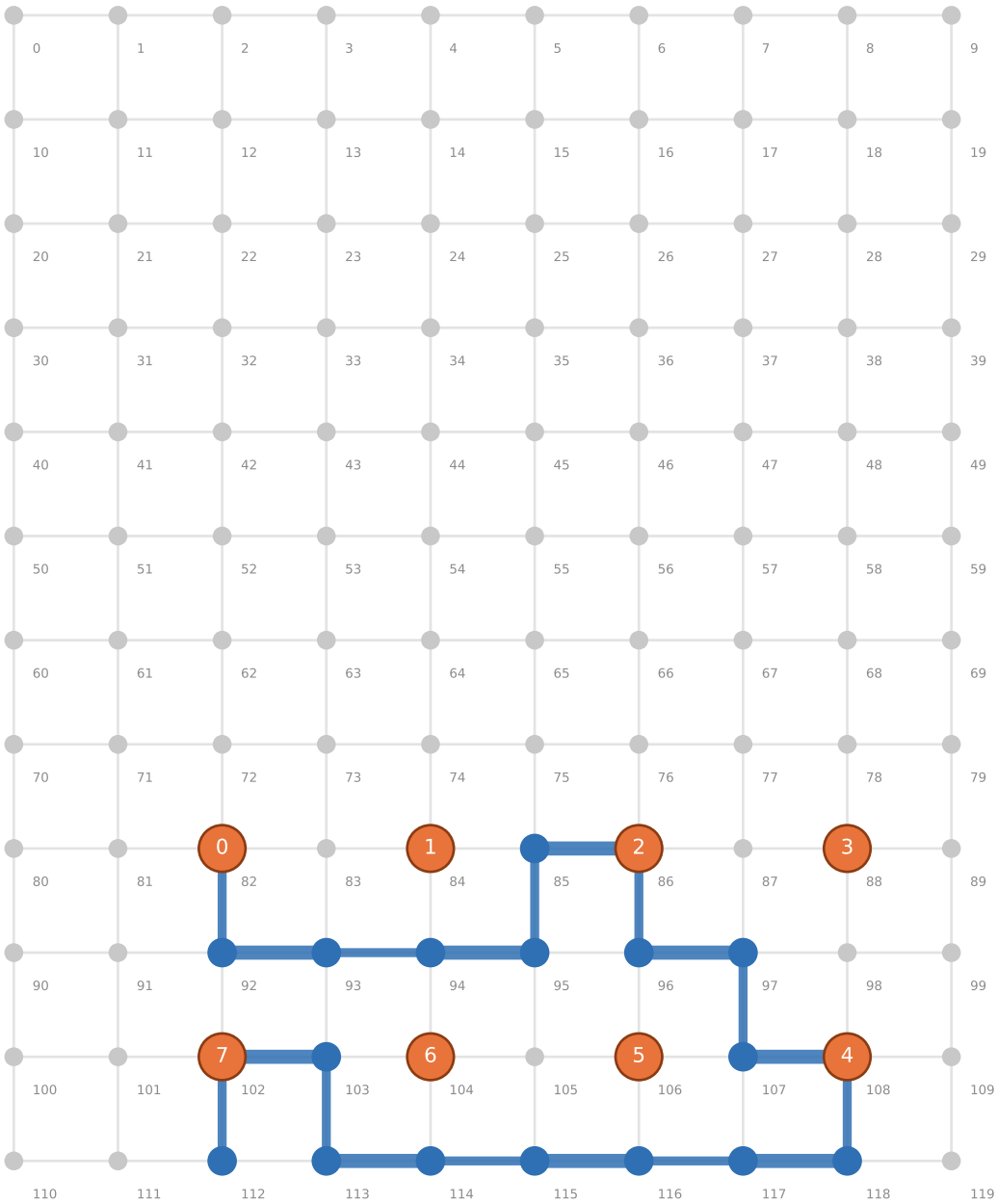
[ZIIXIYIZ_gapped] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers

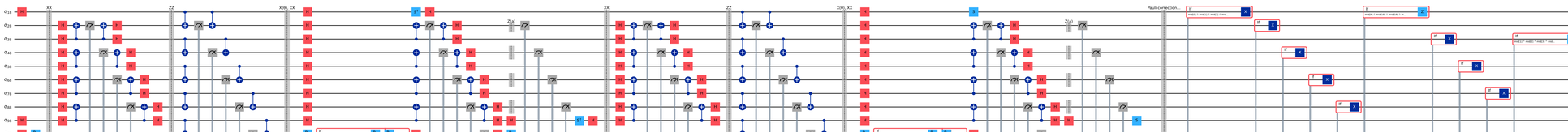



[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 101, 48, 91, 49, 103, 3, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119

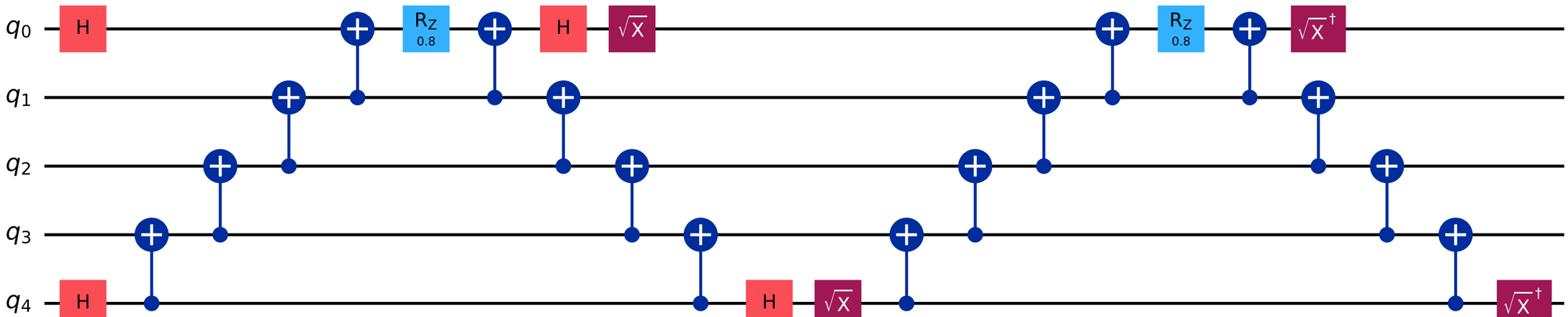
[ZIIXIYIZ_gapped] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 56



[XZZZX+YZZZY] DYNAMIC constant-depth circuit, fault-tolerant (@> ancilla) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 ·Y0 Z1 Z2 Z3 Y4 · (5 logical qubits)
2q depth 12, 36 mid-circuit measurements, logical~physical [18, 38, 58, 78, 98]

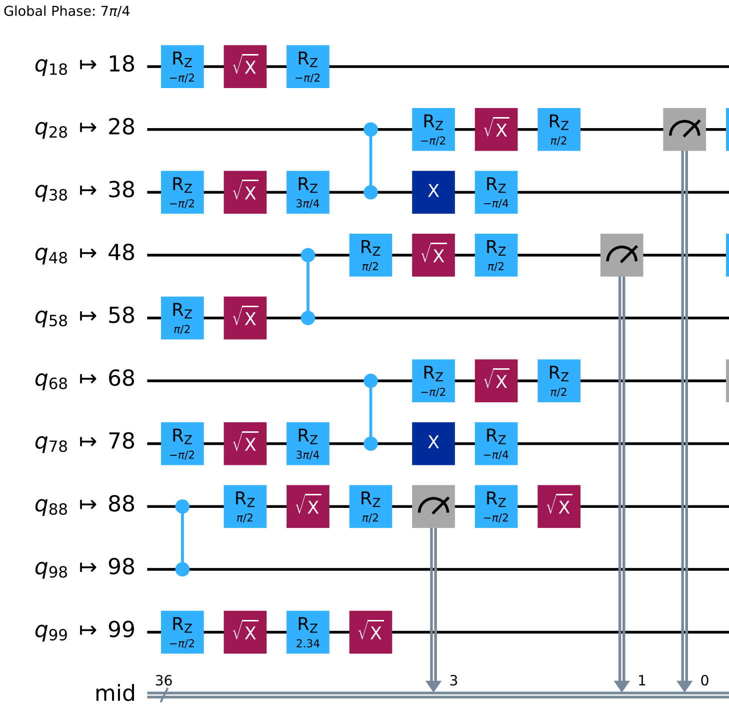


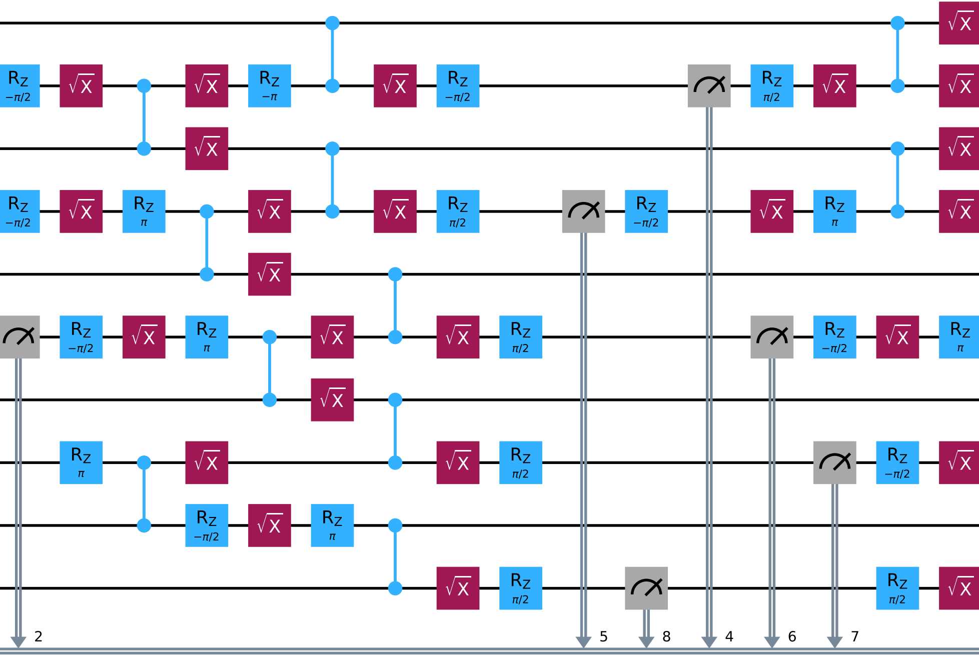
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



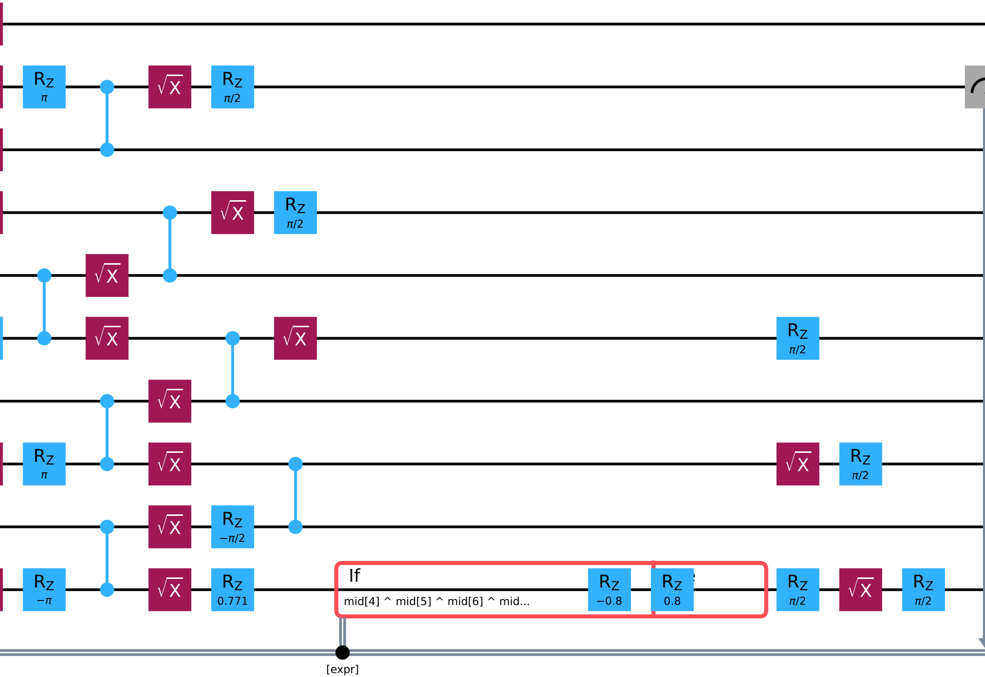
[XZZZX+YZZZY] DYNAMIC circuit, fault-tolerant (|θ> ancilla), TRANSPILED for ibm_phoenix (ISA: rz:122 sx:101 cz:52
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 12, total depth 70

measure:36 if_else:17 x:6)

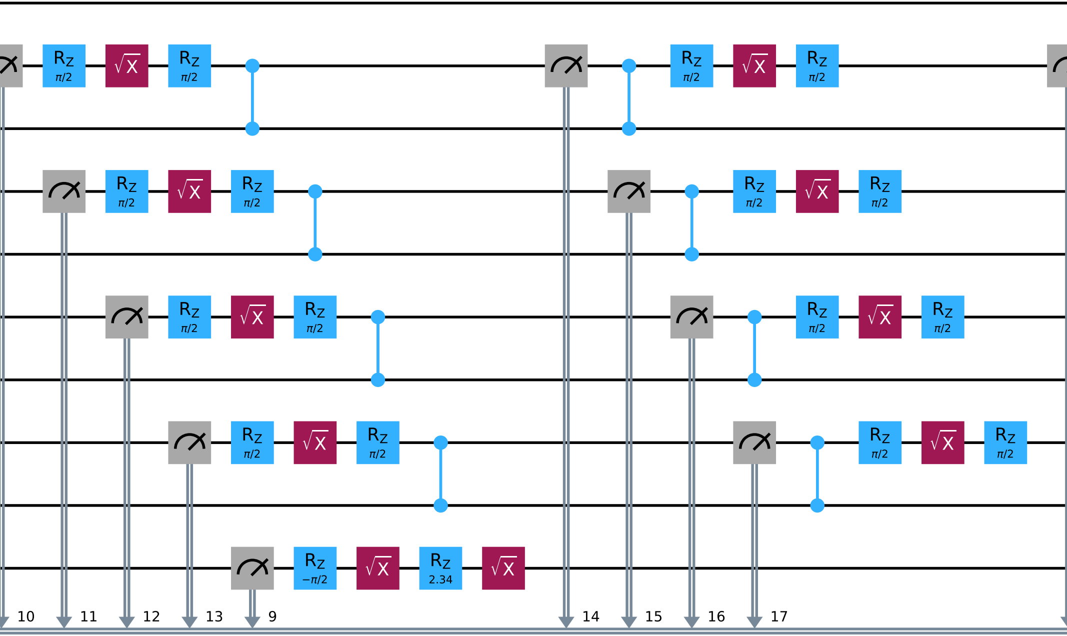

Global Phase: 7π/4
q_{18} ↦ 18
q_{28} ↦ 28
q_{38} ↦ 38
q_{48} ↦ 48
q_{58} ↦ 58
q_{68} ↦ 68
q_{78} ↦ 78
q_{88} ↦ 88
q_{98} ↦ 98
q_{99} ↦ 99
mid
36
R_Z −π/2
√X
R_Z π/2
R_Z 3π/4
R_Z −π/4
R_Z 2.34
X
3
1
0

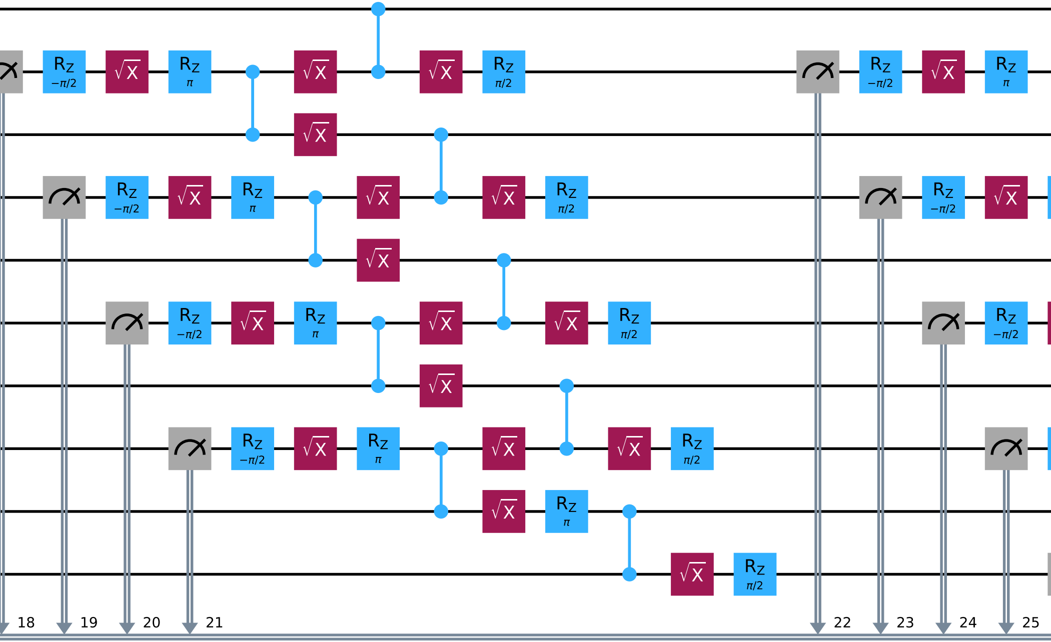


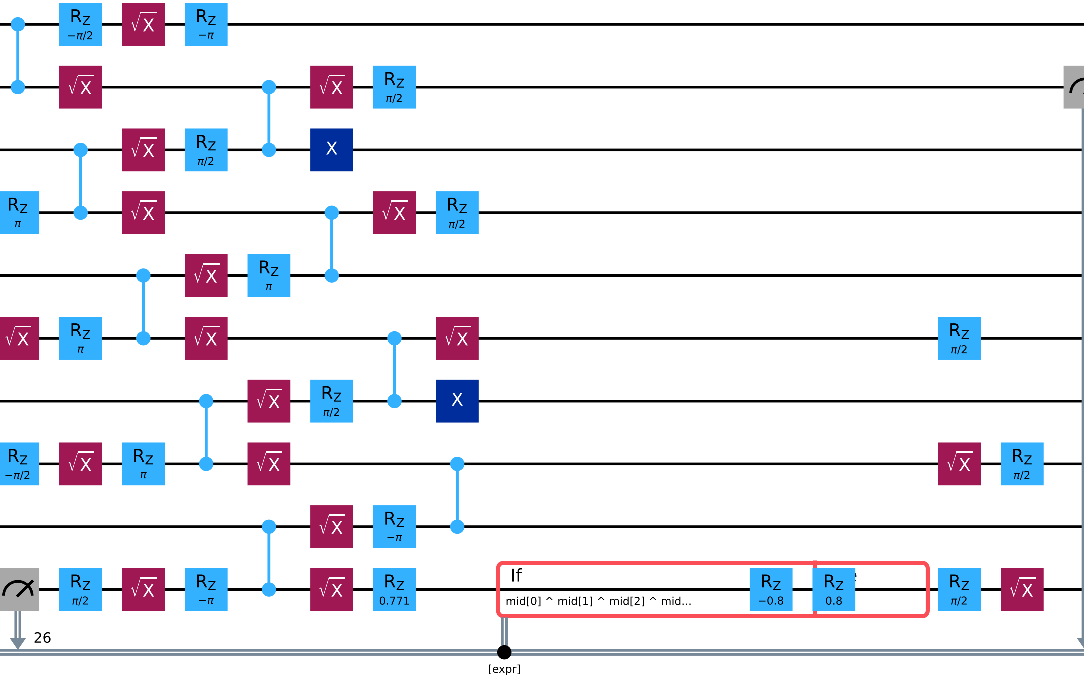
R_Z
$-\pi/2$
$\pi/2$
$-\pi$
π
$\sqrt{X}$
2
5
8
4
6
7



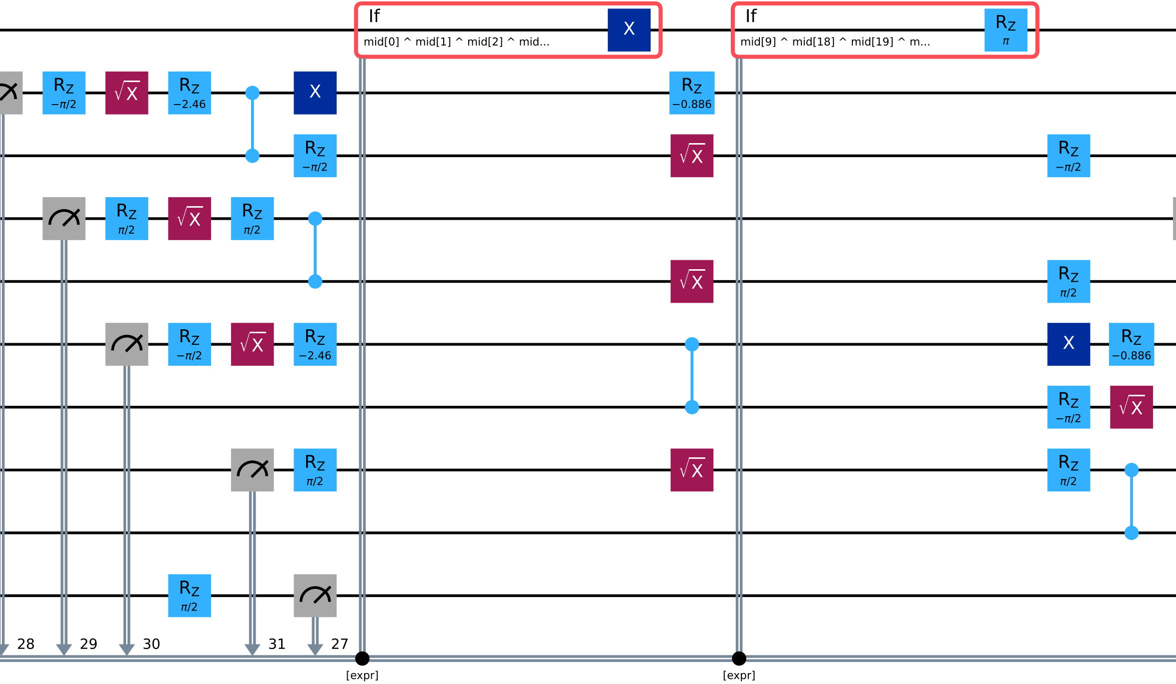
R_Z π
√X
R_Z π/2
√X
R_Z π/2
√X
√X
√X
R_Z π/2
√X
R_Z π
√X
√X
R_Z π/2
√X
R_Z −π/2
R_Z −π
√X
R_Z 0.771
If
mid[4] ^ mid[5] ^ mid[6] ^ mid...
R_Z −0.8
R_Z 0.8
R_Z π/2
√X
R_Z π/2
[expr]

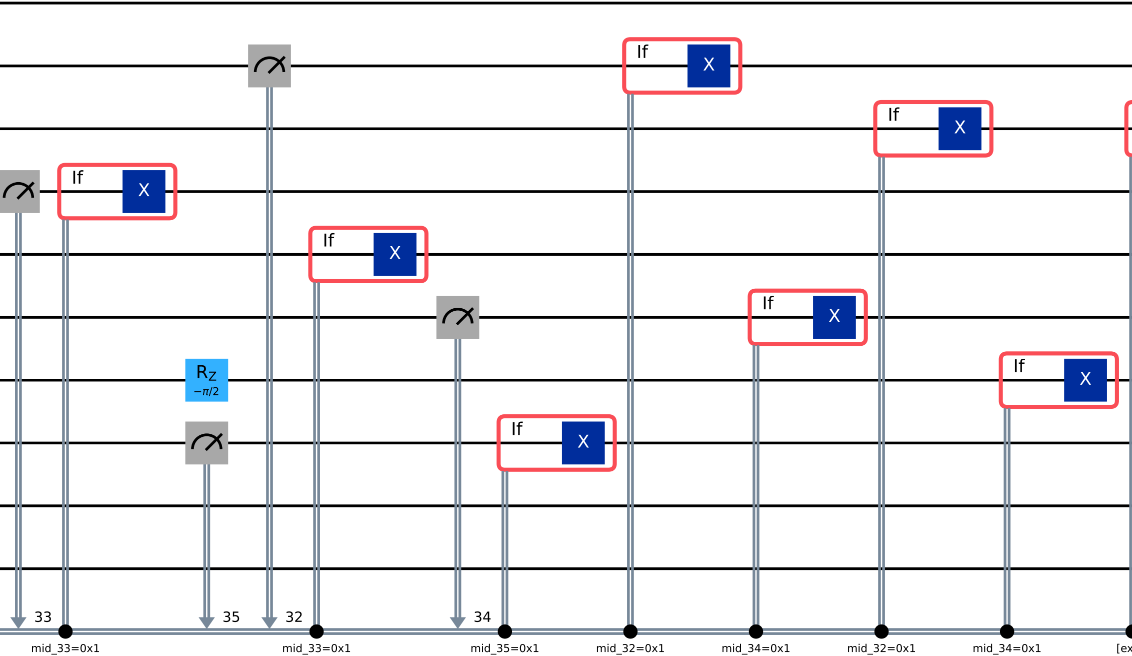

R_Z
π/2
√X
−π/2
2.34
10
11
12
13
9
14
15
16
17

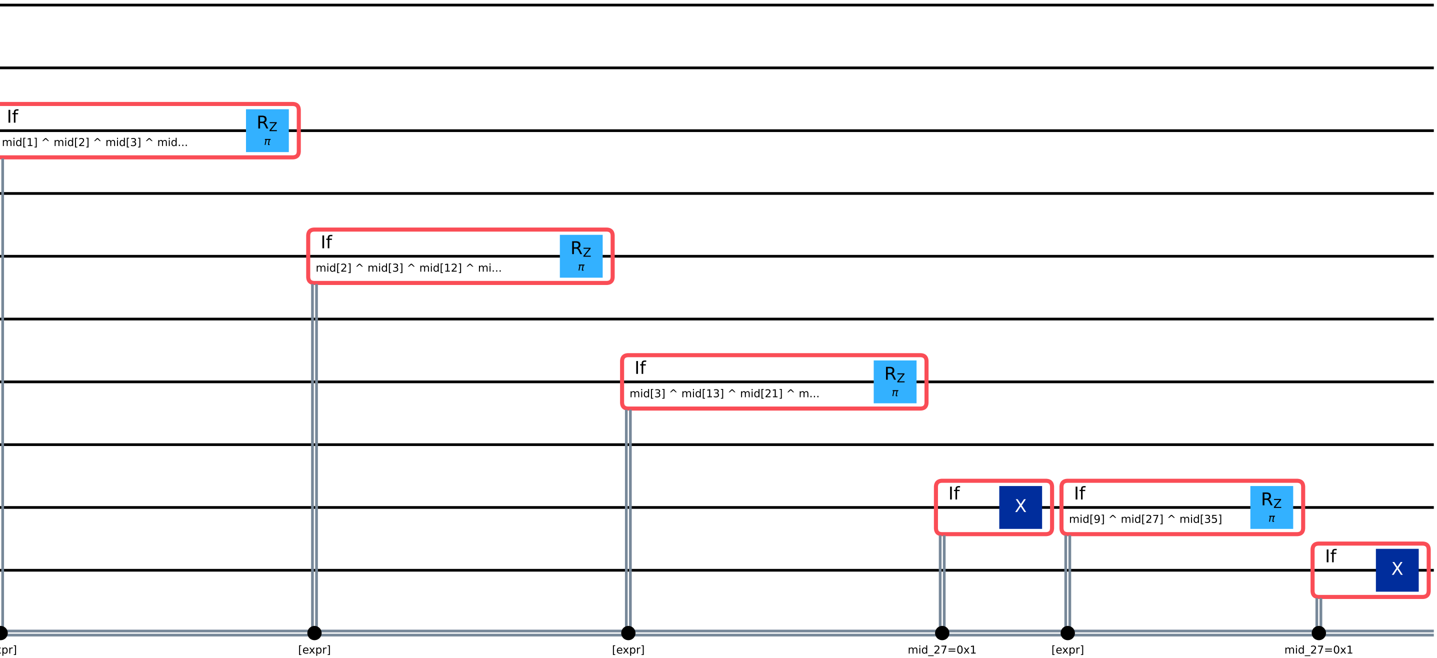

RZ
−π/2
√X
RZ
π
RZ
π/2
18
19
20
21
22
23
24
25



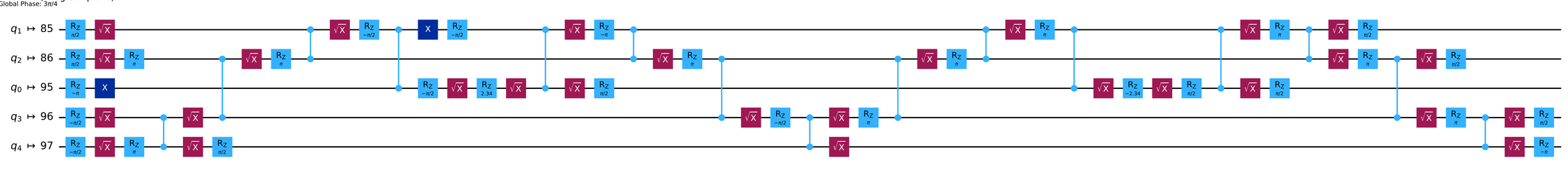
RZ −π/2
√X
RZ −π
√X
√X
RZ π/2
√X
RZ π/2
X
RZ π
√X
√X
RZ π/2
√X
RZ π
√X
RZ π
√X
√X
RZ π/2
√X
RZ π/2
X
RZ −π/2
√X
RZ π
√X
√X
RZ π/2
√X
RZ −π
RZ π/2
√X
RZ −π
√X
RZ 0.771
If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
RZ −0.8
RZ 0.8
RZ π/2
√X
26
[expr]


If
mid[0] ^ mid[1] ^ mid[2] ^ mid...
X
If
mid[9] ^ mid[18] ^ mid[19] ^ m...
R_Z π
R_Z $-\pi/2$
$\sqrt{X}$
R_Z -2.46
X
R_Z -0.886
R_Z $-\pi/2$
$\sqrt{X}$
R_Z $-\pi/2$
R_Z $\pi/2$
$\sqrt{X}$
R_Z $\pi/2$
$\sqrt{X}$
R_Z $\pi/2$
R_Z $-\pi/2$
$\sqrt{X}$
R_Z -2.46
X
R_Z -0.886
R_Z $-\pi/2$
$\sqrt{X}$
R_Z $\pi/2$
$\sqrt{X}$
R_Z $\pi/2$
R_Z $\pi/2$
28
29
30
31
27
[expr]
[expr]


If
X
R$_Z$
−π/2
33
35
32
34
mid_33=0x1
mid_33=0x1
mid_35=0x1
mid_32=0x1
mid_34=0x1
mid_32=0x1
mid_34=0x1
[ex

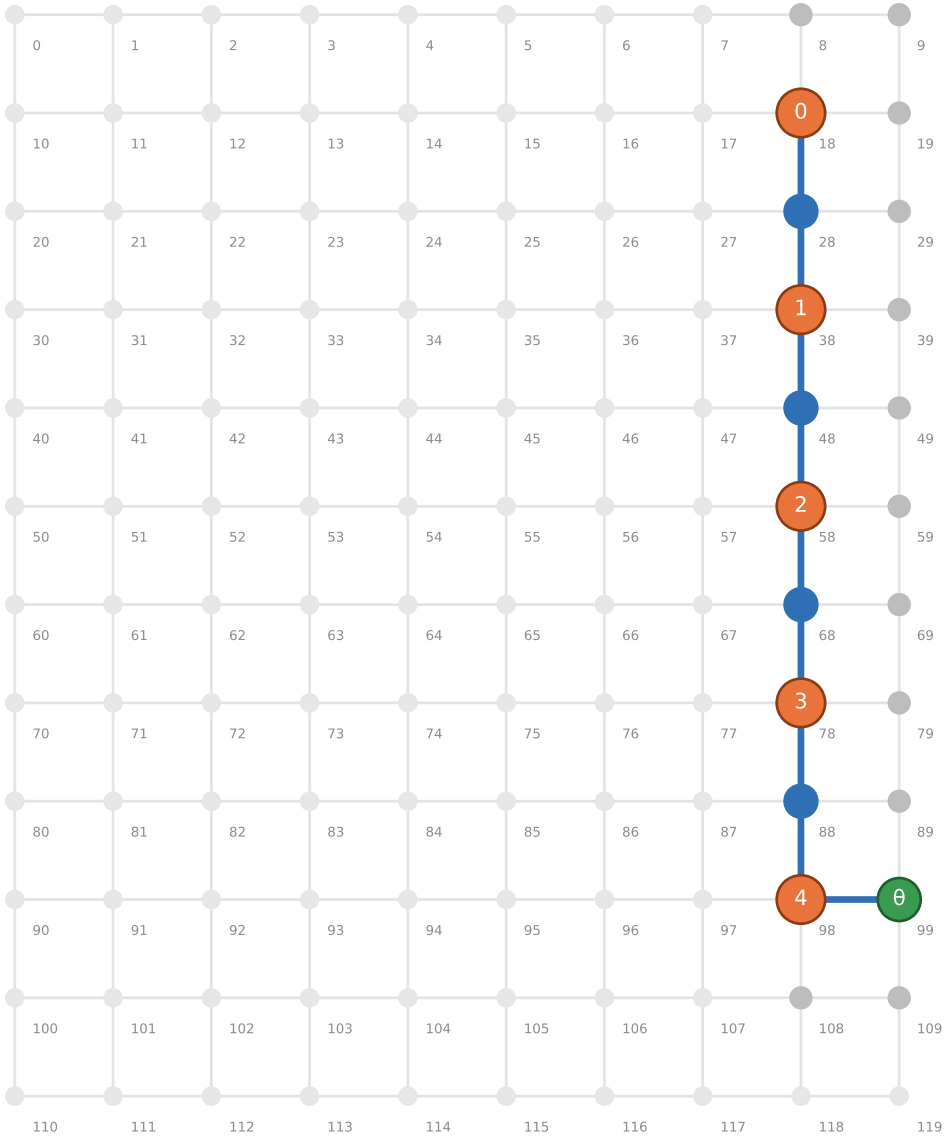

If
mid[1] ^ mid[2] ^ mid[3] ^ mid...
R$_Z$
π
If
mid[2] ^ mid[3] ^ mid[12] ^ mi...
R$_Z$
π
If
mid[3] ^ mid[13] ^ mid[21] ^ m...
R$_Z$
π
If
X
If
mid[9] ^ mid[27] ^ mid[35]
R$_Z$
π
If
X
pr]
[expr]
[expr]
mid_27=0x1
[expr]
mid_27=0x1

[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:30 sx:28 cz:15 x:2)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical qubits)
2q depth 15, total depth 51

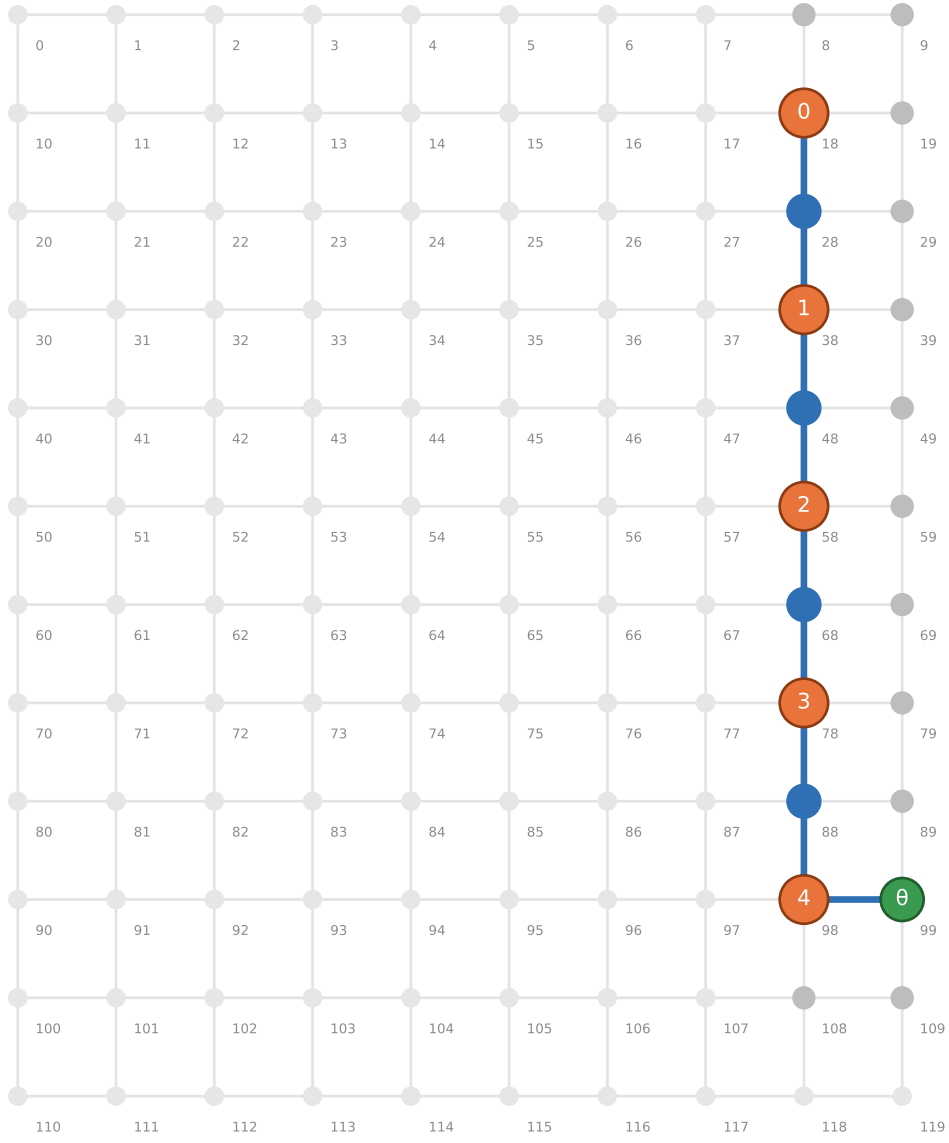


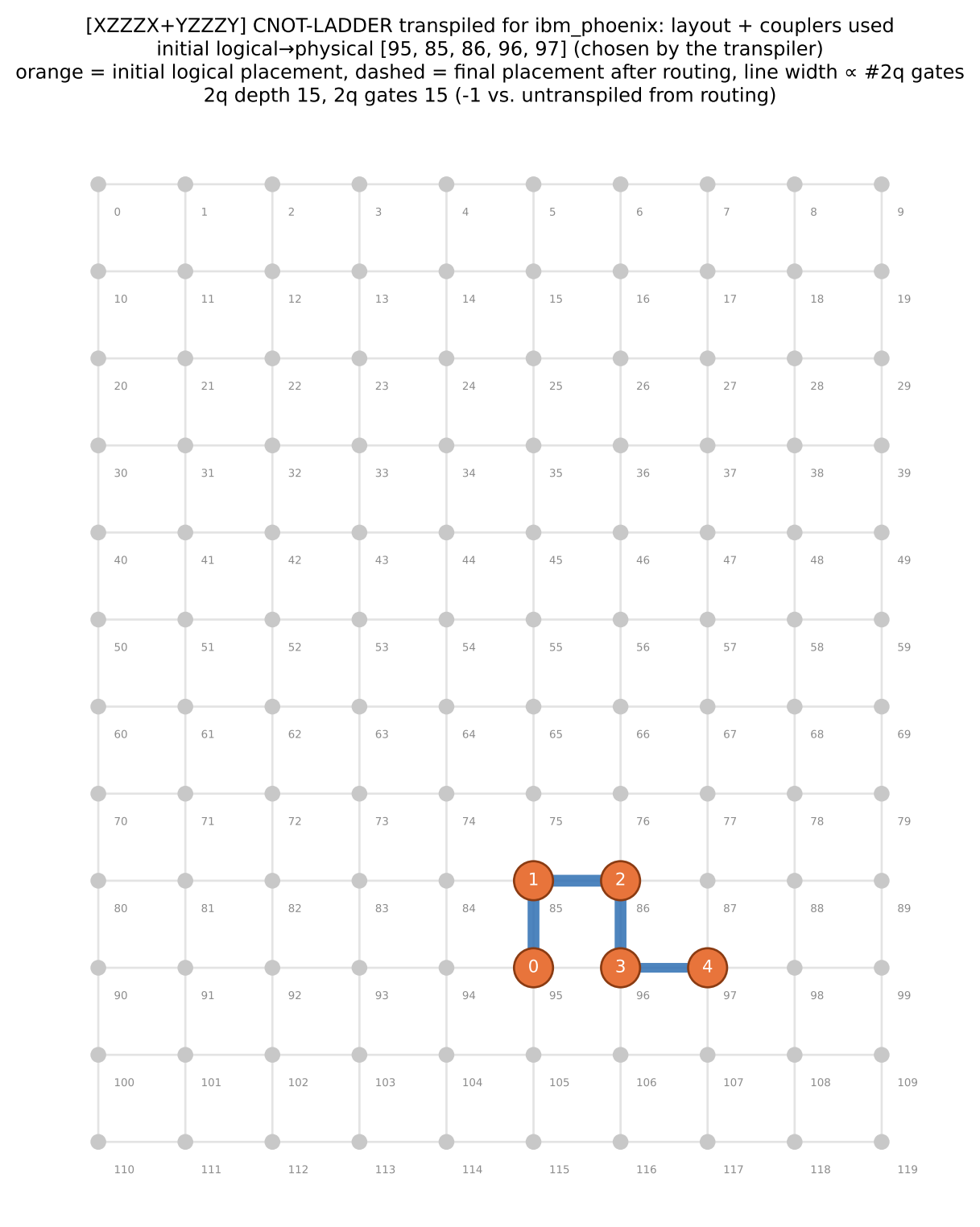
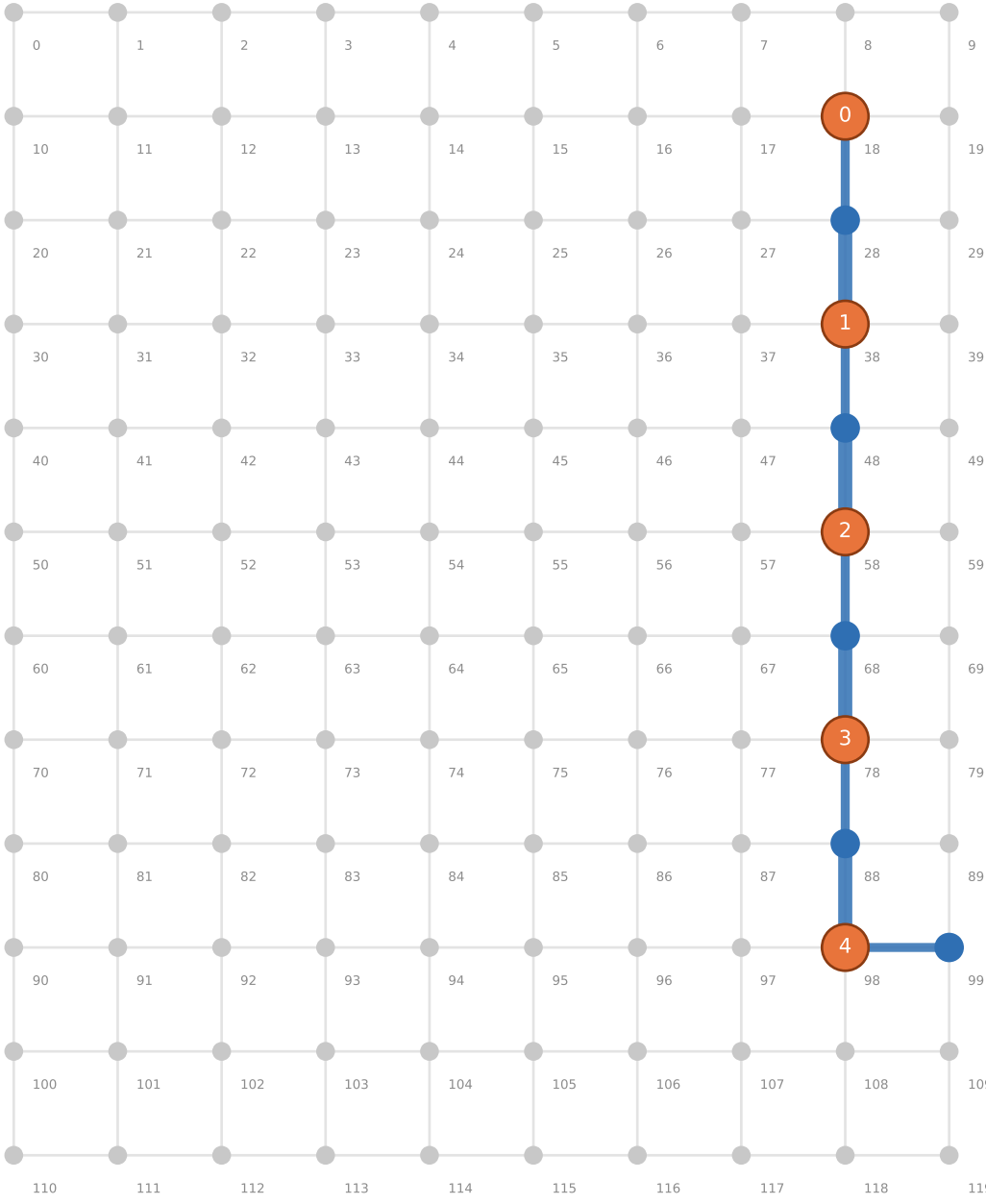
[XZZZX+YZZZY] DYNAMIC circuit layout, fault-tolerant (|θ> ancilla), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, green θ = |θ> ancilla, blue line = chain

term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers




[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 86, 96, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
1
2
0
3
4

[XZZZX+YZZZY] DYNAMIC circuit (ft) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 12, 2q gates 52